

The deubiquitinase Ubp3/Usp10 constrains glucose-mediated mitochondrial repression via phosphate budgeting

Reviewed Preprint

Revised by authors after peer review.

About eLife's process

Reviewed preprint version 2

March 12, 2024 (this version)

Reviewed preprint version 1

November 3, 2023

Posted to preprint server

August 25, 2023

Sent for peer review

August 22, 2023

Vineeth Vengayil, Shreyas Niphadkar, Swagata Adhikary, Sriram Varahan, Sunil Laxman 

Institute for Stem Cell Science and Regenerative Medicine (DBT-inStem) GKV Post Bellary Road Bangalore 560065

• Manipal Academy of Higher Education, Manipal, Karnataka 576104, India

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

Abstract

Many cells in high glucose repress mitochondrial respiration, as observed in the Crabtree and Warburg effects. Our understanding of biochemical constraints for mitochondrial activation is limited. Using a *Saccharomyces cerevisiae* screen, we identified the conserved deubiquitinase Ubp3 (Usp10), as necessary for mitochondrial repression. Ubp3 mutants have increased mitochondrial activity despite abundant glucose, along with decreased glycolytic enzymes, and a rewired glucose metabolic network with increased trehalose production. Utilizing $\Delta ubp3$ cells, along with orthogonal approaches, we establish that the high glycolytic flux in glucose continuously consumes free Pi. This restricts mitochondrial access to inorganic phosphate (Pi), and prevents mitochondrial activation. Contrastingly, rewired glucose metabolism with enhanced trehalose production and reduced GAPDH (as in $\Delta ubp3$ cells) restores Pi. This collectively results in increased mitochondrial Pi and derepression, while restricting mitochondrial Pi transport prevents activation. We therefore suggest that glycolytic-flux dependent intracellular Pi budgeting is a key constraint for mitochondrial repression.

eLife assessment

This **valuable** study informs on the regulation of metabolic flux between glycolysis and respiration in yeast, mediated by inorganic phosphate. The authors propose a novel mechanism involving Ubp3/Usp10 that potentially mitigates the Crabtree effect, based on an array of assays that offer substantial insight into mitochondrial biogenesis under high glucose conditions. However, the evidence supporting the conclusions remains **incomplete** because of a limited number of replicates, particularly in protein blot analyses with insufficient normalization methods, which undermines the robustness of the findings. This work could influence the subfield significantly if the methodological weaknesses are addressed to provide more support for the proposed model.

Introduction

Rapidly proliferating cells have substantial metabolic and energy demands in order to increase biomass (1, 2). This includes a high ATP demand, obtained from cytosolic glycolysis or mitochondrial oxidative phosphorylation (OXPHOS), to fuel multiple reactions (3). Interestingly, many rapidly proliferating cells preferentially rely on ATP from glycolysis/fermentation over mitochondrial respiration even in oxygen-replete conditions, and is the well-known Warburg effect (4, 5). Many such cells repress mitochondrial processes in high glucose, termed glucose-mediated mitochondrial repression or the Crabtree effect (6, 7). This is observed in some tumors (5), neutrophils (8), activated macrophages (9), stem cells (10–12), and famously *Saccharomyces cerevisiae* (7). Numerous studies have identified signaling programs or regulators of glucose-mediated mitochondrial repression. However, biochemical programs and regulatory processes in biology evolve around key biochemical constraints (13). The biochemical constraints for mitochondrial repression remain unresolved (14, 15).

There are two hypotheses on the biochemical principles driving mitochondrial repression. The first proposes direct roles for glycolytic intermediates in driving mitochondrial respiration, by inhibiting specific mitochondrial outputs (16, 17). The second hypothesizes that a competition between glycolytic and mitochondrial processes for mutually required metabolites/co-factors such as pyruvate, ADP or inorganic phosphate (Pi) could determine the extent of mitochondrial repression (14, 15, 18). These are not all mutually exclusive, and a combination of these factors might dictate mitochondrial repression in high glucose. However, any hierarchies of importance are unclear (19), and experimental data for the necessary constraints for mitochondrial repression remains incomplete.

One approach to resolve this question has been to identify regulators of metabolic state under high glucose. Post-translational modifications (PTMs) regulated by signaling systems can regulate mitochondrial repression (20–22). Ubiquitination is a PTM that regulates global proteostasis (23, 24), but the roles of ubiquitination-dependent processes in regulating mitochondrial repression are poorly explored. Ubiquitination itself is determined by the balance between ubiquitination, and deubiquitinase (DUB) dependent deubiquitination (25). Little is known about the roles of DUBs in regulating metabolic states, making the DUBs interesting candidate regulators of mitochondrial repression.

In this study, using an *S. cerevisiae* DUB knock-out library-based screen, we identified the evolutionarily conserved deubiquitinase Ubp3 (mammalian Usp10) as required for mitochondrial repression in high glucose. Loss of Ubp3 resulted in mitochondrial activation, along with a reduction in the glycolytic enzymes-phosphofructokinase 1 (Pfk1) and GAPDH (Tdh2 and Tdh3). This consequently reroutes glucose flux and increases trehalose biosynthesis. This metabolic rewiring increases Pi release from trehalose synthesis, and decreases Pi consumption in glycolysis, to cumulatively increase Pi pools. Using *ubp3Δ* cells along with independent analysis of wild-type cells, and isolated mitochondrial fractions, we establish that glycolytic-flux dependent Pi allocations to mitochondria determines mitochondrial activity. Through these data, we propose how intracellular Pi balance as controlled by glycolytic flux, is a biochemical constraint for mitochondrial repression.

Results

A deubiquitinase deletion screen identifies Ubp3 as a regulator of glucose-mediated mitochondrial repression

In cells such as *S. cerevisiae*, high glucose represses mitochondrial activity as well as OXPHOS-dependent ATP synthesis (7, 26) (illustrated in **Figure 1A**). Our initial objective was to identify proteostasis regulators of glucose-mediated mitochondrial repression. We generated and used a deubiquitinase (DUB) deletion strain library of *S. cerevisiae* (**Figure 1-figure supplement 1A**), to unbiasedly identify regulators of mitochondrial repression by measuring the fluorescence intensity of a potentiometric dye Mitotracker CMXRos (illustrated in **Figure 1B**). Using this screen, we identified DUB mutants with altered mitochondrial membrane potential (**Figure 1C**, **Figure 1-figure supplement 1 A**). Note: WT cells in a respiratory medium (2% ethanol) were used as a control to estimate maximum mitotracker fluorescence intensity (**Figure 1-figure supplement 1B**).

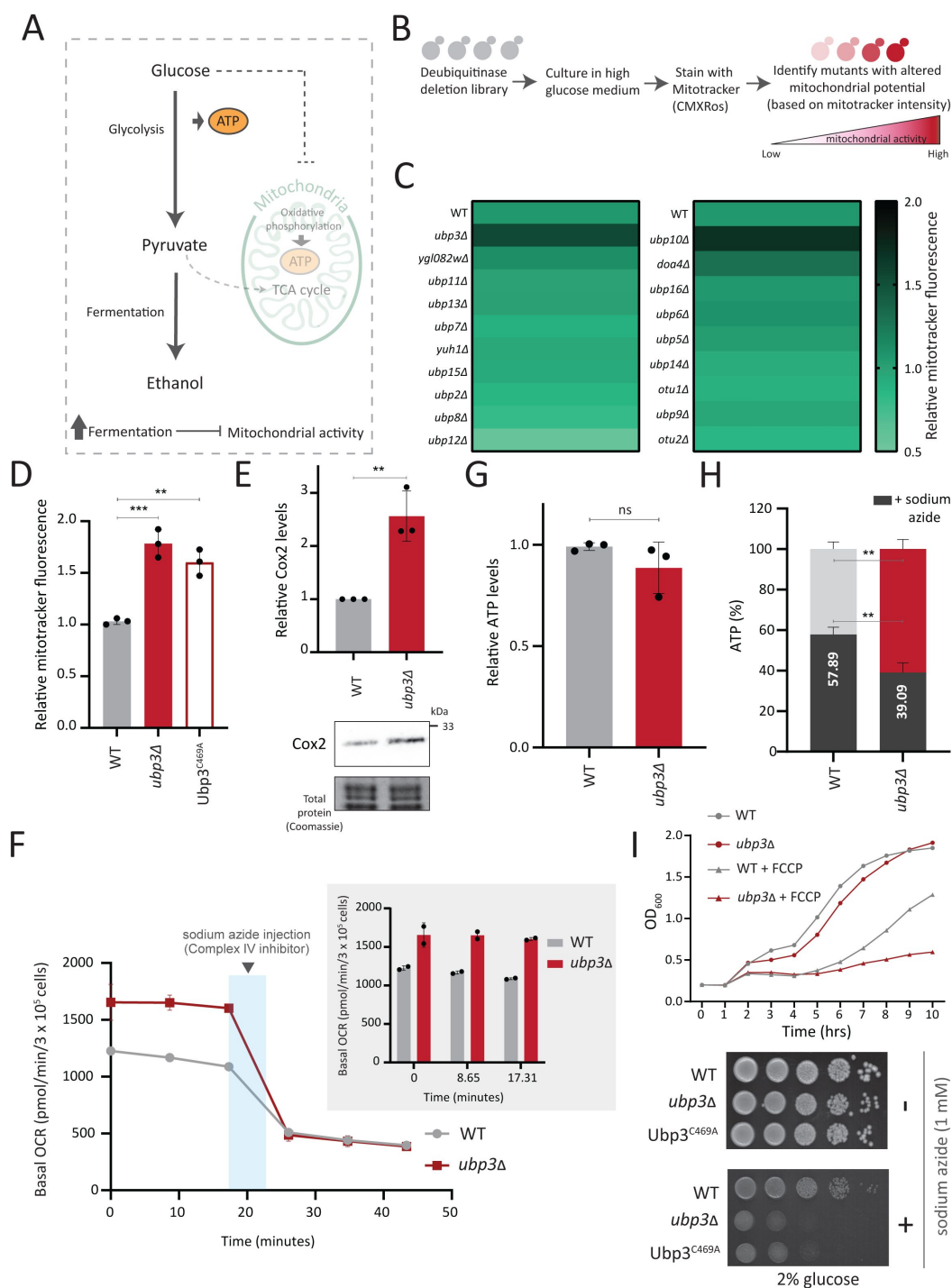


Figure 1

A deubiquitinase deletion screen identifies Ubp3 as a regulator of glucose-mediated mitochondrial repression

A) Schematic depicting glucose-mediated mitochondrial repression (Crabtree effect).

B) Schematic describing the screen with a yeast DUB KO library to identify regulators of Crabtree effect.

C) Identifying DUB knockouts with altered mitochondrial potential. Heat map shows relative mitochondrial membrane potential of 19 deubiquitinase deletions in high glucose, from 2 biological replicates. Also see **figure 1-figure supplements 1A, B** [↗](#).

D) The deubiquitinase activity of Ubp3 and repression of mitochondrial membrane potential. WT, *ubp3Δ*, and Ubp3^{C469A} were grown in high glucose and relative mitochondrial membrane potential was measured. Data represent mean ± SD from three biological replicates (n=3). Also see **figure 1-figure supplement 1D** [↗](#).

E) Effect of loss of Ubp3 on ETC complex IV subunit Cox2. WT and *ubp3Δ* were grown in high glucose, and Cox2 was measured (western blot using an anti-Cox2 antibody). A representative blot (out of 3 biological replicates, n=3) and their quantifications are shown. Data represent mean ± SD.

F) Basal oxygen consumption rate (OCR) in high glucose in *ubp3Δ*. WT and *ubp3Δ* were grown in high glucose, and OCR was measured. Basal OCR corresponding to $\sim 3 \times 10^5$ cells, from two independent experiments (n=2), normalized to the OD₆₀₀ is shown. Bar graph representations are shown in the inset. Data represent mean ± SD.

G) Total ATP levels in *ubp3Δ* and WT. WT and *ubp3Δ* were grown in high glucose, and total ATP were measured. Data represent mean ± SD from three biological replicates (n=3).

H) Dependence of *ubp3Δ* on mitochondrial ATP. WT and *ubp3Δ* cells were grown in high glucose, and treated with 1 mM sodium azide for 45 minutes. Total ATP levels in sodium azide treated and untreated cells were measured. Data represent mean ± SD (n=3).

I) Requirement for mitochondrial respiration in high glucose in *ubp3Δ*. A growth curve of WT and *ubp3Δ* in high glucose in the presence of OXPHOS uncoupler FCCP (10 μM), and serial dilution growth assay in high glucose in the presence/absence of sodium azide (1 mM) are shown. Data represent mean ± SD (n=2). Also see **figure 1-figure supplement 1H, I** [↗](#).

Data information: **p<0.01, ***p<0.001.

A prominent ‘hit’ was the evolutionarily conserved deubiquitinase Ubp3 (**Figure 1C** [↗](#)), (homologous to mammalian Usp10) (**Figure 1-figure supplement 1D** [↗](#)). Due to its high degree of conservation across eukaryotes (**Figure 1-figure supplement 1D** [↗](#)) as well as putative roles in metabolism or mitochondrial function ([27](#) [↗](#)–[29](#) [↗](#)), we focused our further attention on this DUB. Cells lacking Ubp3 showed an ~1.5-fold increase in mitotracker fluorescence (**Figure 1C** [↗](#), **Figure 1-figure supplement 1A, C** [↗](#)). Cells with catalytically inactive Ubp3 (Ubp3^{C469A}), showed increased mitochondrial potential comparable to *ubp3Δ* (**Figure 1D** [↗](#)). This data confirmed that Ubp3 catalytic activity is required to fully repress mitochondrial activity under high glucose. The catalytic site mutation did not affect steady-state Ubp3 levels (**Figure 1-figure supplement 1E** [↗](#)).

Next, to assess the requirement of Ubp3 for mitochondrial function, we quantified the electron transport chain (ETC) complex IV subunit Cox2 ([30](#) [↗](#)) (**Figure 1E** [↗](#)). *ubp3Δ* had higher Cox2 than WT (**Figure 1E** [↗](#)). As a control, we estimated total mitochondrial content in WT and *ubp3Δ* cells, using either estimates of the structural protein Tom70, or measuring the fluorescence-intensity in strains engineered with mitochondrial targeted mNeonGreen ([31](#) [↗](#)). There was no increase in the total mitochondrial volume (estimated by measuring the intensity of mitochondria targeted mNeon green) (**Figure 1-figure supplement 1F** [↗](#)) or mitochondrial outer membrane protein Tom70 (**Figure 1-figure supplement 1G** [↗](#)). This suggests that the increased Cox2 is not merely because of higher total mitochondrial content. We next measured the basal oxygen consumption rate (OCR) of *ubp3Δ*, and basal OCR was higher in *ubp3Δ*, indicating higher respiration (**Figure 1F** [↗](#)).

Next, we asked if mitochondrial ATP synthesis was higher in *ubp3Δ*. The total ATP levels in WT and *ubp3Δ* were comparable (**Figure 1G**). However, upon treatment with the ETC complex IV inhibitor sodium azide, ATP levels in WT were higher than *ubp3Δ*, contributing to ~60% of the total ATP (**Figure 1H**). In contrast, the ATP levels in *ubp3Δ* after sodium azide treatment were ~40% of the total ATP (**Figure 1H**). These data suggest a higher contribution of mitochondrial ATP synthesis towards the total ATP pool in *ubp3Δ*.

We next asked if *ubp3Δ* required higher mitochondrial activity for growth, using a series of mitochondrial activity inhibitors and comparing relative growth. In high glucose, WT cells show minimal growth inhibition in the presence of sodium azide, indicating lower reliance on mitochondrial function (**Figure 1I**). Contrastingly, *ubp3Δ* or Ubp3^{C469A} show a severe growth defect in the presence of the mitochondrial ETC complex inhibitors sodium azide and oligomycin, and the mitochondrial OXPHOS uncoupler FCCP (**Figure 1I**, **Figure 1-figure supplement 1H**). Additionally, the loss of Ubp3 in respiration defective *cox2-62* cells (32), or Rho0 cells (which lacks mitochondrial DNA) resulted in a severe growth defect (**Figure 1-figure supplement 1I**). Deletion of ATP synthase subunits Atp1 and Atp10 also results in a severe growth defect in *ubp3Δ* compared to WT (**Figure 1-figure supplement 1I**). Together, these results indicate that the loss of Ubp3 makes cells dependent on mitochondrial ATP synthesis in high glucose.

Collectively these data show that in 2% glucose, *ubp3Δ* have high mitochondrial activity, respiration, and rely on this mitochondrial function for ATP production and growth. We therefore decided to use *ubp3Δ* cells to start delineating requirements for glucose-mediated mitochondrial repression.

Key glycolytic enzymes decrease and glucose flux is rerouted in *ubp3Δ* cells

Glucose-6 phosphate (G6P) is the central node in glucose metabolism where carbon-allocations are made towards distinct metabolic arms, primarily: glycolysis, the pentose phosphate pathway (PPP), and trehalose biosynthesis (**Figure 2A**). We first compared amounts of two key ('rate-controlling') glycolytic enzymes - Phosphofructokinase 1 (Pfk1), GAPDH isozymes (Tdh2, Tdh3) (3), along with the enolase isozymes (Eno1, Eno2) in WT and *ubp3Δ* cells. Pfk1, Tdh2, and Tdh3 substantially decreased in *ubp3Δ* (but not Eno1 and Eno2) (**Figure 2B**, **Figure 2-figure supplement 1A**). Since deubiquitinases can control protein amounts by regulating proteasomal degradation, we asked if the decrease in Pfk1, Tdh2 and Tdh3 in *ubp3Δ* is due to proteasomal degradation. To test this, we measured the levels of these enzymes in *ubp3Δ* after treatment with proteasomal inhibitor MG132. We did not observe any rescue in the levels of these enzymes in MG132 treated samples, suggesting that the decreased levels of these enzymes were not due to increased proteasomal degradation (**Figure 2-figure supplement 1B**). To further understand if these enzyme transcripts are altered in *ubp3Δ*, we measured the mRNA levels of PFK1, TDH2 and TDH3 in WT, *ubp3Δ* and Ubp3^{C469A} cells. The transcripts of all the three genes in *ubp3Δ* and Ubp3^{C469A} cells decreased (**Figure 2-figure supplement 1C**), suggesting that Ubp3 regulates the transcripts of these glycolytic enzyme genes. The reduction in the Pfk and GAPDH enzyme amounts was intriguing, because the Pfk and GAPDH steps are critical in determining glycolytic flux (3, 33). A reduction in these enzymes could therefore decrease glycolytic flux, and would reroute glucose (G6P) allocations via mass action towards other branches of glucose metabolism, primarily the pentose phosphate pathway as well as trehalose biosynthesis (**Figure 2A**). To assess this, we first measured the steady-state levels of key glycolytic and pentose phosphate pathway (PPP) intermediates, and trehalose in WT or *ubp3Δ* using targeted LC-MS/MS (**Figure 2C**, **Figure 2-figure supplement 1D**). Glucose-6/fructose-6 phosphate (G6P/F6P) increased in *ubp3Δ* (**Figure 2C**). Concurrently, trehalose, and the PPP intermediates ribose 5-phosphate (R5P) and sedoheptulose 7-phosphate (S7P), increased in *ubp3Δ* (**Figure 2C**).

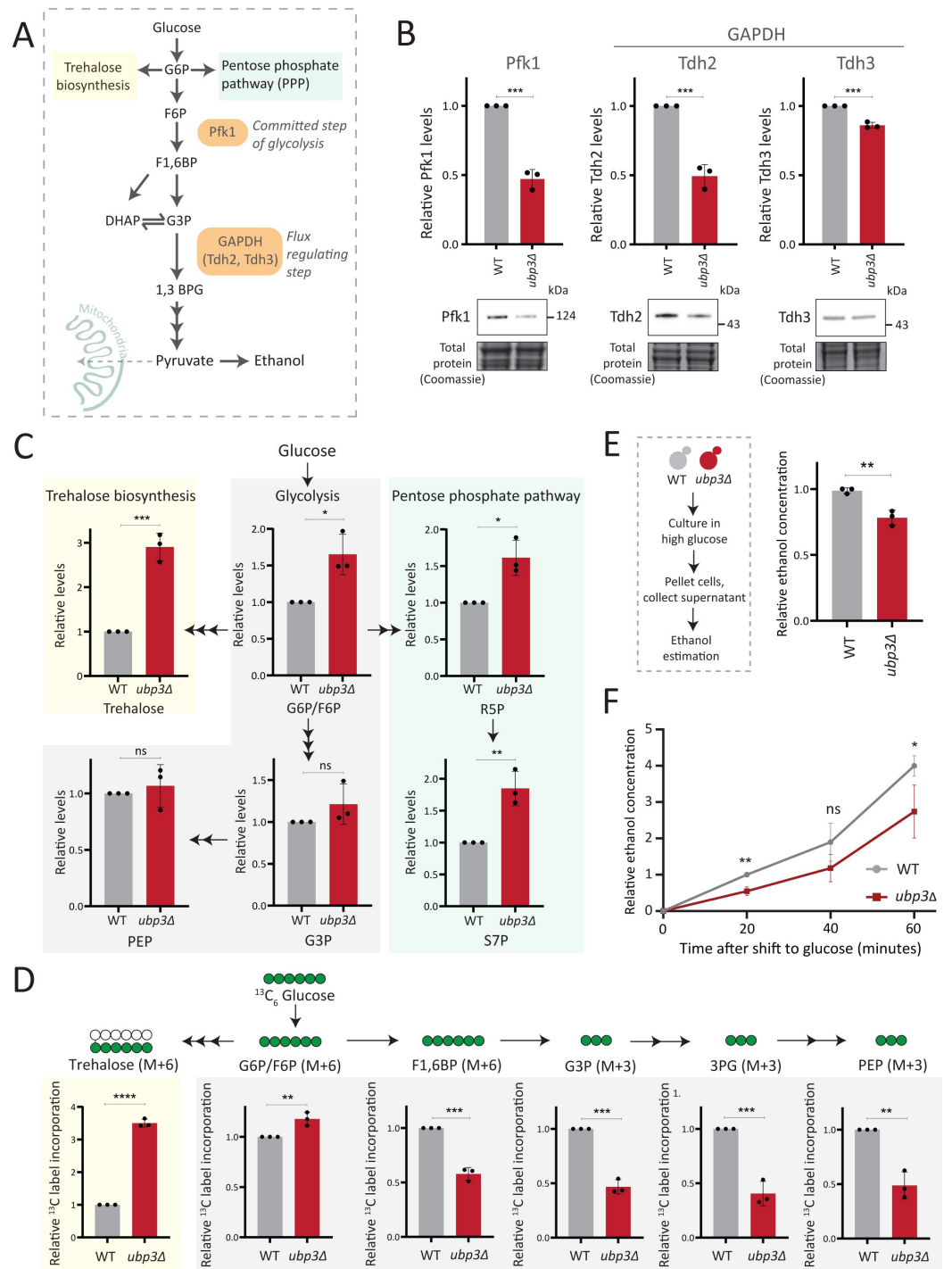


Figure 2

Key glycolytic enzymes decrease and glucose flux is rerouted in *ubp3Δ* cells

A) A schematic illustrating directions of glucose-6-phosphate (G6) flux in cells. Glucose is converted to G6P, a precursor for trehalose, the pentose phosphate pathway (PPP), and glycolysis.

B) Effect of loss of Ubp3 on key glycolytic enzymes. WT and *ubp3Δ* were grown in high glucose and the Pfk1, Tdh2, and Tdh3 levels were measured by western blot using an anti-FLAG antibody. A representative blot (out of three biological replicates, n=3) and their quantification are shown. Data represent mean ± SD. Also see **figure 2-figure supplement 1A** [↗](#).

C) Steady-state metabolite amounts in WT and *ubp3Δ* in high glucose. Relative steady-state levels of trehalose, major glycolytic, and PPP intermediates were estimated in WT and *ubp3Δ*. Data represent mean ± SD from three biological replicates (n=3). Also see Appendix Table S3.

D) Relative glycolytic and trehalose synthesis flux in WT and *ubp3Δ*. Relative ¹³C-label incorporation into trehalose and glycolytic intermediates, after a pulse of 1% ¹³C₆ glucose is shown. Data represent mean ± SD from three biological replicates (n=3). Also see Appendix Table S3, **figure 2-figure supplement 1 D, E** [↗](#).

E) Ethanol production in *ubp3Δ*. WT and *ubp3Δ* were grown in high glucose and ethanol in the media was measured. Data represent mean ± SD from three biological replicates (n=3).

F) Relative rate of ethanol production in WT vs *ubp3Δ*. WT and *ubp3Δ* were grown in high glucose (to OD₆₀₀ ~ 0.6), equal numbers of cells were shifted to fresh medium (high glucose) and ethanol concentration in the medium was measured temporally. Data represent mean ± SD from three biological replicates (n=3)

Data information: *p<0.05, **p<0.01, ***p<0.001.

Since steady-state metabolite amounts cannot separate production from utilization, in order to unambiguously assess if glycolytic flux is reduced in *ubp3Δ* cells, we utilized a pulse labeling of ¹³C₆ glucose, following which the label incorporation into glycolytic and other intermediates was measured. Note that because glycolytic flux is very high in yeast, this experiment would require rapid pulsing and extraction of metabolites in order to stay in a linear range and avoid label saturation. We therefore established a very short time point of label-addition, quenching and metabolite extraction post ¹³C glucose pulse. Since flux saturates/reaches steady-state in seconds, we first ensured that the label incorporation into individual metabolites after the ¹³C glucose pulse was in the linear range, and for early glycolytic intermediates this was seconds after glucose addition. This new methodology is extensively described in materials and methods, with required controls shown in **Figure 2-figure supplement 1F** [↗](#). WT and *ubp3Δ* cells were grown in high glucose, pulsed with ¹³C₆ glucose, and the relative ¹³C label incorporation into glycolytic intermediates and trehalose were measured, as shown in the schematic (**Figure 2-figure supplement 1E** [↗](#)). In *ubp3Δ*, ¹³C label incorporation into G6P/F6P as well as trehalose substantially increased (**Figure 2D** [↗](#)). Contrastingly, ¹³C label incorporation into glycolytic intermediates F1,6BP, G3P, 3PG and PEP decreased, indicating decreased glycolytic flux (**Figure 2D** [↗](#)). We next measured ethanol concentrations and production rates, as an additional output of relative glycolytic rates. We observed decreased steady-state ethanol levels, as well as ethanol production rates in *ubp3Δ* (**Figure 2F** [↗](#)).

Glycolysis-derived pyruvate is transported to mitochondria and fuels the TCA cycle. Therefore, we asked if the decreased glycolytic rate result in a decrease in the TCA cycle flux as well. To test this, we first measured the steady state levels of TCA cycle intermediates in WT or *ubp3Δ* using targeted LC-MS/MS. We did not observe any significant change in the levels of TCA cycle intermediates in *ubp3Δ*, except malate which showed a significant decrease in *ubp3Δ* (**Figure 2-figure supplement 2A** [↗](#)). Next, in order to assess if TCA cycle flux reduces in *ubp3Δ* cells, WT and *ubp3Δ* cells were grown in high glucose, pulsed with ¹³C₆ glucose, and the relative ¹³C label incorporation into TCA cycle intermediates was measured, as shown in the schematic (**Figure 2-figure supplement 2B** [↗](#)). The kinetics of ¹³C label incorporation in TCA cycle intermediates are shown in **Figure 2-figure supplement 2C** [↗](#). We did not observe any significant change in the relative ¹³C label

incorporation in TCA cycle intermediates in *ubp3Δ*. Therefore, these data suggest that the decreased glycolytic flux in *ubp3Δ* does not result in a decrease in TCA cycle flux. The increased respiration and mitochondrial activity in *ubp3Δ* cells is therefore driven via other factors.

Collectively, these results reveal that that reduced Pfk1 and GAPDH (in *ubp3Δ*) decrease glucose flux via glycolysis, which results in rewired glucose flux towards trehalose biosynthesis and the PPP.

Rerouted glucose flux results in phosphate (Pi) accumulation

We therefore asked if the proteomic state, as observed in *ubp3Δ* cells, could provide clues to explain the coupling between mitochondrial derepression and rerouted glucose flux. A recent study by Isasa et al., had systematically quantified the changes in protein levels in *ubp3Δ* cells (28 [↗](#)). We therefore reanalyzed this extensive dataset, looking for changes in proteins that would correlate with these metabolic processes. Notably, we observed increased levels of proteins of the mitochondrial ETC and respiration, and decreased amounts of glucose metabolizing enzymes in *ubp3Δ* (Figure 3A [↗](#)). Additionally, multiple proteins involved in regulating phosphate (Pi) homeostasis were decreased in *ubp3Δ* (Figure 3A [↗](#)) (28 [↗](#)). We had recently uncovered a reciprocal coupling of Pi homeostasis with the different arms of glucose metabolism, particularly trehalose biosynthesis (34 [↗](#), 35 [↗](#)), and therefore wondered if a glycolytic-flux dependent change in Pi homeostasis had any role in mitochondrial respiration. Our hypothesis was refined based on the reasoning given below.

Figure 3

Rerouted glucose flux results in phosphate (Pi) accumulation

A) Changes in protein levels in *ubp3Δ* (dataset from Isasa et al., 2015). *ubp3Δ* cells have an increase in proteins involved in mitochondrial respiration and decrease in proteins involved in glucose and phosphate metabolism.

B) Schematic showing maintenance of Pi balance during glycolysis. Trehalose synthesis from G6P releases Pi, and the conversion of G3P to 1,3BPG by GAPDH consumes Pi. In *ubp3Δ*, trehalose biosynthesis (which releases Pi) increases. *ubp3Δ* have decreased GAPDH, which will decrease Pi consumption. This increase in Pi release along with decreased Pi consumption could increase cytosolic Pi.

C) Intracellular Pi levels in WT and *ubp3Δ*. WT and *ubp3Δ* were grown in high glucose and the total free phosphate (Pi) levels were estimated. WT in high Pi (2% glucose, 10mM Pi) was a positive control. Data represent mean ± SD from three biological replicates (n=3). Also see [figure 3-figure supplement 1A](#).

D) Pho regulon responses in WT and *ubp3Δ*. Protein levels of Pho84-FLAG and Pho12-FLAG were compared between WT grown in high glucose and in high Pi, *ubp3Δ* in high glucose with or without a shift to a no-Pi medium for one hour, by western blot. A representative blot (out of three biological replicates, n=3) and their quantifications are shown. Data represent mean ± SD.

E) Contribution of trehalose synthesis as a Pi source. WT, *tps2Δ*, *ubp3Δ*, and *ubp3Δtps2Δ* were grown in high glucose and the total Pi levels were estimated. Data represent mean ± SD from three biological replicates (n=3). Also see [figure 3-figure supplement 1B](#).

F) Loss of GAPDH isozymes Tdh2 and Tdh3 and effect on Pi. WT, *ubp3Δ*, and *tdh2Δtdh3Δ* were grown in high glucose and total Pi was estimated. Data represent mean ± SD from three biological replicates (n=3).

G) Pi levels in *ubp3Δ* and *tdh2Δtdh3Δ* cells in ethanol medium. WT, *ubp3Δ*, and *tdh2Δtdh3Δ* cells were grown in ethanol medium and the total Pi levels were estimated. Data represent mean ± SD from three biological replicates (n=3).

Data information: *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001.

One explanation for mitochondrial repression can be an internal competition for shared metabolites/co-factors between (cytosolic) glycolytic and mitochondrial processes, that mitochondria might not be sufficiently able to access ([14](#), [18](#), [19](#)). In this context, a plausible role for inorganic phosphate (Pi) in regulating mitochondrial repression can be hypothesized. Cytosolic glycolysis requires rapid, high consumption of net Pi ([19](#), [36](#), [37](#)), and this could possibly limit the Pi that is continuously available for mitochondrial use ([18](#), [38](#)), thereby repressing mitochondria. Contextually, the balance between reactions releasing vs consuming Pi could explain changes in global Pi levels ([35](#)). A continuous hub of Pi consumption is glycolysis. Here, GAPDH catalyzes G3P to 1,3BPG, converting ADP to ATP, while concurrently consuming a molecule of Pi ([37](#), [39](#)). This Pi that goes into ATP will subsequently be used for nucleotide biosynthesis, polyphosphate biosynthesis and protein phosphorylation ([35](#), [40](#)). Therefore, we can surmise that in high glycolytic flux, the production of ATP, nucleotides and polyphosphates is concurrent with Pi consumption ([39](#), [41](#), [42](#)). Could this reaction therefore limit cytosolic Pi for the mitochondria (as illustrated in [Figure 3B](#))? Notably, *ubp3Δ* have reduced GAPDH levels and decreased glycolytic flux. Second, trehalose synthesis is a Pi releasing reaction, and a major source of free Pi that is critical for Pi homeostasis ([34](#), [37](#)). Flux through this reaction is also substantially higher in *ubp3Δ*. Therefore, these cells might have increased Pi release (via trehalose), coupled with decreased Pi consumption (via decreased GAPDH). We therefore asked if total Pi increases in *ubp3Δ* ([Figure 3B](#))?

To test this, we directly assessed total Pi levels in *ubp3Δ* and WT. *ubp3Δ* cells had higher Pi than WT, and this was comparable to Pi in WT grown in excess Pi (**Figure 3C**). Similarly, Pi amounts also increased in *Ubp3^{C469A}* (**Figure 3-figure supplement 1A**). Therefore, the loss of Ubp3 increases intracellular Pi levels. Next, we asked if *ubp3Δ* cells exhibit signatures of a 'high Pi' state. *S. cerevisiae* maintains internal Pi balance by controlling the expression of multiple genes collectively known as the Pho regulon (43). The Pho regulon is induced under Pi limitation, and repressed during Pi sufficiency (34, 43). We assessed two major Pho proteins (Pho84: a high-affinity membrane Pi transporter, and Pho12: an acid phosphatase) in WT and *ubp3Δ* in high glucose. *ubp3Δ* cells have lower amounts of Pho84 and Pho12 (**Figure 3D**). Further, upon shifting to low Pi for one hour, Pho84 and Pho12 increased in *ubp3Δ*. These data suggest that reduced Pho84 and Pho12 amounts in *ubp3Δ* are because of increased Pi, and not due to altered Pho regulon function itself (**Figure 3D**). These data clarify earlier observations from *ubp3Δ* which noted reduced Pho proteins (28). Therefore, *ubp3Δ* constitutively have higher Pi, and likely a consequent decrease in Pho proteins.

We next asked if the increased Pi in *ubp3Δ* is because of altered G6P allocations towards different end-points, particularly trehalose synthesis, which can be a major node of Pi restoration (34, 37). We assessed the contribution of increased trehalose synthesis towards the high Pi in *ubp3Δ*, by estimating Pi levels in the absence of trehalose 6-phosphate phosphatase (Tps2), which catalyzes the Pi-releasing step in trehalose synthesis. Notably, loss of Tps2 in *ubp3Δ* decreased Pi (**Figure 3E**). There was no additive difference in Pi between *tps2Δ* and *ubp3Δtps2Δ* (**Figure 3E**). As an added control, we assessed trehalose in WT and *ubp3Δ* in the absence of Tps2, and found no difference (**Figure 3-figure supplement 1B**). Therefore, increasing G6P flux towards trehalose biosynthesis is a major source of the increased Pi in *ubp3Δ*.

Since the major GAPDH isozymes, Tdh2 and Tdh3, are reduced in *ubp3Δ*, we directly asked if reducing GAPDH can decrease Pi consumption and increase Pi. To assess this, we generated *tdh2Δtdh3Δ* cells, which exhibit a growth defect, but are viable, permitting further analysis. *tdh2Δtdh3Δ* had higher Pi in high glucose (**Figure 3F**). Expectedly, we observed a decrease in ethanol in *tdh2Δtdh3Δ* (**Figure 3-figure supplement 1C**), along with an accumulation of F1,6 BP and G3P, and decreased 3PG and PEP (**Figure 3-figure supplement 1D**). However, G6P and trehalose levels between WT and *tdh2Δtdh3Δ* were comparable (**Figure 3-figure supplement 1D, E**). These data suggest that unlike in *ubp3Δ*, the increased Pi in *tdh2Δtdh3Δ* comes mainly from decreased Pi consumption (GAPDH step). To further assess the role of reduced glycolytic flux in increasing Pi (*ubp3Δ* and *tdh2Δtdh3Δ*), we measured the Pi in these cells growing in a gluconeogenic medium – 2%, ethanol. In this scenario, the GAPDH catalyzed reaction will be reversed, converting 1,3BPG to G3P, which should release and not consume Pi. Compared to WT, the Pi levels decreased in *ubp3Δ* and *tdh2Δtdh3Δ* (**Figure 3G**), suggesting that the changes in Pi in these mutants is driven by the relative change in Pi release vs consumption.

These results collectively indicate that the combined effect of increased Pi release coming from trehalose synthesis, and decreased Pi consumption from reduced GAPDH increase Pi levels in *ubp3Δ*.

Mitochondrial Pi availability correlates with mitochondrial activity in *ubp3Δ*

We therefore wondered if this observed Pi increase from the combined rewiring of glucose metabolism resulted in more Pi becoming accessible to the mitochondria. This would effectively result in Pi budgeting between cytosolic glycolysis and mitochondria based on the relative flux in different arms of glucose metabolism, by increasing Pi availability for the mitochondria (**Figure 4A**). To test if this were so, we first asked if the mitochondria in *ubp3Δ* have increased Pi. Mitochondria were isolated by immunoprecipitation from WT and *ubp3Δ*, the isolation efficiency was analyzed (**Figure 4-figure supplement 1A**), and relative Pi levels compared. Pi levels were

normalised to Idh1 (isocitrate dehydrogenase) in isolated mitochondria, since Idh1 protein levels did not decrease in *ubp3Δ* (Figure 4-figure supplement 1B), and consistent with (28). Mitochondrial Pi was higher in *ubp3Δ* (Figure 4B). We next asked if the increased mitochondrial activity in *ubp3Δ* is a consequence of high Pi. If this were so, *tdh2Δtdh3Δ* should partially phenocopy *ubp3Δ* with respect to mitochondrial activity. Consistently, *tdh2Δtdh3Δ* have higher mitotracker intensity as well as Cox2 levels compared to WT (Figure 4-figure supplement 1C, Figure 4C). Also consistent with this, a high basal OCR in *tdh2Δtdh3Δ* was observed (Figure 4D), together indicating high mitochondrial activity in *tdh2Δtdh3Δ*, which is comparable to *ubp3Δ*.

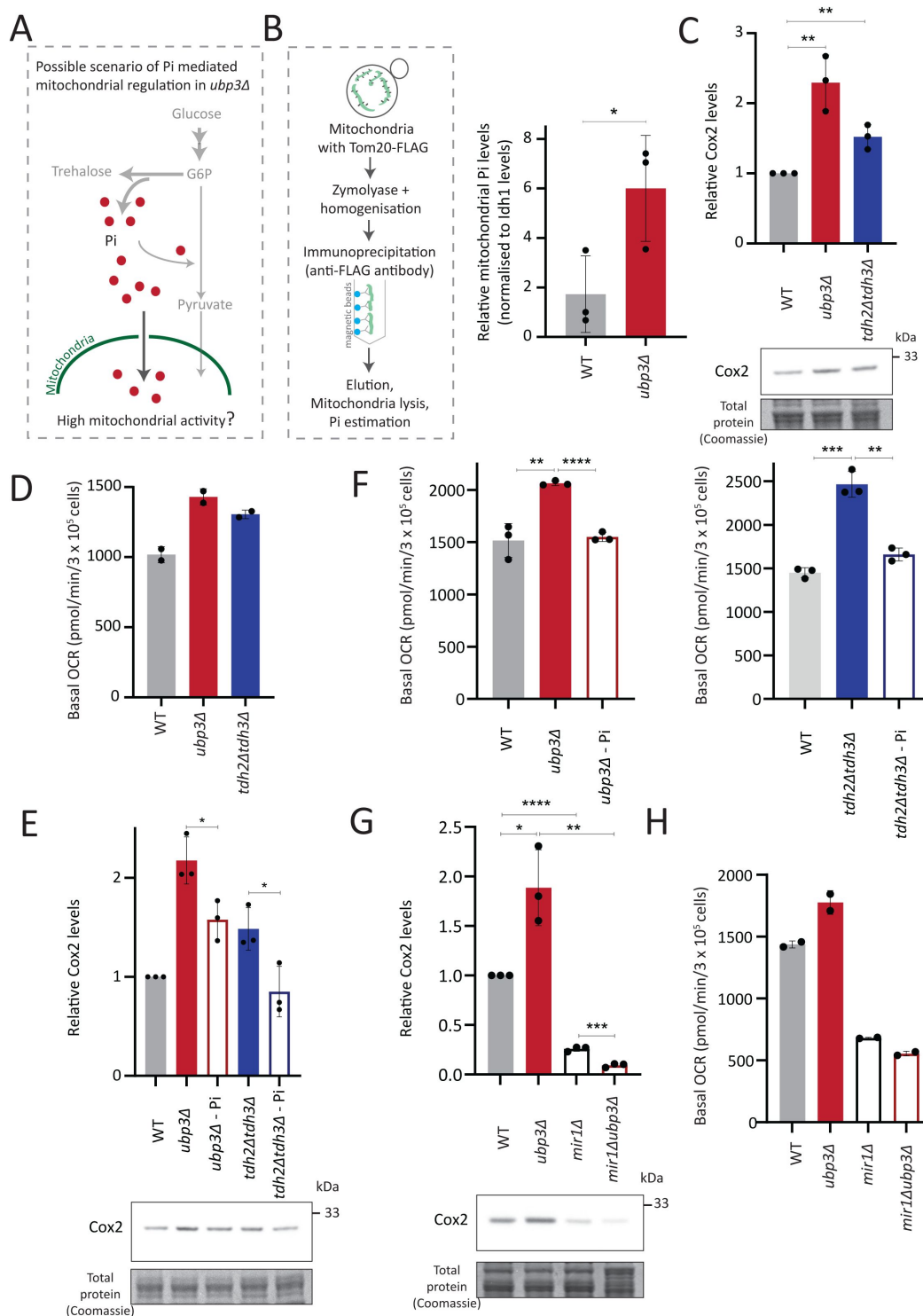


Figure 4

Mitochondrial Pi availability correlates with mitochondrial activity in *ubp3Δ* A) A hypothetical mechanism of cytosolic free Pi controlling mitochondrial activity by regulating mitochondrial Pi availability.

B) Mitochondrial Pi amounts in WT vs *ubp3Δ*. Mitochondria were isolated by immunoprecipitation from WT and *ubp3Δ* and mitochondrial Pi estimated. Mitochondrial Pi levels (normalised to Idh1) are shown. Data represent mean $\pm$ SD from three biological replicates (n=3). Also see **figure 4-figure supplement 1A, B**.

C) Cox2 protein in *tdh2Δtdh3Δ*. WT, *ubp3Δ*, and *tdh2Δtdh3Δ* were grown in high glucose and Cox2 protein was estimated. A representative blot (out of three biological replicates, n=3) and their quantifications are shown. Data represent mean $\pm$ SD.

D) Basal OCR levels in *tdh2Δtdh3Δ*. WT, *ubp3Δ*, and *tdh2Δtdh3Δ* were grown in high glucose and basal OCR was measured from two independent experiments (n=2). Data represent mean $\pm$ SD. Also see **figure 4-figure supplement 1C**.

E) Comparative Pi amounts and Cox2 levels in *ubp3Δ*, *tdh2Δtdh3Δ*, WT cells. WT cells were grown in high glucose, *ubp3Δ* and *tdh2Δtdh3Δ* were grown in high glucose and low Pi, and Cox2 protein was estimated. A representative blot (out of three biological replicates, n=3) and their quantifications are shown. Data represent mean $\pm$ SD. Also see **figure 4-figure supplement 1D, F**.

F) Pi amounts and basal OCR in *ubp3Δ* and *tdh2Δtdh3Δ* vs WT cells. WT cells were grown in high glucose, *ubp3Δ* and *tdh2Δtdh3Δ* were grown in high glucose and low Pi, and basal OCR was measured from three independent experiments (n=3). Data represent mean $\pm$ SD.

G) Effect of loss of mitochondrial Pi transporter Mir1 on Cox2 protein. WT, *ubp3Δ*, *mir1Δ*, and *mir1Δubp3Δ* were grown in high glucose and Cox2 amounts compared. A representative blot (out of three biological replicates, n=3) and their quantifications are shown. Data represent mean $\pm$ SD.

H) Relationship of mitochondrial Pi transport and basal OCR in WT vs *ubp3Δ*. WT, *ubp3Δ*, *mir1Δ*, and *mir1Δubp3Δ* cells were grown in high glucose and basal OCR was measured from two independent experiments (n=2). Data represent mean $\pm$ SD.

Data information: *p<0.05, **p<0.01, ****p<0.0001.

We next asked if higher Pi is necessary to increase mitochondrial activity in *ubp3Δ* and *tdh2Δtdh3Δ*. Since glycolysis is defective in *ubp3Δ* and *tdh2Δtdh3Δ*, it is necessary to distinguish the effect of high Pi vs. only the effect of low glycolysis in activating mitochondria. Logically, if decreased glycolysis (independent of Pi) is sufficient to activate mitochondria, bringing down the Pi levels in *ubp3Δ* to that of WT should not affect mitochondrial activity. Notably, *ubp3Δ* grown in low (1 mM) Pi have Pi levels similar to WT in standard (normal Pi) medium (**Figure 4-figure supplement 1D**). *ubp3Δ* grown in low Pi also had decreased ethanol, suggesting reduced glycolysis (**Figure 4-figure supplement 1E**). Therefore, we used this condition to further understand the role of Pi in inducing mitochondrial activity. Mitotracker fluorescence decreased in both *ubp3Δ* and *tdh2Δtdh3Δ* in low Pi (**Figure 4-figure supplement 1F**). Note: basal mitotracker fluorescence in WT also decreases in low Pi, which is consistent with a required role of Pi for mitochondrial activity (**Figure 4-figure supplement 1F**). Similarly, Cox2 levels were reduced in both *ubp3Δ* and *tdh2Δtdh3Δ* (**Figure 4E**). Consistent with both reduced mitotracker intensity and Cox2 levels, basal OCR also decreased in both *ubp3Δ* and *tdh2Δtdh3Δ* in low Pi (**Figure 4F**). As an additional control, we used Rho0 strains (which have no mitochondrial DNA and therefore lack functional ETC) to compare basal OCR. The expectation in these strains is that

basal OCR will not change if Pi changes. Consistently, we did not observe any significant difference in basal OCR in WT, *ubp3Δ* and *ubp3Δ* in low Pi in a Rho0 strain background (which lacks mitochondrial DNA) (**Figure 4-figure supplement 1G** [↗](#)). These data collectively suggest that high intracellular Pi is necessary to increase mitochondrial activity in *ubp3Δ* and *tdh2Δtdh3Δ*.

Next, we asked how mitochondrial Pi transport regulates mitochondrial activity. Mir1 and Pic2 are mitochondrial Pi transporters, with Mir1 being the major Pi transporter ([44](#) [↗](#), [45](#) [↗](#)). We first limited mitochondrial Pi availability in *ubp3Δ* by knocking-out *MIR1*. In *mir1Δ*, the increased Cox2 observed in *ubp3Δ* was no longer observed (**Figure 4G** [↗](#)). Consistent with this, we observed no further increase in the basal OCR in *mir1Δubp3Δ* compared to *mir1Δ* (**Figure 4H** [↗](#)). Furthermore, *mir1Δ* showed decreased Cox2 as well as basal OCR even in WT cells (**Figure 4G** [↗](#), **Figure 4H** [↗](#)). These data together suggest that mitochondrial Pi transport is critical for increasing mitochondrial activity in *ubp3Δ*, and in maintaining basal mitochondrial activity even in high glucose.

As a control, no significant increase in Mir1 and Pic2 was observed in *ubp3Δ* (**Figure 4-figure supplement 1H** [↗](#)), suggesting that *ubp3Δ* do not increase mitochondrial Pi by merely increasing the Pi transporters, but rather by increasing available Pi pools.

Taken together, these data suggest the possibility that increased cytosolic Pi increases the mitochondrial Pi pool, which increases mitochondrial activity. Decreasing mitochondrial Pi by either reducing total Pi, or by reducing mitochondrial Pi transport decreases mitochondrial activity.

Mitochondrial Pi availability constrains mitochondrial activity under high glucose

So far, these data suggest that the cytosolic Pi available for the mitochondria can determine the extent of mitochondrial activity. Therefore, we further investigated if mitochondrial Pi allocation was a necessary constraint for glucose-mediated mitochondrial repression.

To test this, we first asked how important mitochondrial Pi transport was to switch to increased respiration. In WT yeast, low glucose or glycolytic inhibition will result in increased respiration ([22](#) [↗](#)). What happens therefore if we restrict mitochondrial Pi in this context? For this, we measured the basal OCR in WT and *mir1Δ* after switching from high (2%) to low (0.1%) glucose. We observed a significant increase in the basal OCR in WT but not in *mir1Δ* (**Figure 5A** [↗](#)). The alternate scenario is after glycolytic inhibition. We assessed the role of mitochondrial Pi in this context, by inhibiting glycolysis using 2-deoxyglucose (2DG). WT, but not *mir1Δ*, increased their OCR (respiration) upon a 1-hour treatment with 2DG (**Figure 5B** [↗](#)). Consistent with this, mitotracker fluorescence increased with an increase in 2DG in WT, but not in *mir1Δ* (**Figure 5-figure supplement 1A** [↗](#)). We further asked if the mitochondrial Pi transporter itself glucose repressed, and therefore assessed Mir1 amounts in high and low glucose. Mir1 levels are higher upon a shift to low glucose, and in cells grown in 2% ethanol, suggesting that Mir1 is glucose repressed (**Figure 5C** [↗](#), **Figure 5-figure supplement 1B** [↗](#)). These data suggest that mitochondrial Pi transport is necessary for increasing mitochondrial activity after glucose derepression.

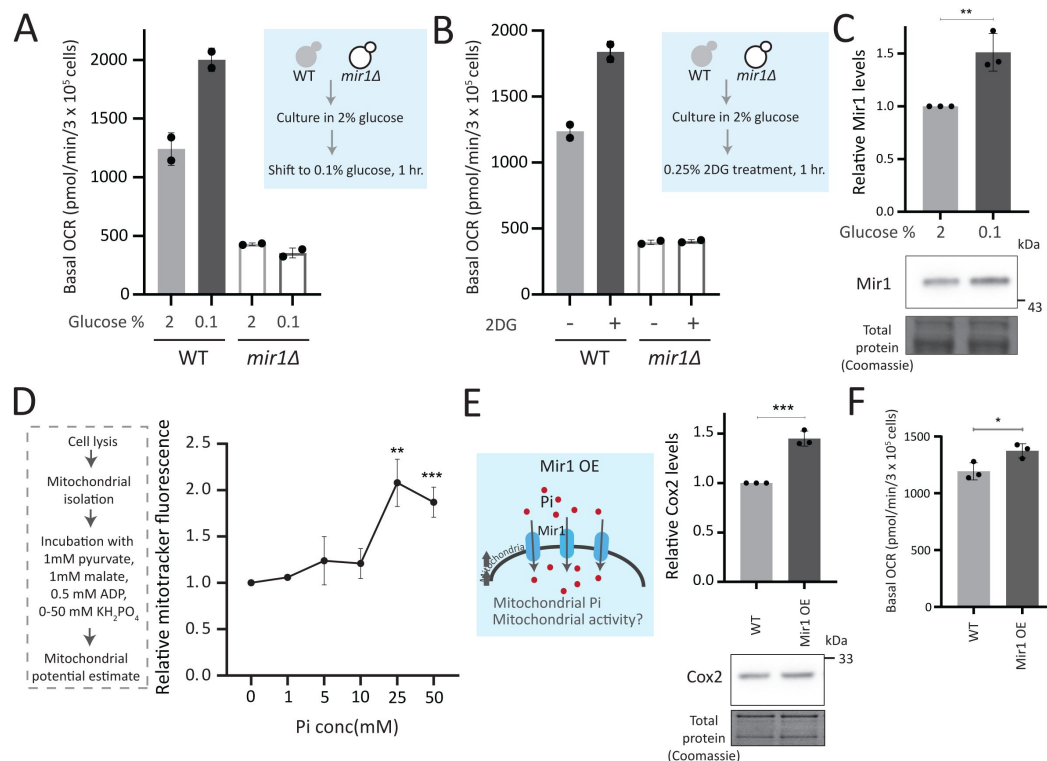


Figure 5

Mitochondrial Pi availability constrains mitochondrial activity under high glucose

A) Relationship of mitochondrial Pi transport and respiration after glucose removal. WT and *mir1Δ* cells were cultured in high (2%) glucose and shifted to low (0.1%) glucose for 1 hour. The normalized basal OCR, from two independent experiments (n=2) are shown. Data represent mean ± SD.

B) Requirement of mitochondrial Pi transport for switch to respiration upon glycolytic inhibition by 2DG. WT and *mir1Δ* cells were cultured in high glucose and treated with or without 0.25% 2DG for 1 hour. Basal OCR was measured from two independent experiments (n=2). Data represent mean ± SD. Also see [figure 5-figure supplement 1A](#).

C) Glucose-dependent regulation of Mir1. Cells (with Mir1-HA) were grown in high glucose and shifted to low glucose (0.1% glucose) for 1 hour, and Mir1 levels compared. A representative blot (out of three biological replicates, n=3) and their quantifications are shown. Data represent mean ± SD. Also see [figure 5-figure supplement 1B](#).

D) Increasing Pi concentrations and mitochondrial activity in isolated mitochondria. Mitochondria were isolated from WT cells grown in high glucose, incubated with 1 mM pyruvate, 1 mM malate, 0.5 mM ADP and 0-50 mM KH₂PO₄. The mitochondrial activity was estimated by mitotracker fluorescence intensity, and intensities relative to the sample with 0 mM KH₂PO₄ is shown. Data represent mean ± SD from three biological replicates (n=3).

E) Effect of overexpressing Mir1 on Cox2 protein. WT (containing empty vector) and Mir1 overexpressing (Mir1OE) cells were grown in high glucose and Cox2 levels were estimated. A representative blot (out of three biological replicates, n=3) and their quantifications are shown. Data represent mean ± SD. Also see [figure S5F](#).

F) Effect of overexpressing Mir1 on basal OCR. The basal OCR in WT (containing empty vector) and Mir1OE in high glucose was measured from three independent experiments (n=3). Data represent mean ± SD.

Data information: *p<0.05, **p<0.01, ***p<0.001.

We next asked whether just adding external Pi was sufficient to increase mitochondrial activity, when cells are in high glucose. In medium supplemented with excess Pi, the internal Pi increases as seen earlier (**Figure 3C**). Therefore, a simplistic assumption would be that the addition of external Pi to cells in high glucose would also increase mitochondrial Pi. However, an alternate possibility presents itself wherein since glycolytic flux is already high in glucose, supplementing Pi will continue to fuel glycolysis. Indeed, this was originally observed by Harden and Young in 1908, where adding Pi increased fermentation (46). In such a scenario, mitochondrial Pi will not increase. We estimated mitochondrial Pi in this condition where excess Pi was externally supplemented. Notably, cells grown in high Pi had decreased mitochondrial Pi (**Figure 5-figure supplement 1C**), without any changes in total mitochondria volume or amounts (**Figure 5-figure supplement 1D**). Furthermore, directly adding Pi to cells growing in high glucose also decreased basal OCR (**Figure 5-figure supplement 1E**), consistent with decreased mitochondrial Pi. These data indicate that in high glucose, simply supplementing Pi will not increase Pi access to the mitochondria. We therefore now asked, if we inhibit glycolytic flux and then supplement Pi, what would happen to mitochondrial activity. For this, we treated cells with 2DG, and subsequently added Pi and measured the OCR (**Figure 5-figure supplement 1F**). In this case, supplementing Pi increased the basal OCR (**Figure 5-figure supplement 1F**). Collectively, these data suggest that a combination of decreasing glycolysis and increasing Pi can together increase respiration. Next, in order to directly test mitochondrial activation based on external Pi availability, we isolated mitochondria, and estimated activity *in vitro* upon adding increasing Pi. Mitochondrial activity increased with increased Pi, with maximum activity observed with 25 mM Pi supplemented (**Figure 5D**). In a complementary experiment, we overexpressed the Mir1 transporter in wild-type cells, to increase Pi within mitochondria (**Figure 5-figure supplement 1G**). Mir1-OE cells have higher Cox2 levels and basal OCR (**Figure 5E**, **5F**). Therefore, increasing Pi transport to mitochondria is sufficient to increase mitochondrial activity in high glucose.

Mitochondrial pyruvate transport is also required for mitochondrial respiration (47). We asked where Pi availability stands in a hierarchy of constraints for mitochondrial derepression, as compared to mitochondrial pyruvate transport. We measured the amounts of the Mpc3 subunit of the mitochondrial pyruvate carrier (MPC) complex (47, 48). Mpc3 protein increases in *ubp3Δ* (**Figure 5-figure supplement 1H**). This also correlated with the unimpaired TCA cycle flux (**Figure 2-figure supplement 2D**) and the increased mitochondrial activity. Interestingly, in *ubp3Δ* grown in low Pi, Mpc3 further increased (**Figure 5-figure supplement 1H**), but as shown earlier this condition cannot increase OCR or mitochondrial activity (**Figure 4F**, **Figure 4-figure supplement 1F**). Basal Mpc3 levels decrease in *mir1Δ*, but upon shifting to 0.1% glucose, Mpc3 increases in both WT and *mir1Δ*, with higher levels in *mir1Δ* (**Figure 5-figure supplement 1I**). Therefore, even where Mpc3 is high (*ubp3Δ* in low Pi, and *mir1Δ* in low glucose), mitochondrial activity remains low if Pi is restricted (**Figure 4F**, **4H**). There was also no decrease in basal OCR in *mpc3Δ* in high glucose, and the basal OCR increased to the same level as of WT after shifting to 0.1% glucose (**Figure 5-figure supplement 1J**). Since Mpc3 changes with mitochondrial Pi availability (Figure S5G, S5H), we also measured Mpc3 in the Mir1OE. No further changes in Mpc3 were observed in Mir1OE (**Figure 5-figure supplement 1K**), indicating that increasing mitochondrial Pi alone need not increase Mpc3. Overall, although Mpc3 levels correlate with decreased glycolysis (*ubp3Δ* - **Figure 5-figure supplement 1H**, *ubp3Δ* in low Pi - **Figure 5-figure supplement 1H**, low glucose - **Figure 5-figure supplement 1I**), increased Mpc3 alone cannot increase mitochondrial activity and respiration in the absence of adequate mitochondrial Pi.

Collectively, mitochondrial Pi availability constrains glucose-mediated mitochondrial repression. Increasing available pools of Pi to enter the mitochondria is sufficient to induce mitochondrial activity.

Repression of mitochondrial respiration via Pi budgeting is conserved in Ubp3 mutants across diverse yeast genetic backgrounds

So far, we have identified a role for intracellular Pi budgeting as a constraint for mitochondrial activity under high glucose. These were all carried out using a robust, prototrophic yeast strain from a CEN.PK background. *S. cerevisiae* however, while Crabtree positive, have tremendous genetic diversity ([49](#)). We therefore asked if this mitochondrial repression through Pi budgeting (mediated by Ubp3 function) is conserved across other strains of *S. cerevisiae* as well. To test this, we generated Ubp3 deletion mutants in different genetic backgrounds of *S. cerevisiae*, including BY4742, W303 and Σ 1278. In all these strain backgrounds, we observed a significant increase in mitotracker fluorescence intensity and Cox2 protein levels in *ubp3Δ* (**Figure 6A, B**). This suggests the role of Ubp3 as a regulator of mitochondrial repression, independent of the genetic background of the yeast (*S. cerevisiae*) strain. To further assess if loss of Ubp3 shows a concurrent increase in Pi levels, we measured the total Pi levels in WT and *ubp3Δ* in a W303 strain background. We observed a significant increase in total Pi levels in *ubp3Δ* cells in this strain (**Figure 6C**), similar to what we observed in the CEN.PK strain (**Figure 3C**). Finally, to test if the altered Pi budgeting regulates the mitochondrial activity in these cells, we measured the basal OCR in WT, *ubp3Δ* and *ubp3Δ* in low (1 mM) Pi medium, in the W303 strain background. Consistent with the increase in mitotracker fluorescence intensity and Cox2 protein levels, we observed a significant increase in the basal OCR in *ubp3Δ* cells (**Figure 6D**). This increase was not observed in *ubp3Δ* in a low Pi medium (**Figure 6D**). This suggests that the role of altered Pi budgeting in regulating mitochondrial respiration is conserved in other genetic backgrounds of *S. cerevisiae*.

Figure 6

Repression of mitochondrial respiration via Pi budgeting is conserved in Ubp3 mutants across diverse yeast genetic backgrounds

A) Effect of loss of Ubp3 on mitochondrial membrane potential in different yeast strains. WT and *ubp3Δ* cells (in CEN.PK, BY4742, W303 and Σ1278 strains of *S. cerevisiae*) were grown in high glucose and relative mitochondrial membrane potential was measured. Data represent mean ± SD from three biological replicates (n=3).

B) Effect of loss of Ubp3 on ETC complex IV subunit Cox2. WT and *ubp3Δ* (in BY4742, W303 and Σ1278 strains of *S. cerevisiae*) were grown in high glucose, and Cox2 was measured. A representative blot (out of 3 biological replicates, n=3) and their quantifications are shown. Data represent mean ± SD.

C) Intracellular Pi levels in WT and *ubp3Δ* in W303 strain background. WT and *ubp3Δ* (in W303 strain background) were grown in high glucose and the total free phosphate (Pi) levels were estimated. Data represent mean ± SD from three biological replicates (n=3).

D) Effect of low Pi on the basal OCR in WT and *ubp3Δ* cells in W303 strain background. WT cells were grown in high glucose and *ubp3Δ* were grown in high glucose and low Pi, and basal OCR was measured. Data represent mean ± SD (n=3).

E) Requirement of mitochondrial Pi transport for growth after 2DG treatment. Shown are serial dilution growth assays in high glucose in the presence and absence of 0.1% 2DG, using WT and *mir1Δ* cells. The results after 40hrs incubation/30°C are shown.

F) A model illustrating how mitochondrial Pi availability controls mitochondrial activity. In high glucose, the decreased Pi due to high Pi consumption in glycolysis, along with the glucose-mediated repression of mitochondrial Pi transporters, decreases mitochondrial Pi availability. This reduces mitochondrial activity. In low glucose, increased mitochondrial Pi transporters and lower glycolytic flux increases mitochondrial Pi, leading to enhanced mitochondrial activity. In *ubp3Δ* cells in high glucose, high trehalose synthesis and lower glycolytic flux results in an increase in Pi. This increases mitochondrial Pi availability and thereby the mitochondrial activity.

Data information: *p<0.05, **p<0.01, ***p<0.001.

Finally, we asked how important mitochondrial Pi transport was for growth, under glycolytic inhibition. Consistent with the requirement for mitochondrial Pi transport to increase mitochondrial respiration upon glycolytic inhibition (**Figure 5B** [↗](#)), WT cells only exhibit a slightly decreased growth in the presence of 2DG (**Figure 6E** [↗](#)). In contrast, *mir1Δ* show a severe growth defect upon 2DG treatment (**Figure 6E** [↗](#)), revealing a synergetic effect of combining 2DG with inhibiting mitochondrial Pi transport. Therefore, the combined inhibition of glycolysis and mitochondrial Pi transport restricts the growth of glycolytic cells.

Collectively, our data suggests a conserved role for intracellular Pi budgeting in regulating mitochondrial repression in high glucose and the role of mitochondrial Pi transport in regulating adaptation for growth under glycolytic inhibition.

Discussion

In this study, we highlight a role for Pi budgeting between cytosolic glycolysis and mitochondrial processes (which compete for Pi) in constraining mitochondrial repression (**Figure 6F**). Ubp3 controls this process (**Figure 1**), by maintaining the amounts of the glycolytic enzymes Pfk1 and GAPDH (Tdh2 and Tdh3) and thereby allows high glycolytic flux. At high fermentation rates, glycolytic enzymes levels at maximal activity maintain high glycolytic rates (effectively following zero-order kinetics), and therefore, changes in the enzyme levels will have a direct, proportionate effect on flux (50). The loss of Ubp3 decreases glycolytic flux, resulting in a systems-level, mass-action based rewiring of glucose metabolism where more G6P is routed towards trehalose synthesis and PPP (**Figure 2**, **Figure 6F**). Indeed, inhibiting just phosphofructokinase can reroute glucose flux from glycolysis to PPP (51–53), and our study now permits contextualized interpretations of these results. Such reallocations of glucose flux will collectively increase overall Pi, coming from the combined effect of increased Pi release from trehalose synthesis, and decreased Pi consumption via reduced GAPDH. This altered intracellular Pi economy increases Pi pools available to mitochondria, and increasing mitochondrial Pi is necessary and sufficient to increase respiration in high glucose (**Figure 4**, **Figure 5**, **Figure 6**). Mitochondrial Pi transport maintains mitochondrial activity in high glucose, and increases mitochondrial activity in low glucose (**Figure 4**, **Figure 5**). Finally, the mitochondrial Pi transporter Mir1 itself decreases in high glucose (**Figure 5**). Therefore, this glucose-dependent repression of Mir1 also restricts mitochondrial Pi availability (and thereby activity) in high glucose.

Traditionally, loss-of-function mutants of metabolic enzymes are used to understand metabolic state regulation. This approach negates nuanced investigations, since metabolic enzymes are often essential for viability. Furthermore, metabolic pathways have multiple, contextually regulated nodes, through which cells maintain their metabolic state. Therefore, alternate approaches to identify global regulators of metabolic states (as opposed to single enzymes), might uncover ways via which multiple nodes are simultaneously tuned, and can reveal unanticipated systems-level principles of metabolic state rewiring. In this study of glucose-mediated mitochondrial repression, the loss of the DUB Ubp3 decreases glycolytic flux by reducing the enzymes at two critical nodes in the pathway - Pfk1 and GAPDH. Unlike loss of function mutants, a reduction in amounts will only rewire metabolic flux. By ‘hitting’ multiple steps in glycolysis simultaneously, *ubp3Δ* have decreased Pi consumption, as well as increased Pi release. Such a cumulative phenomenon reveals more than inhibiting only GAPDH, where increased Pi comes only from reduced Pi consumption, and not from increased trehalose biosynthesis. Our serendipitous identification of a regulator which regulates multiple steps in glucose metabolism to change the metabolic environment, now suggests a general basis of mitochondrial regulation that would have otherwise remained hidden. Separately, finding the substrates of Ubp3 and whether Ubp3 directly regulates glycolytic enzymes are exciting future research questions requiring concurrent innovations in accessible chemical-biological approaches to study DUBs.

Because phosphates are ubiquitous, it is challenging to identify hierarchies of Pi-dependent processes in metabolic state regulation (35). Phosphate transfer reactions are the foundation of metabolism, driving multiple, thermodynamically unfavorable reactions (54, 55). Contextually, the laws of mass action predict that the relative rates of these reactions will regulate overall Pi balance, and contrarily the Pi allocation to Pi-dependent reactions will determine reaction rates (35, 37). Additionally, cells might control Pi allocations for different reactions via compartmentalizing Pi in organelles, to spatially restrict Pi availability (56–58). Our data collectively suggest a paradigm where the combination of factors regulates mitochondrial Pi and thereby activity. In glycolytic yeast cells growing in high glucose, mitochondrial Pi availability becomes restricted due to higher utilization of Pi in glycolysis compared to mitochondria. Consistent with this, a rapid decrease in Pi upon glucose addition has been observed (18, 19, 39). Interesting, *in vitro* studies with isolated mitochondria from tumor cells also find that

decreasing Pi levels decreases respiration (19), which would be consistent with this scenario. Further, supplementing Pi correlates with decreased mitochondrial repression in tumors (18, 59). By increasing Pi through a systems-level rewiring of glucose metabolism (such as in *ubp3Δ* cells), cells can collectively increase mitochondrial access to Pi. This Pi budgeting determines mitochondrial activity. Supplementing Pi under conditions of low glycolysis (where mitochondrial Pi transport is enhanced), as well as directly supplementing Pi to isolated mitochondria, increases respiration (Figure 5, Figure 5-figure supplement 1). Therefore, in order to derepress mitochondria, a combination of increased Pi along with decreased glycolysis is required. An additional systems-level phenomenon that might regulate Pi transport to the mitochondria is the decrease in cytosolic pH upon decreased glycolysis (60, 61). The cytosolic pH in highly glycolytic cells is ~7, and decreasing glycolysis results in cytosolic acidification (60, 61). Therefore, under conditions of decreased glycolysis (2DG treatment, deletion of Ubp3, and decreased GAPDH activity), cytosolic pH becomes acidic. Since mitochondrial Pi transport itself is dependent on the proton gradient, a low cytosolic pH would favour mitochondrial Pi transport (62). Therefore, under conditions of decreased glycolysis (2DG treatment, or loss of Ubp3, or decreased GAPDH activity), where cytosolic pH would be acidic, increasing cytosolic Pi might indirectly increase mitochondria Pi transport, thereby leading to increased respiration. Alternately, increasing mitochondrial Pi transporter amounts can achieve the same result, as seen by overexpressing Mir1 (Figure 5). A similar observation has been reported in *Arabidopsis*, reiterating an evolutionarily conserved role for mitochondrial Pi in controlling respiration (63). Relatedly, glycolytic inhibition can suppress cell proliferation in Warburg-positive tumors (64, 65), or inflammatory responses (66). However, these cells survive by switching to mitochondrial respiration (67, 68), requiring alternate approaches to prevent their proliferation (69). Inhibiting mitochondrial Pi transport in combination with glycolytic inhibition could restrict the proliferation of Warburg/Crabtree positive cells.

Since its discovery in the 1920s, the phenomenon of accelerated glycolysis with concurrent mitochondrial repression has been intensely researched. Yet, the biochemical constraints for glucose-mediated mitochondrial repression remains unresolved. One hypothesis suggests that the availability of glycolytic intermediates might determine the extent of mitochondrial repression. F1,6BP inhibits complex III and IV of the electron transport chain (ETC) in Crabtree positive yeast (14–17). Similarly, the ratio between G6P and F1,6BP regulates the extent of mitochondrial repression (16, 17). Although G6P/F6P accumulates in *ubp3Δ* (Figure 2C), this is not the case in *tdh2Δtdh3Δ* (GAPDH mutant) (Figure 3-figure supplement 1D), suggesting that G6P/F6P accumulation in itself is not the criterion to increase mitochondrial activity. Separately, the competition for common metabolites/co-factors between glycolysis and respiration (such as ADP, Pi or pyruvate) could drive this phenomenon (14, 18). Here we observe that Mpc3 mediated mitochondrial pyruvate transport alone cannot increase respiration. An additional consideration is the possible contribution of changes in ADP in regulating mitochondrial activity, where the use of ADP in glycolysis might limit mitochondrial ADP. Therefore, when Pi changes as a consequence of glycolysis, it could be imagined that a change in ADP balance can coincidentally occur. However, prior studies show that even though cytosolic ADP decreases in the presence of glucose, this does not limit mitochondrial ADP uptake, or decrease respiration, due to the very high affinity of the mitochondrial ADP transporter (14, 19). These collectively reiterate the importance of Pi access and transport to mitochondria in constraining mitochondrial respiration. Indeed, this interpretation can also contextually explain observations from other model systems where mitochondrial Pi transport seems to regulate respiration (70, 71).

We parsimoniously suggest that Pi access to the mitochondria as a key constraint for mitochondrial repression under high glucose. In a hypothetical scenario, a single-step event in evolution, reducing mitochondrial Pi transporter amounts, and/or increasing glycolytic flux (to deplete cytosolic Pi), will result in whole-scale metabolic rewiring to repress mitochondria. More elaborate regulatory events can easily be imagined as subsequent adaptations to enforce mitochondrial repression. Given the central role played by Pi, something as fundamental as access

to Pi will constrain mitochondrial repression. Concurrently, the rapid incorporation of Pi into faster glycolysis can give cells a competitive advantage, while also sequestering Pi in the form of usable ATP. Over the course of evolution, this could conceivably drive other regulatory mechanisms to enforce mitochondrial repression, leading to the currently observed complex regulatory networks and signaling programs observed in the Crabtree effect, and other examples of glucose-dependent mitochondrial repression.

Materials and methods

Statistics and graphing

Unless otherwise indicated, statistical significance for all indicated experiments were calculated using unpaired Student's t-tests (GraphPad prism 9.0.1). Graphs were plotted using GraphPad prism 9.0.1

Yeast strains, media, and growth conditions

A prototrophic CEN.PK strain of *Saccharomyces cerevisiae* (WT) ([72](#)) was used unless mentioned otherwise. Strains are listed in appendix table S1. Gene deletions, chromosomal C-terminal tagged strains were generated by PCR mediated gene deletion/tagging ([73](#)). Mitochondria targeted Mneon strain (Mito-mNeon green) is described in ([31](#)). The cox2-62 strain is described in ([32](#)). Media compositions, growth conditions and CRISPR-Cas9 based mutagenesis are described in extended methods (SI appendix).

Mitotracker fluorescence

Mitotracker fluorescence was measured using Thermo Varioscan™ LUX multimode plate reader (579/599 excitation/emission). Detailed protocol is described in extended methods. Mitotracker fluorescence were normalized using OD₆₀₀ of each sample and relative fluorescence intensity calculated.

Protein extraction and Western blotting

Total protein was precipitated, extracted using trichloroacetic acid (TCA) as described earlier ([74](#)). Blots were quantified using ImageJ software. Detailed protocol is described in extended methods.

Basal OCR measurement

The basal OCR was measured using Agilent Seahorse XFe24 analyzer. Basal OCR readings were normalized for cell number (using OD₆₀₀ of samples) in each well. The detailed methods are described in extended methods.

Mitochondrial volume estimation

High-resolution 3D fluorescence experiments were performed on an inverted confocal laser scanning microscope (Carl Zeiss LSM 780 or Olympus FV3000). For each imaging field of view, sequential z-stacks were acquired for each excitation channel. 488 nm laser excitation for mNeonGreen and 561 nm laser excitation for Mitotracker CMXros dye were used respectively. Images taken were deconvolved and analyzed further in ImageJ software with custom-written routines. Mitochondria segmentation and quantification was done using the Mitochondria Analyzer plugin ([75](#)) in ImageJ. For visualization, maximum intensity projection of 3D images was used.

RNA extraction and RT-qPCR

The RNA extraction was done using the hot phenol extraction method as described in (74 [↗](#)). The isolated RNA was DNase treated, and used for cDNA synthesis. Superscript III reverse transcriptase enzyme (Invitrogen) was used for cDNA synthesis and RT-qPCR was performed using KAPA SYBR FAST qRT PCR kit (KK4602, KAPA Biosystems). Taf10 was used as a control for normalisation and the fold change in mRNA levels were calculated by $2^{-\Delta\Delta Ct}$ method.

ATP, ethanol and Pi measurements

ATP levels were measured by ATP estimation kit (Thermo Fisher A22066). Ethanol concentration in the medium was estimated using potassium dichromate based assay described in (76 [↗](#)) with modifications. Pi was estimated using a malachite green phosphate assay kit (Cayman chemicals, 10009325). Detailed sample collection and assay protocols are described in extended methods.

Metabolite extraction and analysis by LC-MS/MS

The steady state levels and relative ^{13}C label incorporation into metabolites were estimated by quantitative LC-MS/MS methods as described in (77 [↗](#)). Detailed methodology is extensively described in extended methods. Peak area measurements are listed in supplement file 1.

Mitochondrial isolation

Mitochondria was isolated by immunoprecipitation as described in (78 [↗](#), 79 [↗](#)) with modifications. The detailed protocol is described in extended methods (SI appendix). The eluted mitochondria were used in malachite green assay for Pi estimation, boiled with SDS-glycerol buffer for western blots or incubated with mitotracker CMXRos with mitochondria-activation buffer for mitotracker assays.

Data availability

The LC-MS/MS peak area for all the metabolites can be found in supplement file 1. This study includes no data deposited in external repositories.

Acknowledgements

We thank all SL lab members, Benjamin Tu, Anand Bachhawat and Vijay Jayaraman for comments and suggestions. We acknowledge the NCBS/inStem/CCAMP mass spectrometry facility for LC-MS/MS support. We thank Aayushee Khanna for help with experiments and Gaurav Singh for help with microscopy. V.V acknowledges funding support from DST-INSPIRE fellowship (IF170236) from the Department of Science and Technology (DST), Govt. of India, S.N and SA acknowledges intramural funding from the DBT-inStem PhD program. S.L acknowledges support from the DST SERB CRG grant CRG/2019/004772, a DBT-Wellcome India Alliance Senior Fellowship (IA/S/21/2/505922), and institutional support from inStem.

Supplement figure legends

Figure 1-figure supplement 1

DUB screen details and further characterization of Ubp3 functions

A) A list of the known/identified *S. cerevisiae* deubiquitinase enzymes is shown. The DUBs which were used in this study, and the mitotracker intensity of two biological replicates (n=2) relative to WT are shown.

B) WT cells grown in a respiratory, ethanol medium and mitotracker fluorescence intensity compared to glucose medium. WT cells were grown in high glucose medium (2% glucose) and ethanol medium (2% ethanol), and the mitochondrial membrane potential was measured. Data from two independent experiments (n=2) is shown. Data represent mean $\pm$ SD.

C) Representative images of WT and *ubp3Δ* cells treated with mitotracker red CMXRos.

D) Schematic representation of the conserved domains of Ubp3 and its mammalian ortholog Usp10. The catalytic cysteine of Usp10 and Ubp3 is highlighted.

E) The catalytically inactive mutant of Ubp3, Ubp3^{C469A} does not have altered Ubp3 protein levels. WT and Ubp3^{C469A} cells containing endogenous Ubp3 tagged with 3xFLAG (C terminus) were grown in high glucose and the protein levels of Ubp3 were measured. A representative blot (out of three biological replicates, n=3) and their quantification are shown. Data represent mean $\pm$ SD.

F) The total mitochondrial volume does not change in *ubp3Δ* cells. WT and *ubp3Δ* cells with mitochondria targeted mNeon green were imaged, and the total mitochondrial volume per cell (n=104 for WT, n=107 for *ubp3Δ*) was calculated. A representative image and quantification are shown. Data represent mean $\pm$ SD.

G) Tom70 protein amounts do not change in *ubp3Δ* cells. WT and *ubp3Δ* cells containing endogenous Tom70 tagged at the C terminus with a 3xFLAG epitope were grown in high glucose and the protein levels of Tom70 were measured by western blot. A representative blot (out of three biological replicates, n=3) and their quantification are shown. Data represent mean $\pm$ SD.

H) Dependence of *ubp3Δ* on mitochondrial respiration. WT and *ubp3Δ* cells were grown in high glucose with or without mitochondrial inhibitor oligomycin (25 μ M), and the OD₆₀₀ was measured at different time points. Data represent mean $\pm$ SD (n=2).

I) Dependence of *ubp3Δ* on functional mitochondrial respiration. Ubp3 deletions were done in respiration defective strains -Rho0, *cox2-62*, *atp1Δ* and *atp10Δ*, and serial dilution growth assay was done in high glucose. The results after 24 hours of incubation at 30°C are shown.

Figure 2-figure supplement 1

Further estimates of glycolytic enzymes and flux

A) Loss of Ubp3 does not change the protein levels of enolase isozymes Eno1 and Eno2. WT and *ubp3Δ* cells containing Eno1 and Eno2 C terminus endogenously tagged with a 3x FLAG epitope, were grown in high glucose and protein amounts were estimated. A representative blot (out of three biological replicates, n=3) is shown in the lower panel, quantifications shown in the upper panel. Data represent mean ± SD.

B) The decreased levels of glycolytic enzymes in *ubp3Δ* cells is not because of proteasomal degradation. WT and *ubp3Δ* were grown in high glucose and *ubp3Δ* cells were treated with or without MG132 (100 μM) for 30 minutes. Pfk1, Tdh2, and Tdh3 levels were measured by western blot using an anti-FLAG antibody. A representative blot (out of three biological replicates, n=3) and their quantification are shown. Data represent mean ± SD.

C) Loss of Ubp3 results in decreased transcription of PFK1, TDH2 and TDH3 genes. WT, *ubp3Δ* and Ubp3^{C469A} cells were grown in high glucose and the mRNA levels of PFK1, TDH2 and TDH3 were analysed by RT-qPCR. The fold changes in mRNA levels are shown. Data represent mean ± SD.

D) Steady-state pyruvate levels in WT and *ubp3Δ* in high glucose. Data represent mean ± SD from three biological replicates (n=3). Also see Appendix Table S3.

E) Schematic showing the experimental design for measuring ¹³C label incorporation into glycolytic intermediates and trehalose using a ¹³C₆ glucose-pulse.

F) Changes in ¹³C label incorporation into glycolytic intermediates and trehalose with time, and linearity of label incorporation after a pulse of 1% ¹³C₆ glucose is shown.

Figure 2-figure supplement 2

Estimation of steady state levels and flux of TCA cycle intermediates

A) Steady-state TCA metabolite amounts in WT and *ubp3Δ* in high glucose. Relative steady-state levels of TCA cycle intermediates were estimated in WT and *ubp3Δ*. Data represent mean $\pm$ SD from three biological replicates (n=3).

Also see Appendix Table S3.

B) Schematic showing the experimental design for measuring ^{13}C label incorporation into TCA cycle intermediates using a $^{13}\text{C}_6$ glucose-pulse.

C) Changes in ^{13}C label incorporation into TCA cycle intermediates with time, and linearity of label incorporation after a pulse of 1% $^{13}\text{C}_6$ glucose is shown.

D) Relative TCA cycle flux in WT and *ubp3Δ*. Relative ^{13}C -label incorporation into TCA cycle intermediates, after a pulse of 1% $^{13}\text{C}_6$ glucose is shown. Data represent mean $\pm$ SD from three biological replicates (n=3). Also see Appendix Table S3.

Data information: *p<0.05

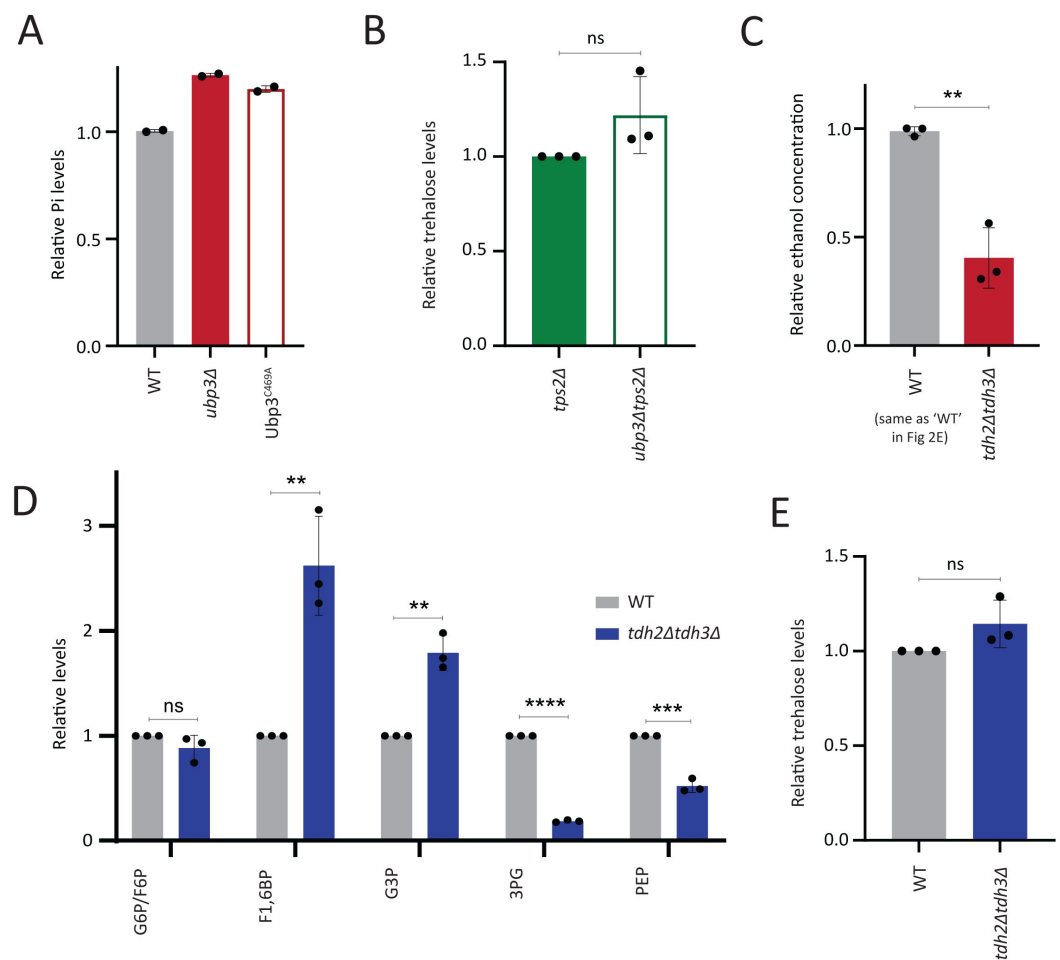


Figure 3-figure supplement 1

Comparisons of phosphate, ethanol and other metabolites in WT, *ubp3Δ* and GAPDH mutants

A) Loss of Ubp3 deubiquitinase activity and Pi levels. WT, *ubp3Δ*, and Ubp3^{C469A} cells were grown in high glucose and the total free phosphate (Pi) levels were estimated. Data from two independent experiments (n=2) is shown. Data represent mean ± SD. B) Trehalose levels in WT and *ubp3Δ* cells in the absence of Tps2. *tps2Δ* and *ubp3Δtps2Δ* cells were grown in high glucose, and the trehalose levels were estimated using targeted LC-MS/MS. Data represent mean ± SD from three biological replicates (n=3).

C) Loss of GAPDH isozymes Tdh2 and Tdh3 and ethanol production. WT and *tdh2Δtdh3Δ* cells were grown in high glucose and the ethanol concentration in the medium was measured. Data are represented as mean ± SD from three biological replicates (n=3). Note: The WT shown here is same as the WT in [Figure 2E](#), the ethanol assays in [Figure 2E](#) and [Figure S3C](#) were done together and a common WT control was used.

D) Steady state levels of glycolytic intermediates: *tdh2Δtdh3Δ* cells have unaltered steady state levels of G6P/F6P, increased levels of F1,6 BP and G3P and decreased levels of 3PG and PEP. WT and *tdh2Δtdh3Δ* cells were grown in high glucose and the steady state amounts of glycolytic intermediates were estimated using targeted LC-MS/MS. Data represent mean ± SD from three biological replicates (n=3). Also see Appendix Table S3.

E) Trehalose levels in *tdh2Δtdh3Δ* cells. WT and *tdh2Δtdh3Δ* cells were grown in high glucose and trehalose amounts were estimated using targeted LC-MS/MS. Data represent mean ± SD from three biological replicates (n=3).

Data information: *p<0.05, **p<0.01.

Figure 4-figure supplement 1

Mitochondrial Pi estimation characterizations and correlations of mitochondrial activity with Pi availability

A) Lack of vacuolar contamination in mitochondria isolated by immunoprecipitation. Mitochondria were isolated from WT cells grown in high glucose and the protein levels of Tom20, Cox2, Idh1 and Vph1 and were measured in cell lysate and immunoprecipitated mitochondria by western blot.

B) Idh1 levels in *ubp3Δ* cells and WT cells grown in high Pi. WT cells were grown in high glucose, or high glucose and high Pi medium (10mM Pi) and *ubp3Δ* cells were grown in high glucose, and the Idh1 protein levels were estimated by western blot. A representative blot (out of three biological replicates, n=3) is shown in the lower panel. Quantification is shown in the upper panel. Data represent mean $\pm$ SD.

C) Mitochondrial membrane potential in *tdh2Δtdh3Δ* cells. WT, *ubp3Δ*, and *tdh2Δtdh3Δ* cells were grown in high glucose and the mitochondrial membrane potential was measured. Data represent mean $\pm$ SD from three biological replicates (n=3).

D) Pi levels in *ubp3Δ* cells grown in low Pi (1mM Pi) vs WT cells. WT cells were grown in high glucose and *ubp3Δ* cells were grown in high glucose or high glucose-low Pi, and the total Pi levels were estimated. Data represent mean $\pm$ SD from three biological replicates (n=3).

E) Ethanol production in *ubp3Δ* cells grown in a low Pi medium. WT cells were grown in high glucose and *ubp3Δ* cells were grown in high glucose or high glucose-low Pi, and the ethanol concentration in the medium was measured. Data represent mean $\pm$ SD from three biological replicates (n=3).

F) Pi amounts and mitochondrial membrane potential in WT, *ubp3Δ*, and *tdh2Δtdh3Δ* cells. The cells were grown in high glucose or high glucose-low Pi, and the mitochondrial membrane potential was measured. Data from two independent experiments (n=2) is shown. Data represent mean $\pm$ SD.

G) The Pi mediated change in basal OCR in *ubp3Δ* is dependent on mitochondrial respiration. WT and Rho0 cells were grown in high glucose, Rho0/*ubp3Δ* were grown in high glucose and low Pi, and basal OCR was measured from three independent experiments (n=3). Data represent mean $\pm$ SD.

H) The protein levels of mitochondrial Pi transporters Mir1 and Pic2 in *ubp3Δ* cells. WT and *ubp3Δ* cells containing endogenously tagged Mir1 and Pic2 at their C terminus with a 6x HA epitope tag were grown in high glucose and Mir1 and Pic2 proteins were measured by western blot. A representative blot (out of three biological replicates, n=3) is shown in the lower panel. Quantification is shown in the upper panel. Data represent mean $\pm$ SD.

Data information: *p<0.05, **p<0.01, ***p<0.001.

Figure 5-figure supplement 1

Mitochondrial phosphate and pyruvate transport relationships with mitochondrial activity

A) Requirement of Mir1 for switching to increased mitochondrial activity upon 2-deoxyglucose (2-DG) treatment. WT and *mir1Δ* cells were grown in high glucose and treated with 0.1% and 0.25% 2-DG for one hour. The mitochondrial membrane potential was measured from two independent experiments (n=2). Data represent mean ± SD.

B) Mir1 protein levels and glucose repression. Cells containing Mir1 endogenously tagged at the C terminus with a 6x HA epitope tag were grown in high glucose or a respiratory medium (2% ethanol). Mir1 protein was estimated by western blot. A representative blot (out of three biological replicates, n=3) is shown in the lower panel. Quantification is shown in the upper panel. Data represent mean ± SD.

C) Effect of supplementing Pi in the medium on mitochondrial Pi. WT cells were grown in high glucose, and high glucose, high Pi medium (10 mM Pi), mitochondria was isolated and the mitochondrial Pi levels were estimated. The mitochondrial Pi levels normalised to Idh1 protein levels is shown. Data represent mean ± SD from three biological replicates (n=3). Also see supplementary figure S4B.

D) Effect of supplementing Pi on the total mitochondrial volume. WT cells with mitochondria targeted mNeon green were grown in high glucose, and high glucose, high Pi medium (10 mM Pi), cells were imaged and the total mitochondrial volume per cell was calculated.

E) Increasing Pi concentration in a high glucose medium results in a decrease in basal OCR. WT cells were grown in high glucose medium (YPD-Pi medium), supplemented with 5 mM Pi and at OD₆₀₀~0.6, the medium was supplemented with Pi (10 mM and 25 mM final concentrations). The basal OCR was measured one hour after Pi supplementation. Data represent mean ± SD from three biological replicates (n=3).

E) Effect of increasing the Pi concentration in a high glucose medium, in the presence of 2DG, on basal OCR. WT cells were grown in high glucose medium (YPD-Pi medium), supplemented with 5 mM Pi and at OD₆₀₀~0.6, the medium was supplemented with Pi (25 mM final concentration) and 2DG (0.25%) for one hour and the basal OCR was measured. Data represent mean ± SD from three biological replicates (n=3).

F) Overexpression of Mir1. C terminal 6x HA epitope-tagged Mir1 was expressed under G6PD promoter in cells with Mir1 endogenously tagged at the C terminus with a 6x-HA epitope tag. WT cells (carrying an empty vector, and expressing Mir1-HA under the endogenous Mir1 promoter) and Mir1-OE cells were grown in high glucose. The protein levels of Mir1 were estimated by western blot, and clones with an ~2-fold increase in Mir1 were selected. A representative blot (out of three biological replicates, n=3) is shown in the lower panel. Quantification is shown in the upper panel. Data represent mean ± SD.

G) Mpc3 protein levels in *ubp3Δ* cells. WT and *ubp3Δ* cells containing endogenously tagged Mpc3 at the C terminus with a 3x FLAG epitope tag, were cultured in high glucose (2% glucose) and high glucose-low Pi (2% glucose, 1 mM Pi). The Mpc3 protein levels were measured by western blot. A representative blot (out of three biological replicates, n=3) is shown in the lower panel, quantifications shown in the upper panel. Data represent mean ± SD.

H) Mpc3 protein amounts upon shifting WT and *mir1Δ* cells to a low-glucose medium. WT and *mir1Δ* cells containing Mpc3-FLAG, were cultured in high glucose (2% glucose) and shifted to low (0.1%) glucose for 1 hour. Mpc3 protein was measured by western blot. A representative blot (out of three biological replicates, n=3) is shown in the lower panel, quantifications shown in the upper panel. Data represent mean ± SD.

I) Requirement of Mpc3 for the increase in basal OCR upon shifting to low glucose. WT and *mpc3Δ* cells were cultured in high glucose (2% glucose) and shifted to low (0.1%) glucose for 1 hour. The basal OCR from two independent experiments (n=2), normalized to the OD₆₀₀ is shown. Data represent mean ± SD.

J) Mpc3 levels and Mir1 overexpression. WT cells and Mir1-OE cells containing endogenously tagged Mpc3 at the C terminus with a 3x FLAG tag were grown in high glucose. The protein levels of Mpc3 were estimated by western blot. A representative blot (out of three biological replicates, n=3) is shown in the panel. Quantification is shown in the upper panel. Data represent mean $\pm$ SD. Data information: *p<0.05, **p<0.01, ***p<0.001.

References

1. Zhu J., Thompson C. B. (2019) **Metabolic regulation of cell growth and proliferation** *Nat. Rev. Mol. Cell Biol* **20**:436–450
2. Cai L., Tu B. P. (2011) **Driving the Cell Cycle Through Metabolism** *Annu. Rev. Cell Dev. Biol* **28**:59–87
3. David L., Nelson D. L., Nelson David L., Lehninger A. L., Cox M. M. (2008) **D. L. (David L. Nelson, D. L. (David L. Nelson, A. L. Lehninger, M. M. Cox, Lehninger principles of biochemistry (2008) (March 10, 2022). Lehninger principles of biochemistry**
4. Warburg O. (1925) **The metabolism of carcinoma cells 1** *J. Cancer Res* **9**:148–163
5. Vander Heiden M. G., Cantley L. C., Thompson C. B. (2009) **Understanding the Warburg Effect: The Metabolic Requirements of Cell Proliferation** *Science (80-.)* **324**:1029–1033
6. Crabtree H. G. (1929) **Observations on the carbohydrate metabolism of tumours** *Biochem. J* **23**:536–545
7. De Deken R. H. (1966) **The Crabtree effect: a regulatory system in yeast** *J. Gen. Microbiol* **44**:149–156
8. Xia L., et al. (2021) **The cancer metabolic reprogramming and immune response** *Mol. Cancer* **20**:1–21
9. Kornberg M. D. (2020) **The immunologic Warburg effect: Evidence and therapeutic opportunities in autoimmunity** *Wiley Interdiscip. Rev. Syst. Biol. Med* **12**:1–17
10. Pacini N., Borziani F. (2014) **Cancer stem cell theory and the warburg effect, two sides of the same Coin?** *Int. J. Mol. Sci* **15**:8893–8930
11. Abdel-Haleem A. M., et al. (2017) **The emerging facets of non-cancerous Warburg effect** *Front. Endocrinol. (Lausanne)* **8**:1–7
12. Tsogtbaatar E., Landin C., Minter-Dykhouse K., Folmes C. D. L. (2020) **Energy Metabolism Regulates Stem Cell Pluripotency** *Front. Cell Dev. Biol* **8**:1–16
13. Cornish-Bowden A. (2016) **Biochemical evolution : the pursuit of perfection** *Garl. Sci*
14. Diaz-Ruiz R., Rigoulet M., Devin A. (2011) **The Warburg and Crabtree effects: On the origin of cancer cell energy metabolism and of yeast glucose repression** *Biochim. Biophys. Acta - Bioenerg* **1807**:568–576
15. Hammad N., Rosas-Lemus M., Uribe-Carvajal S., Rigoulet M., Devin A. (2016) **The Crabtree and Warburg effects: Do metabolite-induced regulations participate in their induction?** *Biochim. Biophys. Acta - Bioenerg* **1857**:1139–1146
16. Díaz-Ruiz R., et al. (2008) **Mitochondrial Oxidative Phosphorylation Is Regulated by Fructose 1,6-Bisphosphate: A POSSIBLE ROLE IN CRABTREE EFFECT INDUCTION? *** *J. Biol. Chem* **283**:26948–26955

17. Lemus M. R., et al. (2018) **The role of glycolysis-derived hexose phosphates in the induction of the Crabtree effect** *J. Biol. Chem* **293**:12843–12854
18. Koobs D. H. (1972) **Phosphate mediation of the Crabtree and Pasteur effects** *Science (80-.)* **178**:127–133
19. Rodríguez-Enríquez S., Juárez O., Rodríguez-Zavala J. S., Moreno-Sánchez R. (2001) **Multisite control of the Crabtree effect in ascites hepatoma cells** *Eur. J. Biochem* **268**:2512–2519
20. Tripodi F., Nicastro R., Reghellin V., Coccetti P. (2015) **Post-translational modifications on yeast carbon metabolism: Regulatory mechanisms beyond transcriptional control** *Biochim. Biophys. Acta - Gen. Subj* **1850**:620–627
21. Hitosugi T., Chen J. (2014) **Post-translational modifications and the Warburg effect** *Oncogene* **33**:4279–4285
22. Broach J. R. (2012) **Nutritional control of growth and development in yeast** *Genetics* **192**:73–105
23. Hershko A., Ciechanover A. (1998) **THE UBIQUITIN SYSTEM** *Annu. Rev. Biochem* **67**:425–79
24. Komander D., Rape M. (2012) **The Ubiquitin Code** *Annu. Rev. Biochem* **81**:203–229
25. Pickart C. M., Eddins M. J. (2004) **Ubiquitin: Structures, functions, mechanisms** *Biochim. Biophys. Acta - Mol. Cell Res* **1695**:55–72
26. Postma E., Verduyn C., Scheffers W. A., Van Dijken J. P. (1989) **Enzymic analysis of the crabtree effect in glucose-limited chemostat cultures of *Saccharomyces cerevisiae*** *Appl. Environ. Microbiol* **55**:468–77
27. Ossareh-Nazari B., et al. (2010) **Cdc48 and Ufd3, new partners of the ubiquitin protease Ubp3, are required for ribophagy** *EMBO Rep* **11**:548–554
28. Isasa M., et al. (2015) **Multiplexed, Proteome-Wide Protein Expression Profiling: Yeast Deubiquitylating Enzyme Knockout Strains** *J. Proteome Res* **14**:5306–5317
29. Nostramo R., Varia S. N., Zhang B., Emerson M. M., Herman P. K. (2016) **The Catalytic Activity of the Ubp3 Deubiquitinating Protease Is Required for Efficient Stress Granule Assembly in *Saccharomyces cerevisiae*** *Mol. Cell. Biol* **36**
30. Fontanesi F., Soto I. C., Horn D., Barrientos A. (2006) **Assembly of mitochondrial cytochrome c-oxidase, a complicated and highly regulated cellular process** *Am. J. Physiol. - Cell Physiol* **291**:1129–1147
31. Dua N., Seshadri A., Badrinarayanan A. (2022) **DarT-mediated mtDNA damage induces dynamic reorganization and selective segregation of mitochondria** *J. Cell Biol* **221**
32. Bonnefoy N., Bsai N., Fox T. D. (2001) **Mitochondrial Translation of *Saccharomyces cerevisiae* COX2 mRNA Is Controlled by the Nucleotide Sequence Specifying the Pre-Cox2p Leader Peptide** *Mol. Cell. Biol* **21**:2359–2372
33. Shestov A. A., et al. (2014) **Quantitative determinants of aerobic glycolysis identify flux through the enzyme GAPDH as a limiting step** *Elife* **3**:1–18

34. Gupta R., et al. (2019) **A tRNA modification balances carbon and nitrogen metabolism by regulating phosphate homeostasis** *Elife* **8**:1–33
35. Gupta R., Laxman S. (2021) **Cycles, sources, and sinks: Conceptualizing how phosphate balance modulates carbon flux using yeast metabolic networks** *Elife* **10**:1–14
36. Mason P. W., Carbone D. P., Cushman R. A., Waggoner A. S. (1981) **The importance of inorganic phosphate in regulation of energy metabolism of *Streptococcus lactis*** *J. Biol. Chem* **256**:1861–1866
37. Van Heerden J. H., et al. (2014) **Lost in transition: Start-up of glycolysis yields subpopulations of nongrowing cells** *Science (80-.)* **343**
38. Brazy P. C., Mandel L. J., Dennis V. W., Gullans S. R. (1982) **Metabolic Requirement for Inorganic Phosphate by the Rabbit Proximal Tubule: EVIDENCE FOR A CRABTREE EFFECT** **Metabolic Requirement for Inorganic Phosphate by the Rabbit Proximal Tubule EVIDENCE FOR A CRABTREE EFFECT** *J Clin Invest* **70**
39. Hohmann S., Bell W., Neves M. J., Valckx D., Thevelein J. M. (1996) **Evidence for trehalose-6-phosphate-dependent and-independent mechanisms in the control of sugar influx into yeast glycolysis** *Mol. Microbiol* **20**:981–991
40. Hunter T. (2012) **Why nature chose phosphate to modify proteins** *Philos. Trans. R. Soc. B Biol. Sci* **367**:2513–2516
41. Ljungdahl P. O., Daignan-Fornier B. (2012) **Regulation of amino acid, nucleotide, and phosphate metabolism in *Saccharomyces cerevisiae*** *Genetics* **190**:885–929
42. Austin S., Mayer A. (2020) **Phosphate Homeostasis – A Vital Metabolic Equilibrium Maintained Through the INPHORS Signaling Pathway** *Front. Microbiol* **11**
43. Mouillon J. M., Persson B. L. (2006) **New aspects on phosphate sensing and signalling in *Saccharomyces cerevisiae*** *FEMS Yeast Res* **6**:171–176
44. Murakami H., Blobel G., Pain D. (1990) **Isolation and characterization of the gene for a yeast mitochondrial import receptor** *Nat. 1990* **347**:488–491
45. Zara V., et al. (1996) **Yeast mitochondria lacking the phosphate carrier/p32 are blocked in phosphate transport but can import preproteins after regeneration of a membrane potential** *Mol. Cell. Biol* **16**:6524–6531
46. Harden A., Young W. J. (1908) **The alcoholic Ferment of Yeast-juice.Part III.-The Function of Phosphates in the Fermentation o f Glucose hy Yeast-juice** *Proc. R. Soc. London. Ser. B* **80**:299–311
47. Timoń-Gómez A., Proft M., Pascual-Ahuir A. (2013) **Differential Regulation of Mitochondrial Pyruvate Carrier Genes Modulates Respiratory Capacity and Stress Tolerance in Yeast** *PLoS One* **8**
48. Bender T., Pena G., Martinou J.-C. (2015) **Regulation of mitochondrial pyruvate uptake by alternative pyruvate carrier complexes** *EMBO J* **34**:911–924
49. Peter J., et al. (2018) **Genome evolution across 1,011 *Saccharomyces cerevisiae* isolates** *Nature* **556**:339–344

50. Grigaitis P., Teusink B. (2022) **An excess of glycolytic enzymes under glucose-limited conditions may enable *Saccharomyces cerevisiae* to adapt to nutrient availability** *FEBS Lett* <https://doi.org/10.1002/1873-3468.14484>
51. Yi W., et al. (2012) **PFK1 Glycosylation Is a Key Regulator of Cancer Cell Growth and Central Metabolic Pathways** *Science* **337**
52. Miyazawa H., et al. (2017) **Rewiring of embryonic glucose metabolism via suppression of PFK-1 and aldolase during mouse chorioallantoic branching** *Dev* **144**:63–73
53. Hollinshead W. D., et al. (2016) **Examining *Escherichia coli* glycolytic pathways, catabolite repression, and metabolite channeling using *Δpfk* mutants** *Biotechnol. Biofuels* **9**:1–13
54. Westheimer F. H. (1987) **Why Nature Chose Phosphates** *Science (80-.)* **235**:1173–1178
55. Kamerlin S. C. L., Sharma P. K., Prasad R. B., Warshel A. (2013) **Why nature really chose phosphate** *Q. Rev. Biophys* **46**:1–132
56. Booth J. W., Guidotti G. (1997) **Phosphate Transport in Yeast Vacuoles** *J. Biol. Chem* **272**:20408–20413
57. Vila T., Frases S., Gomesid F. M. (2022) **Lessons from protozoans: Phosphate sensing and polyphosphate storage in fungi** *PLOS Pathog* **18**
58. Solesio M. E., et al. (2021) **Depletion of mitochondrial inorganic polyphosphate (polyP) in mammalian cells causes metabolic shift from oxidative phosphorylation to glycolysis** *Biochem. J* **478**:1631–1646
59. Brin M., McKee R. W. (1956) **The Inhibition of Respiration by Glucose, Fructose, and Mannose in the Ehrlich Mouse Ascites Tumor** *Cancer Res* **16**:364–368
60. Orij R., Brul S., Smits G. J. (2011) **Intracellular pH is a tightly controlled signal in yeast** *Biochim. Biophys. Acta - Gen. Subj* **1810**:933–944
61. Dechant R., et al. (2010) **Cytosolic pH is a second messenger for glucose and regulates the PKA pathway through V-ATPase** *EMBO J* **29**:2515–2526
62. Hamel P., et al. (2004) **Redundancy in the function of mitochondrial phosphate transport in *Saccharomyces cerevisiae* and *Arabidopsis thaliana*** *Mol. Microbiol* **51**:307–317
63. Jia F., et al. (2015) **Overexpression of Mitochondrial Phosphate Transporter 3 Severely Hampers Plant Development through Regulating Mitochondrial Function in *Arabidopsis*** *PLoS One* **10**
64. Pelicano H., Martin D. S., Xu R. (2006) **Huang Glycolysis inhibition for anticancer treatment** :4633–4646
65. O'Neill S., Porter R. K., McNamee N., Martinez V. G., O'Driscoll L. (2019) **2-Deoxy-D-Glucose inhibits aggressive triple-negative breast cancer cells by targeting glycolysis and the cancer stem cell phenotype** *Sci. Rep* **9**:1–11
66. Soto-Herederó G., Gómez de las Heras M. M., Gabandé-Rodríguez E., Oller J., Mittelbrunn M. (2020) **Glycolysis – a key player in the inflammatory response** *FEBS J* **287**:3350–3369

67. Shiratori R., et al. (2019) **Glycolytic suppression dramatically changes the intracellular metabolic profile of multiple cancer cell lines in a mitochondrial metabolism-dependent manner** *Sci. Rep* **9**:1–15
68. Lu C. L., et al. (2015) **Tumor Cells Switch to Mitochondrial Oxidative Phosphorylation under Radiation via mTOR-Mediated Hexokinase II Inhibition - A Warburg-Reversing Effect** *PLoS One* **10**
69. Cheng G., et al. (2012) **Mitochondria-targeted drugs synergize with 2-deoxyglucose to trigger breast cancer cell death** *Cancer Res* **72**:2634–2644
70. Scheibye-Knudsen M., Quistorff B. (2009) **Regulation of mitochondrial respiration by inorganic phosphate; comparing permeabilized muscle fibers and isolated mitochondria prepared from type-1 and type-2 rat skeletal muscle** *Eur. J. Appl. Physiol* **105**:279–287
71. Seifert E. L., Ligeti E., Mayr J. A., Sondheimer N., Hajnóczky G. (2015) **The mitochondrial phosphate carrier: Role in oxidative metabolism, calcium handling and mitochondrial disease** *Biochem. Biophys. Res. Commun* **464**:369–375
72. van Dijken J. B. J.P., et al. (2000) **An interlaboratory comparison of physiological and genetic properties of four *Saccharomyces cerevisiae* strains** *Enzyme Microb. Technol* **26**:706–714
73. Longtine M. S., et al. (1998) **Additional modules for versatile and economical PCR-based gene deletion and modification in *Saccharomyces cerevisiae*** *Yeast* **14**:953–961
74. Vengayil V., Rashida Z., Laxman S. (2019) **The E3 ubiquitin ligase Pib1 regulates effective gluconeogenic shutdown upon glucose availability** *J. Biol. Chem* **294**:17209–17223
75. Chaudhry A., Shi R., Luciani X. D. S. (2019) **INNOVATIVE METHODOLOGY A pipeline for multidimensional confocal analysis of mitochondrial morphology, function, and dynamics in pancreatic β -cells** *Am. J. Physiol. - Endocrinol. Metab* **318**:87–101
76. Sriariyanun M., Mutrakulcharoen P., Tapaamorndech S., Cheenkachorn K., Rattanaporn K. (2019) **A Rapid Spectrophotometric Method for Quantitative Determination of Ethanol in Fermentation Products** *Orient. J. Chem* **35**:744–750
77. Walvekar A., Rashida Z., Maddali H., Laxman S. (2018) **A. Walvekar, Z. Rashida, H. Maddali, S. Laxman, A versatile LC-MS / MS approach for comprehensive, quantitative analysis of central metabolic pathways [version 1 ; peer review : 2 approved]. 1–15 (2018). A versatile LC-MS / MS approach for comprehensive, quantitative analysis of central metabolic pathways [version 1 ; peer review : 2 approved] :1–15**
78. Liao P. C., Boldogh I. R., Siegmund S. E., Freyberg Z., Pon L. A. (2018) **Isolation of mitochondria from *Saccharomyces cerevisiae* using magnetic bead affinity purification** *PLoS One* **13**:1–15
79. Chen W. W., Freinkman E., Sabatini D. M. (2017) **Rapid immunopurification of mitochondria for metabolite profiling and absolute quantification of matrix metabolites** *Nat. Protoc* **12**

Article and author information

Vineeth Vengayil

Institute for Stem Cell Science and Regenerative Medicine (DBT-inStem) GKV Post Bellary Road Bangalore 560065, Manipal Academy of Higher Education, Manipal, Karnataka 576104, India

ORCID iD: [0000-0002-1078-5476](https://orcid.org/0000-0002-1078-5476)

Shreyas Niphadkar

Institute for Stem Cell Science and Regenerative Medicine (DBT-inStem) GKV Post Bellary Road Bangalore 560065, Manipal Academy of Higher Education, Manipal, Karnataka 576104, India

Swagata Adhikary

Institute for Stem Cell Science and Regenerative Medicine (DBT-inStem) GKV Post Bellary Road Bangalore 560065, Manipal Academy of Higher Education, Manipal, Karnataka 576104, India

Sriram Varahan

Institute for Stem Cell Science and Regenerative Medicine (DBT-inStem) GKV Post Bellary Road Bangalore 560065

ORCID iD: [0000-0002-3609-4032](https://orcid.org/0000-0002-3609-4032)

Sunil Laxman

Institute for Stem Cell Science and Regenerative Medicine (DBT-inStem) GKV Post Bellary Road Bangalore 560065

For correspondence: sunil@instem.res.in

ORCID iD: [0000-0002-0861-5080](https://orcid.org/0000-0002-0861-5080)

Copyright

© 2023, Vengayil et al.

This article is distributed under the terms of the [Creative Commons Attribution License](https://creativecommons.org/licenses/by/4.0/), which permits unrestricted use and redistribution provided that the original author and source are credited.

Editors

Reviewing Editor

Weiwei Dang

Baylor College of Medicine, Houston, United States of America

Senior Editor

Carlos Isales

Augusta University, Augusta, United States of America

Reviewer #1 (Public Review):

The study by Vengayil et al. presented a role for Ubp3 for mediating inorganic phosphate (Pi) compartmentalization in cytosol and mitochondria, which regulates metabolic flux between cytosolic glycolysis and mitochondrial processes. Although the exact function of increased Pi in mitochondria is not investigated, findings have valuable implications for understanding

the metabolic interplay between glycolysis and respiration under glucose-rich conditions. They showed that UBP3 KO cells regulated decreased glycolytic flux by reducing the key Pi-dependent-glycolytic enzyme abundances, consequently increasing Pi compartmentalization to mitochondria. Increased mitochondria Pi increases oxygen consumption and mitochondrial membrane potential, indicative of increased oxidative phosphorylation. In conclusion, the authors reported that the Pi utilization by cytosolic glycolytic enzymes is a key process for mitochondrial repression under glucose conditions.

Comments on revised version:

This reviewer appreciates the author's responses addressing some of the concerns.

However, the concern of reproducibility and experimental methods applied to the study is still valid, particularly considering that many conclusions were drawn from western blot analysis. The authors used separate gel loading controls for western blot analysis, which is not a valid method. Considering loading and other errors/discrepancies during the transfer phase of the assay, the direct control should be analyzing the membrane after transfer or using an internal control antibody on the same membrane. None of the western blots are indicated with marker sizes, and it isn't very clear how many repeats there are and whether those repeats are biological or technical repeats.

Concern regarding citing the Ouyang et al. paper is still valid. This paper is an essential implication in phosphate metabolism and is directly related to some of the findings associated with mitochondrial function, along with conflicting results, which should be discussed in the discussion section. As a reviewer, I do not request citing any paper from the authors in general; however, considering some of the conflicting results here, citing and discussing paper from Ouyang et al. will improve the interoperation/value of their findings.

Considering these factors, the presented results do not fully support the findings.

<https://doi.org/10.7554/eLife.90293.2.sa1>

Reviewer #2 (Public Review):

Summary:

Cells cultured in high glucose tend to repress mitochondrial biogenesis and activity, a prevailing phenotype type called Crabtree effect that observed in different cell types and cancer. Many signaling pathways have been put forward to explain this effect. Vengayil et al proposed a new mechanism involved in Ubp3/Ubp10 and phosphate that controls the glucose repression of mitochondria. The central hypothesis is that Δ ubp3 shift the glycolysis to trehalose synthesis, therefore lead to the increase of Pi availability in the cytosol, then mitochondria received more Pi and therefore the glucose repression is reduced.

Strengths:

The strength is that the authors used an array of different assays to test their hypothesis. Most assays were well designed and controlled.

Weaknesses:

I think the main conclusions are not strongly supported by the current dataset. Here are my comments on authors' response and model.

(1) The authors addressed some of my concerns related to Δ ubp3. But based on the results they observed and discussed, the Δ ubp3 redirect some glycolytic flux to gluconeogenesis while the 0.1% glucose in WT does not. Similarly, the shift of glycolysis to trehalose synthesis

is also not relevant to the WT cells cultured in low glucose situation. This should be discussed in the manuscript to make sure readers are not misled to think Δ ubp3 mimic low glucose. It is likely that Δ ubp3 induce proteostasis stress, which is known to activate respiration and trehalose synthesis.

(2) Pi flux: it is known that vacuole can compensate the reduction of Pi in the cytosol. The paper they cited in the response, especially the Van Heerden et al., 2014 showed that the pulse addition of glucose caused transient Pi reduction and then it came back to normal level after 10min or so. If the authors mean the transient change of glycolysis and respiration, they should point that out clearly in the abstract and introduction. If the authors are trying to put out a general model, then the model must be reconsidered.

The cytosol has ~50mM Pi (van Eunen et al., 2010 FEBSJ), while only 1-2mM of glycolysis metabolites, not sure why partial reduction of several glycolysis enzymes will cause significant changes in cytosolic Pi level and make Pi the limiting factor for mitochondrial respiration. In response to this comment, the authors explained the metabolic flux that the rapid, continuous glycolysis will drain the Pi pool even each glycolytic metabolite is only 1-2mM. However, the metabolic flux both consume and release Pi, that's why there is such measurement of overall free Pi concentration amid the active metabolism. One possibility is that the observed cytosolic Pi level changes was caused by the measurement fluctuation, as they showed in "Reviewer response image 3".

Importantly, the authors measured Pi inside mito for ethanol and glucose, but not the cytosolic Pi, which is the key hypothesis in their model. The model here is that the glycolysis competes with mito for free cytosolic Pi, so it needs to inhibit glycolysis to free up cytosolic Pi for mitochondrial import to increase respiration. I don't see measurement of cytosolic Pi upon different conditions, only the total Pi or mito Pi. The fact is that in Fig.3C they saw WT+Pi in the medium increase total free Pi more than the Δ ubc3, while WT decrease mito Pi compared to WT control and Δ ubc3 and therefore decrease basal OCR upon Pi supplement. A simple math of $Pi_{total} = Pi_{cyto} + Pi_{mito}$ tells us that if WT has more Pi_{total} (Fig.3C) but less Pi_{mito} (fig.5 supp 1C), then it has higher Pi_{cyto} . This is contradictory to what the authors tried to rationalize. Furthermore, as I pointed out previously, the isolated mitochondria can import more Pi when supplemented, so if there is indeed higher Pi_{cyto} , then the mito in WT should import more Pi. So, to address these contradictory points, the authors must measure Pi in the cytosol, which is a critical experiment not done for their model. For example, they hypothesized that adding 2-DG, or Δ ubp3, suppress glycolysis and thus increase the supply of cytosolic Pi for mito to import, but no cytosolic Pi was measured (need absolute value, not the relative fold changes). It is also important to specify how the experiments are done, was the measurement done shortly after adding 2-DG. Given that the cells response to glucose changes/pulses differently in transient vs stable state, the authors are encouraged to specify that.

The most likely model to me is that, which is also the consensus in the field, is that no matter 2-DG or Δ ubp3, the cells re-wiring metabolism in both cytosol and mitochondria, and it is the total network shift that cause the mitochondrial respiration increase, which requires the increase of mito import of Pi, ADP, O₂, and substrates, but not caused/controlled by the Pi that singled out by the authors in their model.

(3) The explanation that cytosolic pH reduction upon glucose depletion/2DG is a mistake. There are a lot of data in the literature showing the opposite. If the authors do think this is true, then need to show the data. Again, it is important to distinguish transient vs stable state for pH changes.

<https://doi.org/10.7554/eLife.90293.2.sa0>

Author Response

The following is the authors' response to the previous reviews.

We thank the reviewers for their reading of the manuscript, and their suggestions. We have extensively addressed all these concerns in the text, and also included several new data and figures in the revised version of the manuscript. We hope that our response and the new experimental data fully address the concerns raised by the reviewers. We include a detailed, point-by-point response to each of the reviewer concerns, pointing to new data and specific changes made in the main manuscript.

Note: Do note that these new data have resulted in a new figure-figure 6, a new supplementary figure -figure 2-figure supplement 2, and an increase in the number of panels in each figure, as well as supplementary figures.

General response comments, highlighting a few aspects missed by the reviewers

This manuscript has an enormous amount of data in it. This is understandable, since in part we are proposing an entirely new hypothesis, and way to think about mitochondrial repression, built around substantial circumstantial evidences from diverse literature sources. But to keep the narrative readable and the main idea understandable, a lot of information had to be only very briefly mentioned in the text, and is therefore included as supplemental information. Due to that, it may not always be apparent that this study has set several technical benchmarks. These experiments are extremely challenging to perform, took many iterations to standardize, and in themselves are a first in the field. Yeast cells have the highest known rate of glycolytic flux for any organism. Measuring this glycolytic rate using the formation of intermediates is hard, and all current estimates have been in vitro, and using a stop-flow type set up. In this study, we optimized and directly measured the glycolytic flux using isotope labelled glucose (^{13}C -glucose), which has never been reported before in highly glycolytic cells such as yeast. This is due to the very rapid label saturation (within seconds) after ^{13}C glucose pulse (as is now shown in the figure 2-figure supplement 1). For brevity, this is summarized in this study with sufficient information to reproduce the method, but we will put out a more detailed, associated methodology paper describing several challenges, infrastructure requirements, and resources to be able to carry out these types of experiments using yeast. An added highlight of these experiments with WT and Ubp3 deletion strains is the most direct till date experimental demonstration that glycolytic flux in yeast in high glucose follows zero-order kinetics, and depends entirely on the amounts of the glycolytic enzymes (presumably operating at maximal activity). This nicely complements the recent study by Grigatis 2022 (cited in the discussion), that suggests this possibility.

Separately, this study required the estimation of total inorganic phosphates, as well as mitochondrial pools of phosphates. Till date, there are no studies that have estimated mitochondrial pools of phosphate (for a variety of reasons). In this study, we also experimentally determined the changes in mitochondrial phosphate pools. For this, we had to establish and standardize a rapid mitochondrial isolation method in yeast. Thus, this study provides the first quantitative estimates of mitochondrial Pi amounts (in the context of measured mitochondrial outputs), as shown now in Figure 4. This component on mitochondrial isolation in yeast to assess metabolites may also be explored in future as a methods paper.

Specific responses to the Reviews:

Reviewer #1 (Public Review):

The study by Vengayil et al. presented a role for Ubp3 for mediating inorganic phosphate (Pi) compartmentalization in cytosol and mitochondria, which regulates metabolic flux

between cytosolic glycolysis and mitochondrial processes. Although the exact function of increased Pi in mitochondria is not investigated, findings have valuable implications for understanding the metabolic interplay between glycolysis and respiration under glucose-rich conditions. They showed that UBP3 KO cells regulated decreased glycolytic flux by reducing the key Pi-dependent-glycolytic enzyme abundances, consequently increasing Pi compartmentalization to mitochondria. Increased mitochondria Pi increases oxygen consumption and mitochondrial membrane potential, indicative of increased oxidative phosphorylation. In conclusion, the authors reported that the Pi utilization by cytosolic glycolytic enzymes is a key process for mitochondrial repression under glucose conditions.

(1) However, the main claims are only partially supported by the low number of repeats and utilizing only one strain background, which decreased the overall rigor of the study. The full-power yeast model could be utilized with testing findings in different backgrounds with increased biological repeats in many assays described in this study. In the yeast model, it has been well established that many phenotypes are genotype/strain dependent (Liti 2019, Gallone 2016, Boekout 2021, etc...). with some strains utilizing mitochondrial respiration even under high glucose conditions (Kaya 2021). It would be conclusive to test whether wild strains with increased respiration under high glucose conditions would also be characterized by increased mitochondrial Pi.

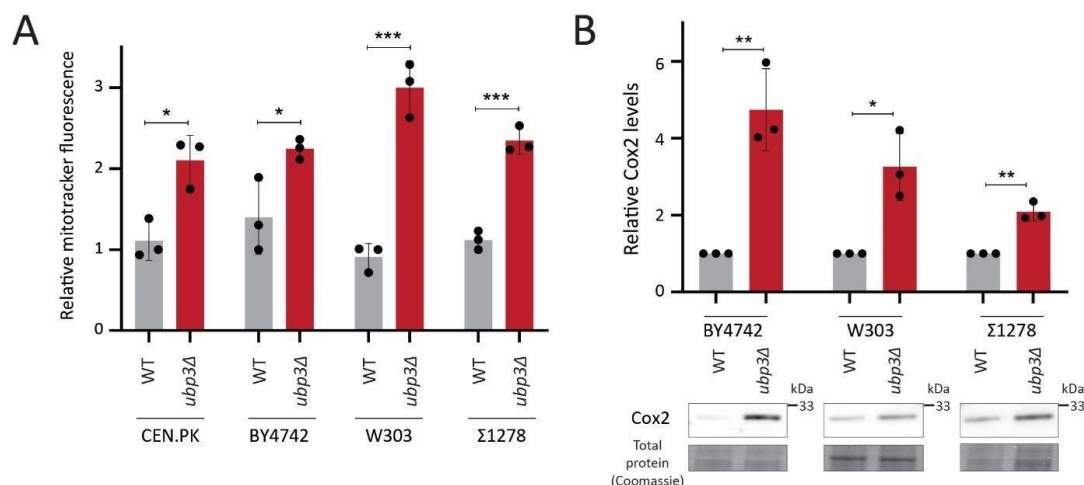
"However, the main claims are only partially supported by the low number of repeats and utilizing only one strain background, which decreased the overall rigor of the study. The full-power yeast model could be utilized with testing findings in different backgrounds with increased biological repeats in many assays described in this study."

Thank you for the suggestion. We agree that a larger, universal statement cannot be made with data from a single strain, since yeasts do have substantial diversity. In this study, we had originally used a robust, prototrophic industrial strain (CEN.PK background). We have now utilized multiple, diverse strains of *S. cerevisiae* to test our findings. This includes strains from the common laboratory backgrounds – W303 and BY4742 – which have different auxotrophies, as well as another robust, highly flocculent strain from a prototrophic Σ 1278 background. Using all these strains, we now comprehensively find that the role of altered Pi budgeting as a constraint for mitochondrial respiration, and the role of Ubp3 as a regulator of mitochondrial repression is very well conserved. In all tested strains of *S. cerevisiae* the loss of Ubp3 increases mitochondrial activity (as shown by increased mitochondrial membrane potential and increased Cox2 levels in Figure 6A, B). These data now expand the generality of our findings, and strengthen the manuscript. These results are included in the revised manuscript as a new figure- Figure 6 and the associated text.

Some of the included data in the revised manuscript are shown below:

Author response image 1.

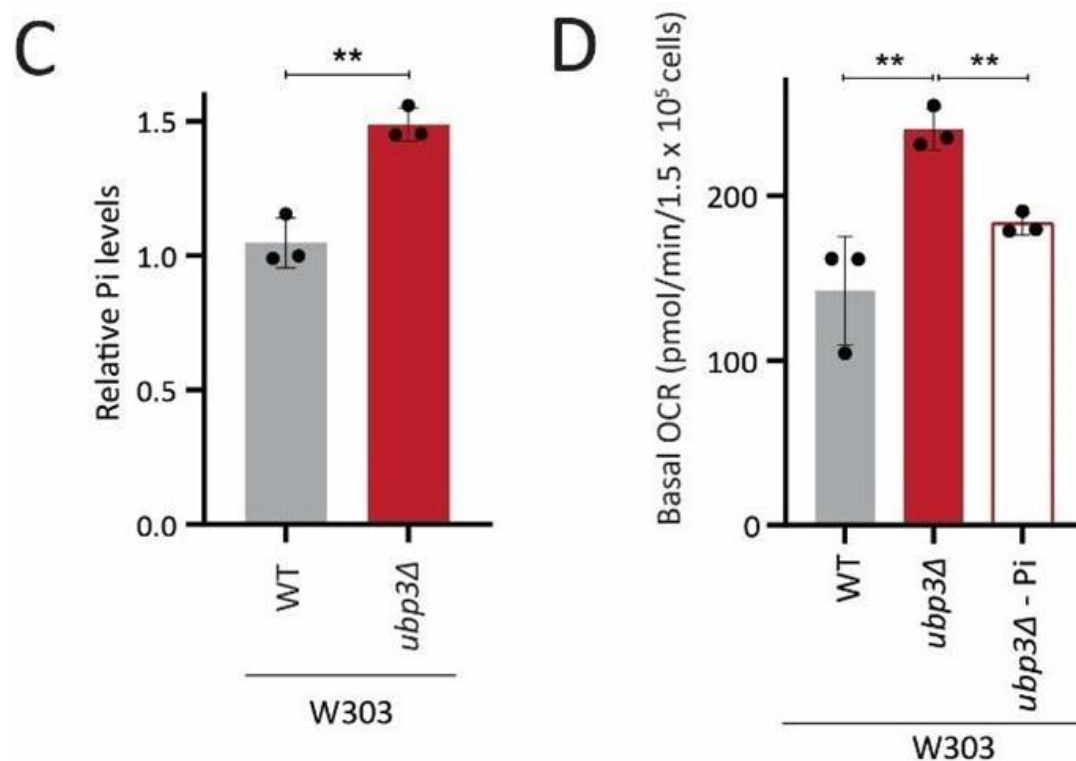
Mitochondrial activity and Cox2 levels in *ubp3Δ* in different genetic backgrounds



We also used the W303 strain to assess Pi levels, and its role in increasing mitochondrial respiration. We find that the loss of Ubp3 in this genetic background also increases Pi levels and that the increased Pi is necessary for increasing mitochondrial respiration (Figure 6C, D).

Author response image 2.

Basal OCR in WT vs *ubp3Δ* (W303 strain background) in normal vs low Pi



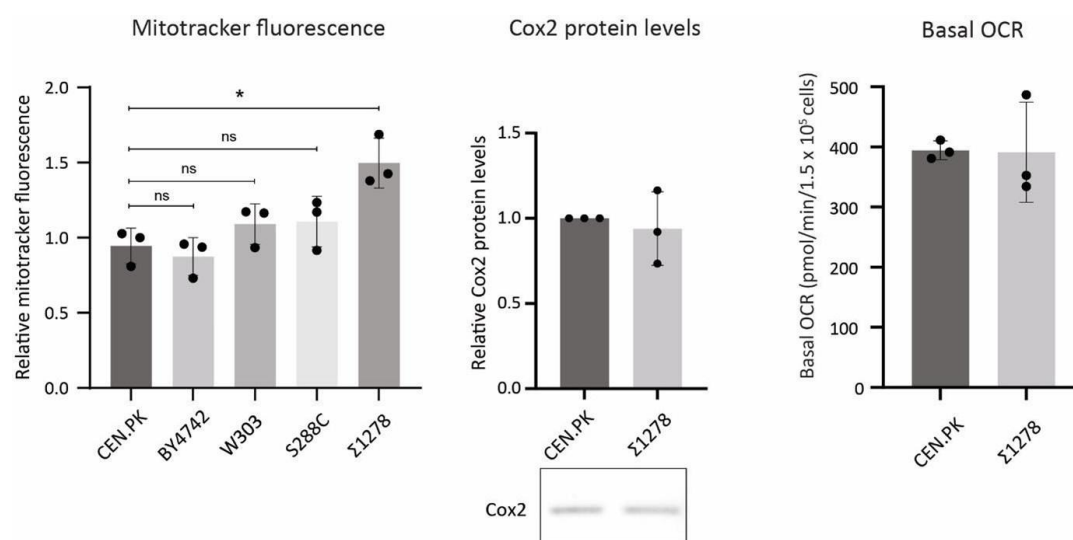
These experiments collectively have strengthened our findings on the critical role of intracellular Pi budgeting as a general constraint for mitochondrial respiration in high glucose.

"It would be conclusive to test whether wild strains with increased respiration under high glucose conditions would also be characterized by increased mitochondrial Pi."

Addressed partially above. Right now the relative basal respiration in glucose across different strains is not well known. We measured mitotracker activity in high glucose in multiple WT strains of *S. cerevisiae* (W303, Σ 1278, S288C and BY4742, compared to the CEN.PK strain). These strains all largely had similar mitotracker potential, except for a slight increase in mitochondrial membrane potential in Σ 1278 strain, but not in other strains. We further characterized this using Cox2 protein levels as well as basal OCR, and found that these do not increase. These data is shown below, and is not included in the main text since it does not add any new component to the study.

Author response image 3.

Mitochondrial respiration in different WT strains



We did find this suggestion very interesting though, and are exploring directions for future research based on this suggestion. Since we have now identified a role for intracellular Pi allocation in regulating the Crabtree effect, an interesting direction can be to understand the glucose dependent mitochondrial Pi transport in Crabtree negative yeast strains. We will have to bring in a range of new tools and strains for this, so these experiments are beyond the focus of this current study.

We hope that these new experiments in different genetic backgrounds increases the breadth and generality of our findings, and stimulates new lines of thinking to address how important the role of Pi budgeting as a constraint for mitochondrial repression in high glucose might be.

(2) It is not described whether the drop in glycolytic flux also affects TCA cycle flux. Are there any changes in the pyruvate level? If the TCA cycle is also impaired, what drives increased mitochondrial respiration?

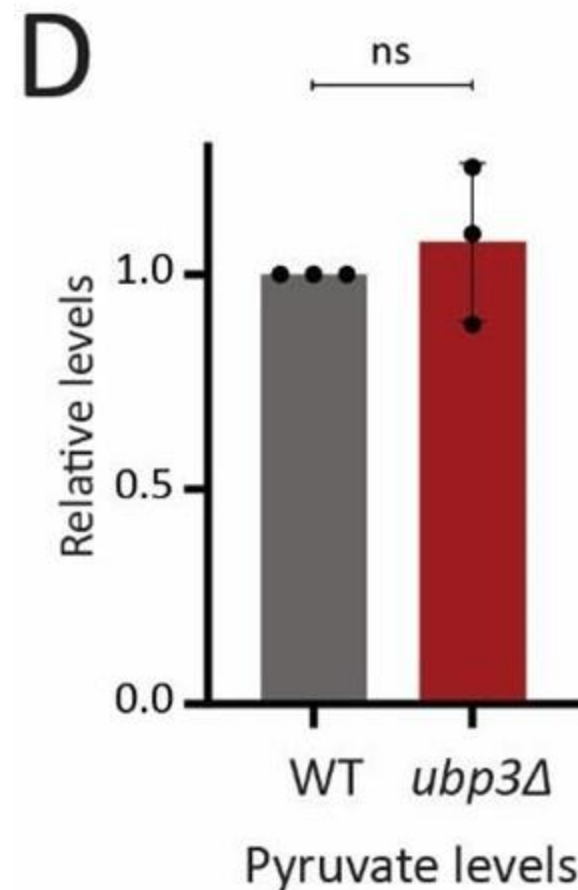
Thank you for pointing this out, and we agree this should be included. We have addressed these concerns in the revised version of the manuscript

Since glucose derived pyruvate must enter the mitochondrial TCA cycle, one possibility is that a decrease in glycolytic rate could decrease the TCA flux. An alternate possibility is that the

cells coincidentally increase the pyruvate transport to mitochondria, to thereby maintain the TCA cycle flux comparable to that of WT cells. To test both these possibilities, we first measured the steady state levels of pyruvate and TCA cycle intermediates in WT vs *ubp3Δ* cells. We do not observe any significant change in the levels of pyruvate, or TCA cycle intermediates (except malate, which showed a significant decrease in *ubp3Δ* cells). This data is now included in the revised manuscript as Figure 2 – figure supplement 1D and figure supplement 2 A, along with associated text.

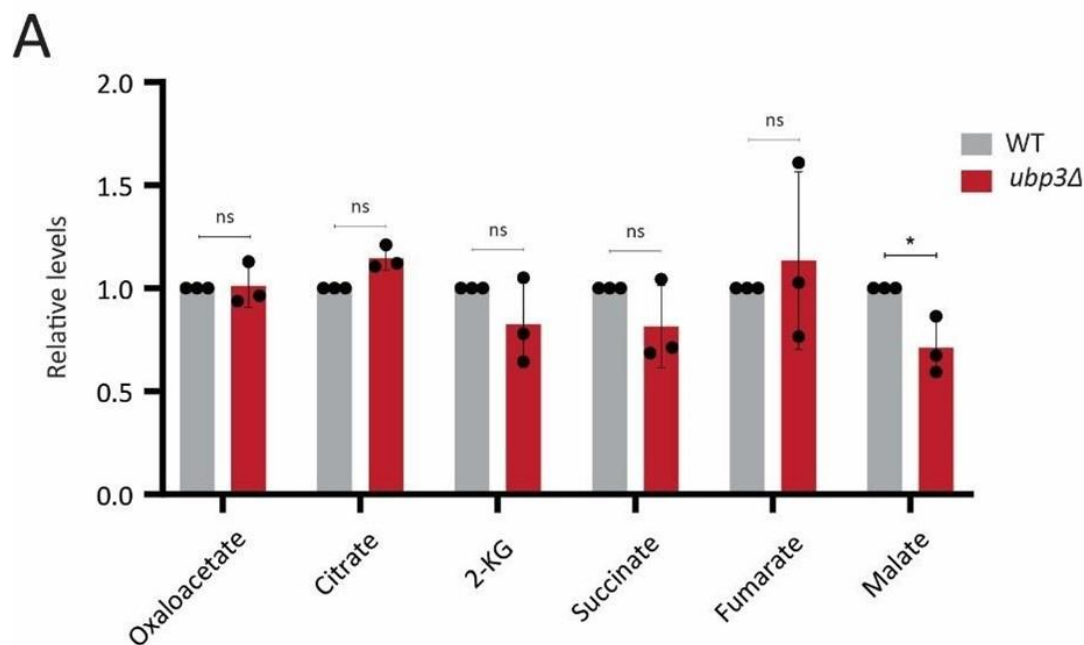
Author response image 4.

Pyruvate levels in WT vs *ubp3Δ*



Author response image 5.

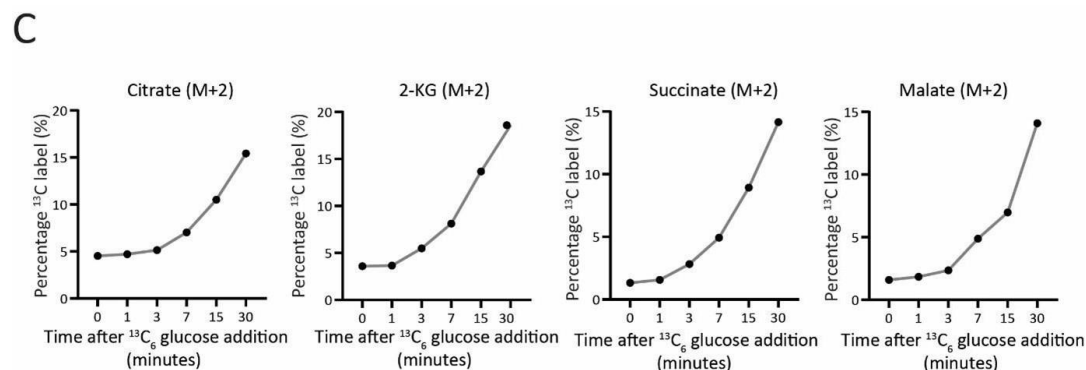
Steady state TCA cycle intermediate levels



Next, in order to address if the TCA cycle flux is impaired in *ubp3Δ* cells, we also measured the TCA cycle flux in WT vs *ubp3Δ* cells by pulsing the cells with ^{13}C glucose and tracking ^{13}C label incorporation from glucose into TCA cycle intermediates. This experiment first required substantial standardization, for the time of cell collection and quenching post ^{13}C glucose addition, by measuring the kinetics of ^{13}C incorporation into TCA cycle intermediates at different time points after ^{13}C glucose addition. The standardization of this method is now included in the revised manuscript as Figure 2 – figure supplement 2 C, along with associated text, and is shown below for reference.

Author response image 6.

Kinetics of ^{13}C labelling in TCA cycle intermediates



Actual TCA cycle flux results: For measuring the TCA cycle flux, cells were treated with 1% ^{13}C glucose, quenched and samples were collected at 7 mins post glucose addition which is in the linear range of ^{13}C label incorporation (Figure 2- Figure 2 – figure supplement 2 C).

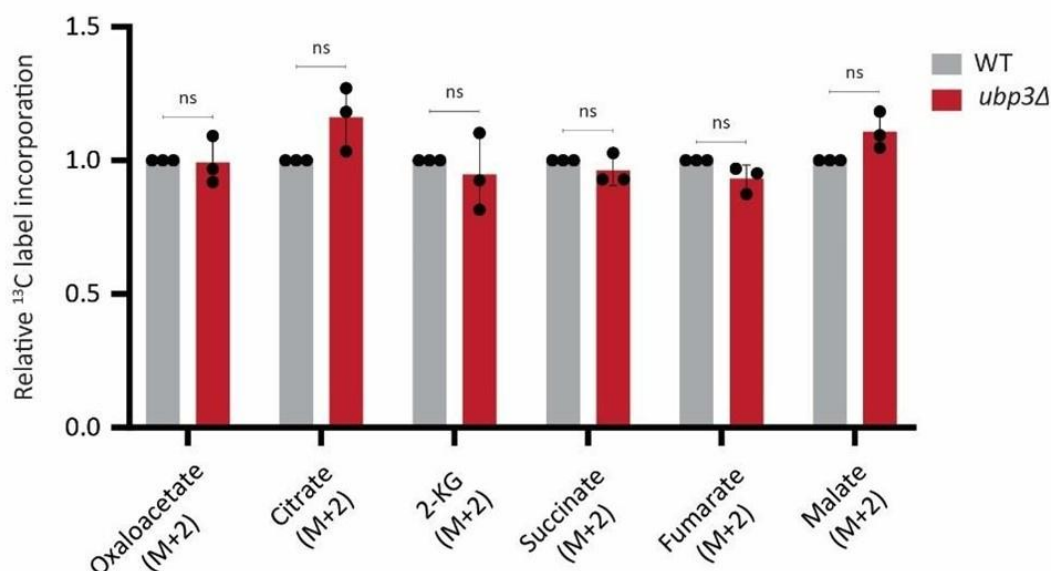
Result: We did not observe any significant changes in the relative ^{13}C label incorporation in TCA cycle intermediates. This data is included in the revised manuscript as Figure 2 – figure

supplement 2 D, along with associated text, and is below for your reference.

Author response image 7.

TCA cycle flux

D



What these data show is that the TCA cycle flux itself is not altered in *ubp3Δ*. A likely interpretation of this data is that this is due to the increase in the pyruvate transport to mitochondria in *ubp3Δ* cells, as indicated by the ~10-fold increase in Mpc3 (mitochondrial pyruvate transporter) protein levels (shown in Figure 5-figure supplement 5H), allowing the net same amount of pyruvate into the mitochondria. This increased mitochondrial pyruvate transport could support maintaining the TCA flux in *ubp3Δ* cells, and supporting the increased respiration. Putting a hierarchy together, the increased respiration in *ubp3Δ* cells could therefore be primarily due to increased Pi transport, followed by a consequent increase in ETC proteins. We leave it to the readers of this study to make this conclusion.

We hope that we have addressed all concerns that the reviewer has with respect to TCA cycle flux in *ubp3Δ* cells.

(3) In addition, some of the important literature was also missed in citation and discussion. For example, in a recent study (Ouyang et al., 2022), it was reported that phosphate starvation increases mitochondrial membrane potential independent of respiration in yeast and mammalian cells, and some of the conflicting results were presented in this study.

We are very aware of the recent study by Ouyang et al, which reports that Pi starvation increases mitochondrial membrane potential independent of respiration. However, this study is distinct from the context of our case due to the reasons listed below.

(a) The reviewer may have misinterpreted our low Pi condition as Pi starvation. There is no Pi ‘starvation’ in this study. Here, we cultured *ubp3Δ* and *tdh2Δtdh3Δ* cells in a low Pi medium with 1 mM Pi concentration in order to bring down the intracellular free Pi to that of WT levels. These cells are therefore not Pi-starved, but have been manipulated to have the same

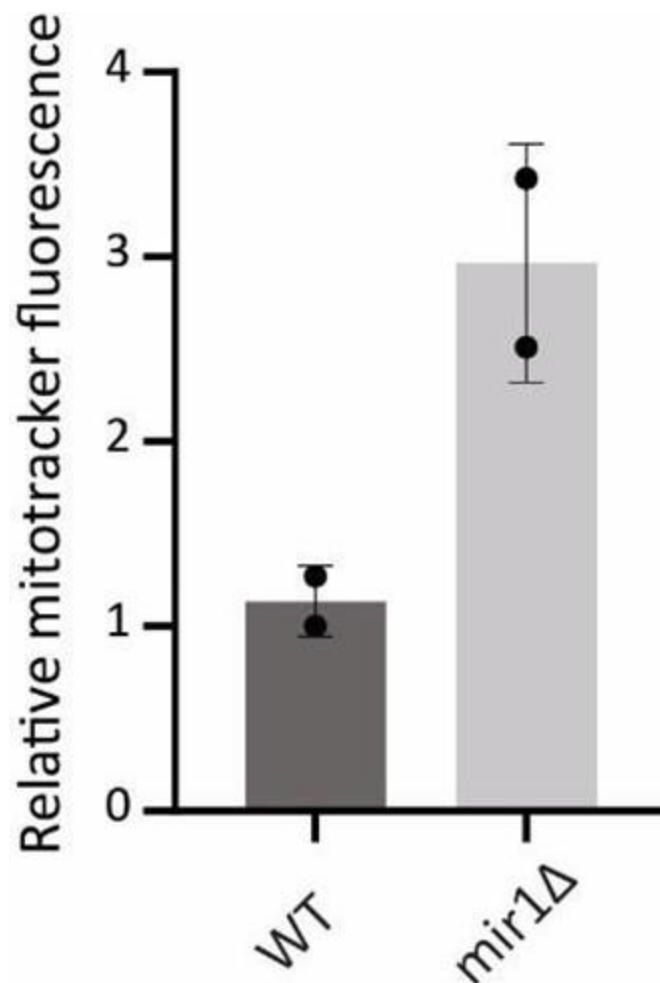
intracellular Pi levels as that of WT cells, as shown in Figure 4-figure supplement 1D. The Pi concentration in the medium is still in the millimolar range, and the cells are grown in this medium for a short time (~4 hrs) till they reach OD600 ~ 0.8. This is entirely different from the conditions used in Ouyang et al., 2022, where the cells were grown in a Pi-starvation condition with 1-100 micromolar Pi in the medium for a time duration of 6-8 hrs. Since cells respond differentially to changes in Pi concentrations over time (Vardi et al., 2014), the response to low Pi vs Pi starvation will be completely different.

(b) In our study, mitochondrial membrane potential is used as only one of the readouts for mitochondrial activity. Our estimations of mitochondrial respiration are established by including other measurements such as Cox2 protein levels (as an indicator of the ETC) and basal OCR measurements (measuring respiration), all of which provide distinct information. The mitochondrial membrane potential can be regulated independent of mitochondrial respiration state (Liu et al., 2021), using membrane potential alone as a readout to estimate mitochondrial respiration can therefore be limiting in the information it provides. As indicated earlier, mitochondrial membrane potential can change, independent of mitochondrial respiration (Ouyang et al., 2022) and ATP synthesis (Liu et al., 2021). Since the focus of our study is mitochondrial respiration, and not just the change in membrane potential, making conclusions based on potential alone are ambiguous. Most studies in the field have in fact not used the comprehensive array of distinct estimates that we use in this study, and we believe the standards set in this study should become a norm for the field.

(c) The only mutant that is similar to the Ouyang et al study is the Mir1 deletion mutant, which results in acute Pi starvation in mitochondria. In this strain, we find an increase in mitochondrial membrane potential. The data is not included in the manuscript but is shown below.

Author response image 8.

Mitochondrial potential in WT vs mir1Δ



As clear from this data, mitochondrial membrane potential is significantly high in mir1Δ cells. However, the basal OCR and Cox2 protein levels clearly show decreased mitochondrial respiration which is expected in this mutant (Figure 5 A,B). This in fact highlights the limitations of solely relying on mitochondrial membrane potential measurements to draw conclusions, as doing so will lead to a misinterpretation of the actual mitochondrial activity in these cells. We do not wish to highlight limitations in other studies, but hope we make our point clear.

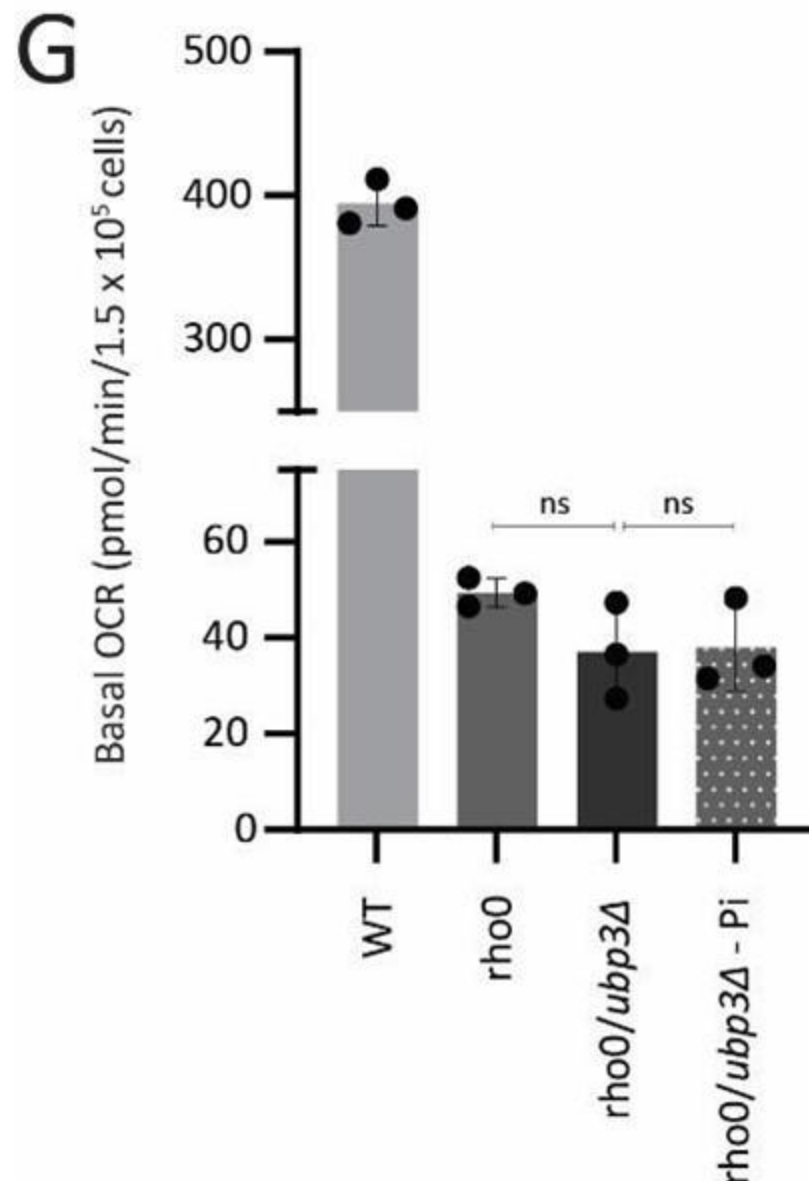
(4) An additional experiment with strains lacking mitochondrial DNA under phosphate-rich and restricted conditions would further strengthen the result.

Strains lacking mitochondrial DNA (Rho0 cells) cannot express the mitochondrially encoded ETC subunit proteins. These strains are therefore incapable of performing mitochondrial respiration. Since Rho0 cells are known to utilize alternate mechanisms to maintain their mitochondrial membrane potential (Liu et al., 2021), using mitotracker fluorescence as a readout of mitochondrial respiration in these strains under different Pi conditions is inconclusive and misleading due to the reasons mentioned in point number 3(b and c). However, since this was a concern raised by the reviewer, we now measured basal OCR in WT and Rho0 strains with Ubp3 deletion under normal vs low Pi medium. As expected, Rho0 cells show extremely low basal OCR values, an entire order of magnitude lower than WT cells. At these very low (barely detectable) levels the deletion of Ubp3 or change in Pi

concentration in the medium does not change basal OCR, since these strains are not capable of respiration. We have included this data as Figure 4-figure supplement 1G.

Author response image 9.

Basal OCR in Rho0 cells



(5) Western blot control panels should include entire membrane exposure, and non-cut western blots should be submitted as supplementary.

The non-cut western blot images and the loading controls are now included in the revised manuscript as a supplementary file 2.

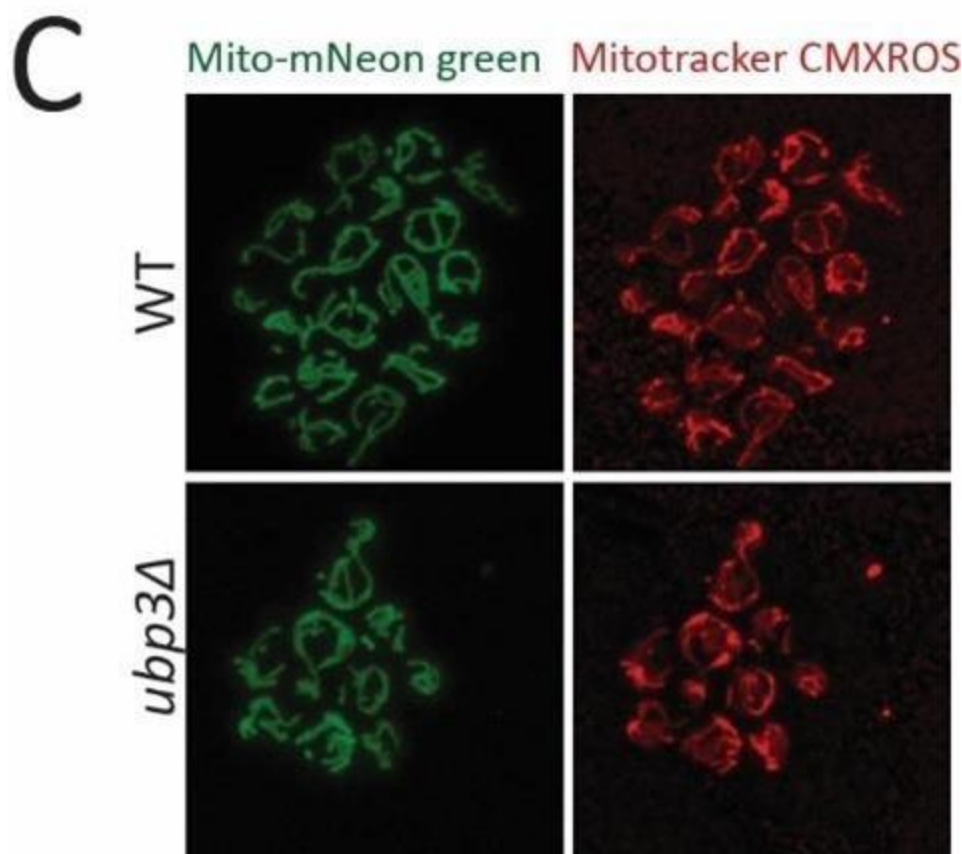
(6) In Figure 4, it is shown that Pi addition decreases basal OCR to the WT level. However, the Cox2 level remains significantly higher. This data is confusing as to whether mitochondrial Pi directly regulates respiration or not.

As described in the previous point, the Cox2 levels and the OCR provide distinct pieces of information. In figure 4, we show that culturing *ubp3Δ* in low Pi significantly decreases both Cox2 protein levels and basal OCR. Since Cox2 protein levels and basal OCR are different readouts for mitochondrial activity, there could be differences in the extent by which Pi availability controls each of these factors. Basal OCR is a direct readout for mitochondrial respiration, and is regulated by multiple factors including ETC protein levels, rate of ATP synthesis, rate of Pi transport etc. In figure 4, we find that culturing *ubp3Δ* in low Pi decreases basal OCR to WT level. This strongly suggests that high Pi levels are necessary to increase basal OCR in *ubp3Δ*.

(7) Representative images of *Ubx3* KO and wild-type strains stained with CMXRos are missing.

Thank you for noticing this. This data is now included in the revised manuscript as Figure 1figure supplement 1C.

Author response image 10.



(8) Overall, mitochondrial copy number and mtDNA copy number should be analyzed in WT and *Ubo3* KO cells as well as Pi-treated and non-treated cells, and basal OCR data should be normalized accordingly. The reported normalization against OD is not appropriate.

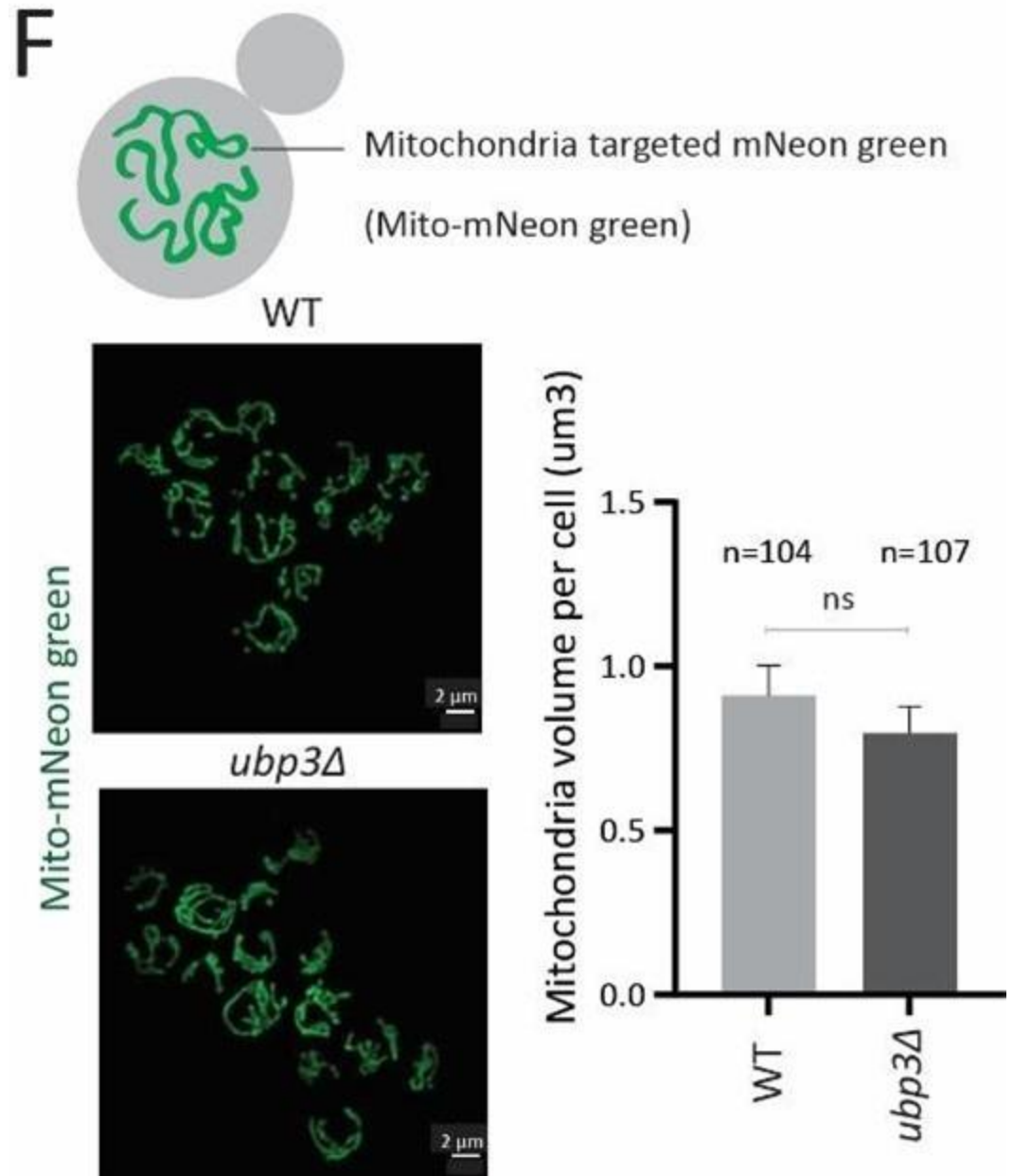
This is a valid concern raised by the reviewer, and something we had extensively considered during the study. To normalize the total mitochondrial amounts in each strain, we always measure the protein levels of the mitochondrial outer membrane protein Tom70. While we

had described this in the methods, it may not have been obvious in the text. But this information is included in Figure 1-figure supplement 1G. We did not observe any significant change in Tom70 levels, suggesting that the total mitochondrial amount does not change in *ubp3Δ*, and we have noted this in the manuscript (results section relevant to Figure 1). As an additional control, to directly measure the mitochondrial amount in these conditions, we have now measured the mitochondrial volume in *ubp3Δ* cells and WT cells treated with Pi. For this, we used a strain which encodes mitochondria targeted with mNeon green protein (described in Dua et al., JCB, 2023), and which can therefore independently assess total mitochondrial amount. We do not observe any changes in mitochondrial volume or amounts in *ubp3Δ* cells or WT+Pi, compared to that of WT cells. Therefore, the change in mitochondrial respiration in Ubp3 deletion and Pi addition are not due to changes in total amounts of mitochondria in these conditions. Given all these, the normalization of basal OCR using total cell number is therefore the most appropriate way for normalization. This is also conventionally used for basal OCR normalization in multiple studies.

We have now included these additional data on mitochondrial volumes and amounts in the revised manuscript as Figure1-figure supplement 1F and Figure5-figure supplement 1D, and associated text, and is shown below.

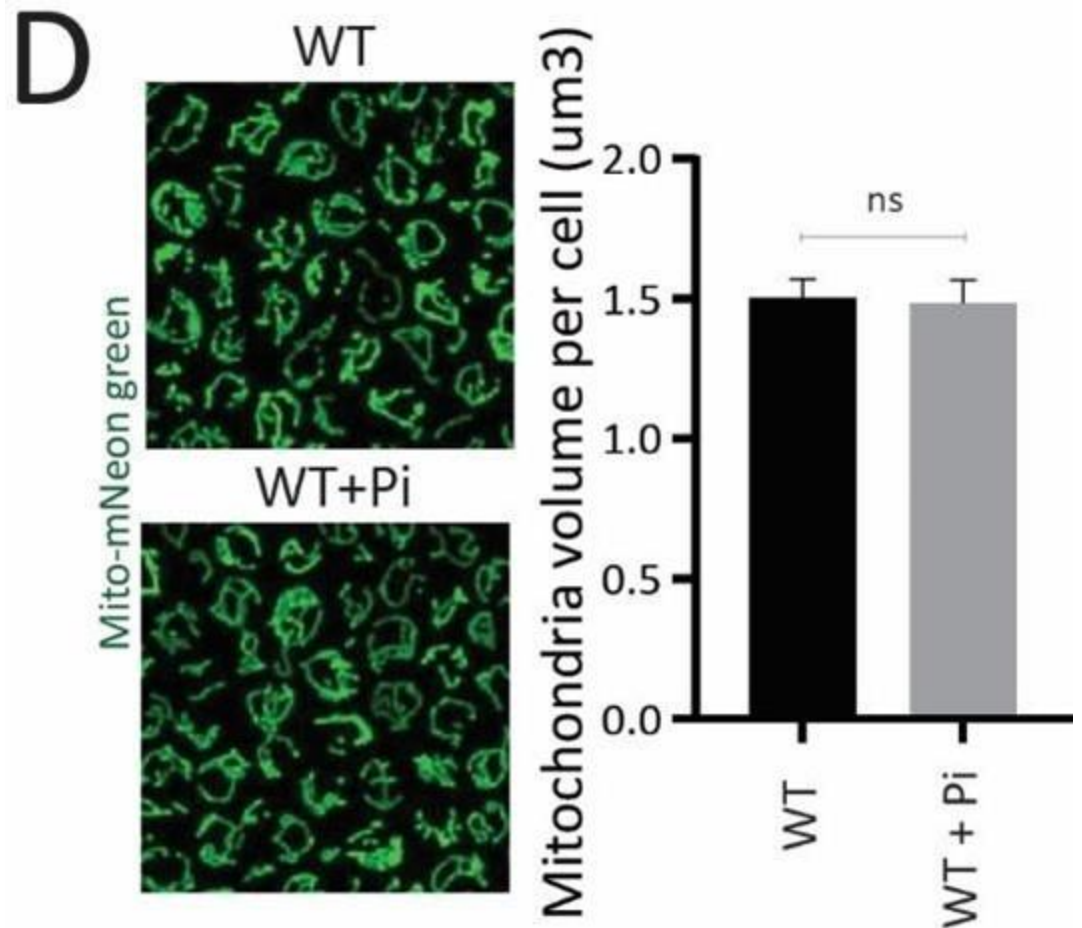
Author response image 11.

Mitochondrial volume in WT vs *ubp3Δ* cells



Author response image 12.

Mitochondrial volume in WT and WT+Pi



These data collectively address the reviewer's concerns regarding changes in mitochondrial amounts in all the conditions and strains used in this study.

Reviewer #2 (Public Review):

Summary:

Cells cultured in high glucose tend to repress mitochondrial biogenesis and activity, a prevailing phenotype type called Crabtree effect that is observed in different cell types and cancer. Many signaling pathways have been put forward to explain this effect. Vengayil et al proposed a new mechanism involved in Ubp3/Ubp10 and phosphate that controls the glucose repression of mitochondria. The central hypothesis is that Δ ubp3 shifts the glycolysis to trehalose synthesis, therefore leading to the increase of Pi availability in the cytosol, then mitochondria receive more Pi, and therefore the glucose repression is reduced.

Strengths:

The strength is that the authors used an array of different assays to test their hypothesis. Most assays were well-designed and controlled.

Weaknesses:

I think the main conclusions are not strongly supported by the current dataset.

(1) Although the authors discovered $\Delta ubp3$ cells have higher Pi and mitochondrial activity than WT in high glucose, it is not known if WT cultured in different glucose concentration also change Pi that correlate with the mitochondrial activity. The focus of the research on $\Delta ubp3$ is somewhat artificial because $\Delta ubp3$ not only affects glycolysis and mitochondria, but many other cellular pathways are also changed. There is no idea whether culturing cells in low glucose, which derepress the mitochondrial activity, involves Ubp3 or not. Similarly, the shift of glycolysis to trehalose synthesis is also not relevant to the WT cells cultured in a low-glucose situation. "The focus of the research on $\Delta ubp3$ is somewhat artificial because $\Delta ubp3$ not only affects glycolysis and mitochondria, but many other cellular pathways are also changed. There is no idea whether culturing cells in low glucose, which de-repress the mitochondrial activity, involves Ubp3 or not."

We would like to clarify that the focus of this research is not on Ubp3, or to address mechanistic aspects of how Ubp3 regulates mitochondrial activity, or to identify the targets of Ubp3. That would be an entirely distinct study, with a very different approach.

In this study, while carrying out a screen, we serendipitously found that $ubp3\Delta$ cells showed an increase in mitochondrial activity in high glucose. Subsequently, we used this observation, bolstered by diverse orthogonal approaches, to identify a general, systems-level principle that governs mitochondrial repression in high glucose. Through this, we identify a role of phosphate budgeting as a controller of mitochondrial repression in high glucose. In this study, our entire focus has been to use orthogonal approaches, as well as parsimonious interpretations, to establish this new hypothesis as a possibility. We hope this idea, supported by these data, will now enable researchers to pursue other experiments to establish the generality of this phenomenon.

We have not focused our effort in identifying the role of Ubp3, or its regulation upon changes in glucose concentration in this context. That is a very specific, and separate effort, and misses the general point we address here. It is entirely possible that Ubp3 might also regulate mitochondrial activity by additional mechanisms other than mitochondrial Pi availability (such as via the reduction of key glycolytic enzymes at nodes of glycolysis, resulting in reduced glycolytic flux and rerouted glucose metabolism). Had the goal been to identify Ubp3 substrates, it is very likely that we would not have found the role of Pi homeostasis in controlling mitochondrial respiration. This is particularly because the loss of Ubp3 does not result in an acute disruption of glycolysis, unlike say a glycolytic enzyme mutant, which would have resulted in severe effects on growth and overall metabolic state. This would have made it difficult to dissect out finer details of metabolic principles that regulate mitochondrial respiration.

In order to further corroborate our findings, we used the glycolysis defective mutant $tdh2\Delta tdh3\Delta$ cells, where we find a similar change in Pi balance. This complements the key observations made using $ubp3\Delta$ cells. Distinctly, we utilized the glycolytic inhibitor 2DG to independently assess the role of mitochondrial Pi transport in regulating respiration. Together, in this study we do not just relying on genetic mutants, but combine the Ubp3 deletion strain with a reduced GAPDH activity strain, and pharmacologic inhibition of glycolysis. Distinctly, we find that mitochondrial Pi transporter levels are repressed under high glucose (Figure 5C, Figure 5-figure supplement 1B). Further, we find that mitochondrial Pi transport is important in increasing mitochondrial respiration upon shift to low glucose and glycolytic inhibition by 2-DG. Therefore, we collectively unravel a more systems level principle that regulates glucose mediated mitochondrial repression, as opposed to a mechanistic study of Ubp3 targets.

Of course, given the conservation of Ubp3, we are very excited to pursue a mechanistic study of Ubp3 targets in future. This is a general challenge for deubiquitinase enzymes, and till date there are very few bona fide substrates known for any deubiquitinase enzyme, from any

cellular system (due to challenges in the field that we discuss separately, and have included in the discussion section of this text).

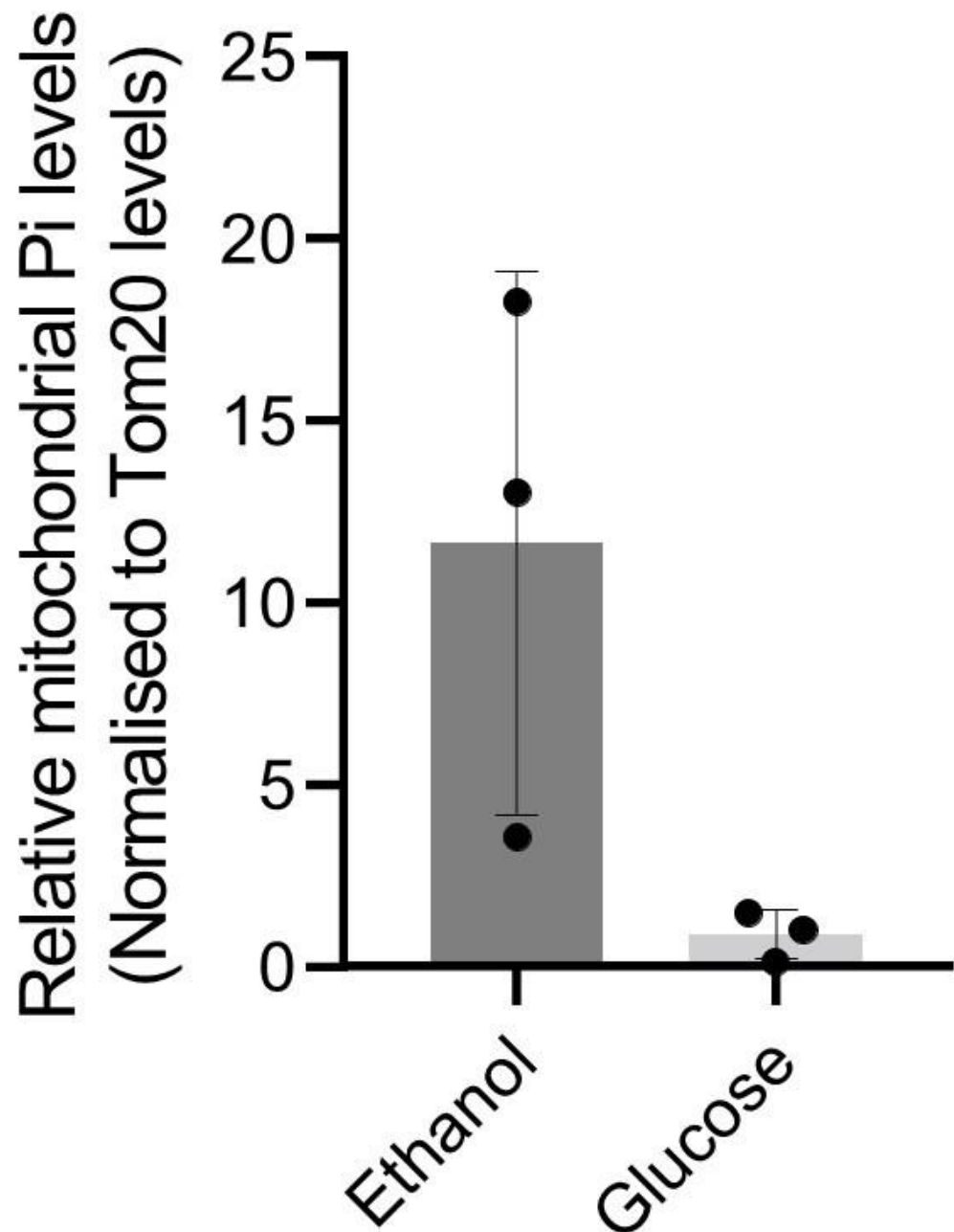
“Similarly, the shift of glycolysis to trehalose synthesis is also not relevant to the WT cells cultured in a low-glucose situation”

The reviewer is correct in pointing out that in low-glucose, the shift to trehalose synthesis might not be as relevant. We observe that the glycolysis defective mutant *tdh2Δtdh3Δ* cells does not show an increase in trehalose synthesis (Figure 3-figure supplement 1E). However, in this context, the decrease in the rate of GAPDH catalysed reaction alone appears to be sufficient to increase the Pi levels (Figure 3F) even without an increase in trehalose. Therefore, there might be differences in the relative contributions of these two arms towards Pi balance, based on whether it is low glucose in the environment, or a mutant such as *ubp3* that modulates glycolytic flux. In *ubp3Δ* cells, the combination of low rate of GAPDH catalyzed reaction and high trehalose will happen (based on how glycolytic flux is modulated), vs only the low rate of GAPDH catalyzed reaction in *tdh2Δtdh3Δ* cells. As an end point the increase in Pi happens in both cases, but with slightly differing outcomes. It is also to be noted that in terms of free Pi sources a low-glucose condition (with low glycolytic rate) is very different from a no-glucose, respiratory condition (where cells perform very high gluconeogenesis). In high respiration conditions such as ethanol, cells switch to high gluconeogenesis, where there is a huge increase trehalose synthesis as a default (eg see Varahan et al 2019). In this condition, trehalose synthesis could be a major source for Pi (eg see Gupta 2021), and could support the increased mitochondrial respiration. In an ethanol medium, the directionality of GAPDH reaction is reversed. Therefore, this reaction will also now become an added source of Pi, instead of a consumer of Pi (see illustration in Figure 3G). Therefore, a reasonable interpretation is that a combination of increased trehalose and increased 1,3 BPG to G3P conversion can be a major Pi source to increasing mitochondrial respiration in a non-glucose, respiratory medium.

“it is not known if WT cultured in different glucose concentration also change Pi that correlate with the mitochondrial activity”

This is valid point raised by the reviewer. We have already found that the protein levels of mitochondrial Pi transporter is increased in a non-glucose respiratory (ethanol) medium and a low (0.1%) glucose medium (see Figure 5C, Figure5-figure supplement 1B). In addition, we have tried measuring mitochondrial Pi levels in cells grown in a high glucose medium vs a respiratory, ethanol medium. The results are shown below for the reviewer’s reference. Reviewer response image 3 – Mitochondrial Pi levels in ethanol vs glucose

Author response image 13.



We observe a clear trend where mitochondrial Pi levels are high in cells grown in ethanol medium compared to that of cells grown in high glucose. However, the estimation of Pi, and normalising the Pi levels in isolated mitochondria is extremely difficult in this condition (note that this has never been done before). This is likely due to a rapid rate of conversion of ADP and Pi to ATP (in ethanol) which increases the variation in the estimation of steady state Pi levels, and the high amounts of mitochondria in ethanol grown cells. Since the data shows

high variation, we have not included this data in the manuscript, but we are happy to include it here in the response.

Indeed, this study opens up the exciting question of addressing how intracellular Pi allocation is regulated in different conditions of glucose. This can be further extended to Crabtree negative strains such as *K. lactis* which do not show mitochondrial repression in high glucose. All of these are rich future research programs.

*(2) The central hypothesis that Pi is the key constraint behind the glucose repression of mitochondrial biogenesis/activity is supported by the data that limiting Pi will suppress mitochondrial activity increase in these conditions (e.g., $\Delta ubp3$). However, increasing the Pi supply failed to increase mitochondrial activity. The explanation put forward by the authors is that increased Pi supply will increase glycolysis activity, and somehow even reduce the mitochondrial Pi. I cannot understand why only the increased Pi supply in $\Delta ubp3$, but not the increased Pi by medium supplement, can increase mitochondrial activity. The authors said "...that $ubp3\Delta$ do not increase mitochondrial Pi by merely increasing the Pi transporters, but rather by increasing available Pi pools". They showed that $\Delta ubp3$ mitochondria had higher Pi but WT cells with medium Pi supplement showed lower Pi, it is hard to understand why the same Pi increase in the cytosol had a different outcome in mitochondrial Pi. Later on, they showed that the isolated mito exposed to higher Pi showed increased activity, so why can't increased Pi in intact cells increase mito activity? Moreover, they first showed that $\Delta ubp3$ had a *Mir1* increase in Fig3A, then showed no changes in FigS4G. It is very confusing.*

"I cannot understand why only the increased Pi supply in $\Delta ubp3$, but not the increased Pi by medium supplement, can increase mitochondrial activity."

This is an interesting point, that requires a nuanced explanation, which we try to provide below.

For mitochondrial respiration to increase in the presence of high Pi, the cytosolic Pi has to be transported to the mitochondria sufficiently. In $ubp3\Delta$ the increased free Pi (as a consequence of rewired glycolysis) is transported to the mitochondria (Figure 4). This increased mitochondrial Pi can therefore increase mitochondrial respiration in $ubp3\Delta$.

In case of WT+Pi, the externally supplemented Pi cannot further enter mitochondria (as shown in Figure 5-Figure supplement 1C) and is most likely restricted to the cytosol. Because of this inability of the Pi to access mitochondria, the mitochondrial respiration does not increase in WT+Pi (Figure 5-Figure supplement 1E).

The likely reason for this difference in mitochondrial Pi transport in $ubp3\Delta$ vs WT+Pi is the relative difference in their glycolytic rate. The glycolytic rate is inherently decreased in $ubp3\Delta$, but not in WT+Pi. To dissect this possibility of glycolytic rate itself contributing to the Pi availability in the mitochondria, we inhibited glycolysis in WT cells (using 2DG), and then supplemented Pi. Compared to cells in the same glucose condition (with 2DG, but without supplementing excess Pi), now the WT+Pi (+2DG) has higher mitochondrial respiration (Figure 5-Figure supplement 1F). This suggests that a combination of low glycolysis and high Pi is required for increasing mitochondrial respiration (as elaborated in the discussion section of the manuscript).

An obvious question that arises out of this observation is how does the change in glycolytic rate regulate mitochondrial Pi transport. One consequence of altering the glycolytic rate is a change in cytosolic pH. This itself will bear on the extent of Pi transport into mitochondria, as discussed in detail below.

In mitochondria, Pi is co-transported along with protons. Therefore, changes in cytosolic pH (which changes the proton gradient) can control the mitochondrial Pi transport (Hamel et al., 2004). Glycolytic rate is a major factor that controls cytosolic pH. The cytosolic pH in highly glycolytic cells is ~7, and decreasing glycolysis results in cytosolic acidification (Orij et al., 2011). Therefore, under conditions of decreased glycolysis (such as loss of Ubp3), cytosolic pH becomes acidic. Since mitochondrial Pi transport is dependent on the proton gradient, a low cytosolic pH would favour mitochondrial Pi transport. Therefore, under conditions of decreased glycolysis (2DG treatment, or loss of Ubp3), where cytosolic pH would be acidic, increasing cytosolic Pi might indirectly increase mitochondria Pi transport, thereby leading to increased respiration.

To explain this and integrate all these points, we have extended a discussion section in this manuscript. We include this section below:

“Supplementing Pi under conditions of low glycolysis (where mitochondrial Pi transport is enhanced), as well as directly supplementing Pi to isolated mitochondria, increases respiration (Figure 5, Figure 5-figure supplement 1). Therefore, in order to derepress mitochondria, a combination of increased Pi along with decreased glycolysis is required. An additional systems-level phenomenon that might regulate Pi transport to the mitochondria is the decrease in cytosolic pH upon decreased glycolysis (60, 61). The cytosolic pH in highly glycolytic cells is ~7, and decreasing glycolysis results in cytosolic acidification (60, 61). Therefore, under conditions of decreased glycolysis (2DG treatment, deletion of Ubp3, and decreased GAPDH activity), cytosolic pH becomes acidic. Since mitochondrial Pi transport itself is dependent on the proton gradient, a low cytosolic pH would favour mitochondrial Pi transport (62). Therefore, under conditions of decreased glycolysis (2DG treatment, or loss of Ubp3, or decreased GAPDH activity), where cytosolic pH would be acidic, increasing cytosolic Pi might indirectly increase mitochondria Pi transport, thereby leading to increased respiration. Alternately, increasing mitochondrial Pi transporter amounts can achieve the same result, as seen by overexpressing Mir1 (Figure 5).”

This possibility of changes in cytosolic pH regulating mitochondrial Pi transport and thereby respiration is a really interesting future research question, and an idea that has not yet been explored till date. This can stimulate new lines of thinking towards finding conserved biochemical principles that control mitochondrial repression in high glucose.

“Moreover, they first showed that $\Delta ubp3$ had a Mir1 increase in Fig3A, then showed no changes in Fig54G. It is very confusing”

increase in Mir1 in $\Delta ubp3$ shown in figure 3A comes from the analysis of the proteomics dataset from a previous study (Isasa et al., 2015). Subsequently, we more systematically experimentally assessed Mir1 levels directly, and did not observe an increase in Mir1 (Figure 4figure supplement 1H in revised manuscript). It is entirely possible that in a large-scale study (as in Isasa 2015), some specific proteomic targets might not fully reproduce when tested very specifically (as is described in Handler et al., 2018 and Mehta et al., 2022). We do clearly indicate this in the text, but given the density of information in this study, it is understandable that this point was missed by the reviewer.

(3) Given that there is no degradation difference for these glycolytic enzymes in $\Delta ubp3$, and the authors found transcriptional level changes, suggests an alternative possibility where $\Delta ubp3$ may signal through unknown mechanisms to parallelly regulate both mitochondrial biogenesis and glycolytic enzyme expression. The increase of trehalose synthesis usually happens in cells under proteostasis stress, so it is important to rule out whether $\Delta ubp3$ signals these metabolic changes via proteostasis dysregulation. This

echoes my first point that it is unknown whether wild-type cells use a similar mechanism as Δ ubp3 cells to regulate the glucose repression of mitochondria.

We appreciate this point raised by the reviewer, but this again requires some clarification (as made earlier). The goal of this study was to identify systems-level principles that explain mitochondrial repression in high glucose. Although we started by performing a screen to identify proteostatic regulators of mitochondrial activity in high glucose, and identified Ubp3 as a mediator of mitochondrial activity, our approach was to use ubp3 Δ cells as a model to understand the metabolic principles that regulate mitochondrial repression. This has been reiterated repeatedly in the manuscript – for example lines 123-124 “We therefore decided to use ubp3 Δ cells to start delineating requirements for glucose-mediated mitochondrial repression.” and again in the discussion section – lines 442-460, where we discuss some unique advantages of using ubp3 Δ cells to understand a general basis of mitochondrial regulation. To test this hypothesis, we also used orthogonal approaches, as well as other mutants and conditions with defective glycolysis, such as tdh2 Δ tdh3 Δ cells and 2DG treatments. Only with these multiple converging evidences do we infer that there might be a role of the change in Pi balance (due to changes in glycolytic rate) in regulating mitochondrial activity.

We certainly agree that there is great value in identifying the mechanistic details of how Ubp3 regulates mitochondria. But this requires very distinct approaches not pursued in this study. This is not the question that we are addressing in this story. Separately, identifying targets of DUBs is one of the exceptional challenges in biology, since there are currently no straightforward chemicalbiology approaches to do so for this class of proteins. Unlike kinase/phosphatase systems, or even ubiquitin ligases, substrate trapping mutants etc have proven to be abject failures in identifying direct targets of DUBs. A quantitative proteomics study might suggest some proteins/cellular processes regulated by Ubp3. This has been attempted for several DUBs, but rarely have any direct substrates of DUBs every been identified, in any system. A high quality quantitative, descriptive proteome dataset of ubp3 Δ cells is already available from a previous study (Isasa et al., 2015), which we cite extensively in this manuscript, and indeed was invaluable for this study. We cannot improve the outstanding quality dataset already available. Interestingly, the findings of this study actually help substantiate our idea of an increased mitochondrial activity and change in Pi homeostasis in ubp3 Δ cells. The Isasa et al dataset finds proteins involved in mitochondrial respiration that are high in ubp3 Δ cells, and the glycolytic enzymes and PHO regulon proteins are reduced. In our study, using these data references, we were able to conceptually piece together how changes in glycolytic flux can alter Pi balance.

Apart from identifying changes in protein levels, a separate challenge in making sense of this quantitative proteomics data is the difficulty in pinpointing any target of Ubp3 that specifically regulates these processes. A single DUB can have multiple substrates, and this could regulate the cellular metabolic state in a combinatorial manner. This is the essence of all signaling regulators in how they function, and it is therefore important to understand what their systems-level regulation of cell states are (separate from their specific individual substrates). Therefore, identifying the specific target of Ubp3 responsible for this metabolic rewiring can be very challenging. These experiments are well beyond the scope or interest of the current manuscript.

If we had pursued that road in this study, we would not have made any general findings related to Pi balance, nor would this more general hypothesis have emerged.

(4) Other major concerns:

(a) The authors selectively showed a few proteins in their manuscript to support their conclusion. For example, only Cox2 and Tom70 were used to illustrate mitochondrial

biogenesis difference in line 97. Later on, they re-analyzed the previous MS dataset from Isasa et al 2015 and showed a few proteins in Fig3A to support their conclusion that Δ ubp3 increases mitochondrial OXPHOS proteins. However, I checked that MS dataset myself and saw that many key OXPHOS proteins do not change, for example, both ATP1 and ATP2 do not change, which encode the alpha and beta subunits of F1 ATPase. They selectively reported the proteins' change in the direction along with their hypothesis.

To clarify, we observe an increase in Cox2 protein levels but not in Tom70 levels which suggests that there is no increase in mitochondrial biogenesis. The increase is specific to some respiration related mitochondrial proteins such as Cox2 (Figure 1E, Figure 3A). We have clearly pointed out this in the manuscript. We used Cox2 protein levels as an additional readout for ETC activity, to validate our observations coming from the potentiometric mitotracker readouts, and basal oxygen consumption rate (OCR) measurements. This was for 3 reasons: Cox2 is a mitochondrial genome encoded subunit of the complex IV (cytochrome c oxidase) in the ETC, and has a redox centre critical for the cytochrome c oxidase activity. The biogenesis and assembly of complex IV subunits have been studied with respect to multiple conditions such as glucose availability and hypoxia and the expression and stability of the mitochondrial encoded complex IV subunits are exceptionally well correlated to changes in mitochondrial respiration (Fontanesi et al., 2006). Cox2 is very well characterised in *S. cerevisiae*, and the commercially available Cox2 antibodies are outstanding, which makes estimating Cox2 levels by western blotting unambiguous and reproducible.

We re-analyzed the proteomic dataset from Isasa et al to find out additional information regarding the key nodes that are differentially regulated in *ubp3 Δ* . We have not claimed at any point in the manuscript that all OXPHOS related proteins are upregulated in *ubp3 Δ* , nor is there any need for that to be so. We identified Ubp3 from our screen, observed an increase in mitochondrial potential, basal OCR, and Cox2 levels. We later found out that the proteomic data set for *ubp3 Δ* also supports our observations that mitochondrial respiration is upregulated in *ubp3 Δ* . The reviewer points out that we “showed a few proteins in Fig3A to support their conclusion that Δ ubp3 increases mitochondrial OXPHOS proteins”. Our conclusion is that the deletion of Ubp3 increases mitochondrial respiration. The combined readouts which we used to reach this conclusion (OCR, mitochondrial potential, mitochondrial ATP production, Cox2 levels) are far more direct, comprehensive and conclusive than showing an increase in a few proteins related to OXPHOS, as also explained earlier toward a distinct reviewer query. Since different mitochondrial proteins are regulated by different mechanisms, we need not see an increase in all the OXPHOS proteins in a mutant like *ubp3 Δ* where mitochondrial respiration is high. An increase in some key proteins would be sufficient to increase the respiration as seen in our case.

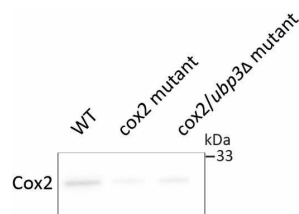
To summarise, the proteomic dataset supports our observation, but our conclusions are not dependent on the increase in OXPHOS proteins observed in the dataset.

(b) The authors said they deleted ETC component Cox2 in line 111. I checked their method and table S1, I cannot figure out how they selectively deleted COX2 from mtDNA. This must be a mistake.

Yes, we understand that for mitochondrially encoded proteins, a simple knock-out strategy has limitations. However, we first tried to generate the Cox2 deletion mutant by a standard PCR mediated gene deletion strategy (Longtine 1998), with the optimistic assumption that even if all Cox2 is not lost, a substantial fraction of the Cox2 genes would be lost via recombination. We selected the transformants after strong antibiotic selection, and then we measured the Cox2 protein levels. Gratifyingly, we found that the mutant strain had substantially decreased Cox2 protein levels (but not a complete loss), and this was retained across generations. The data is shown below.

Author response image 14.

Cox2 levels in WT vs Cox2 mutants

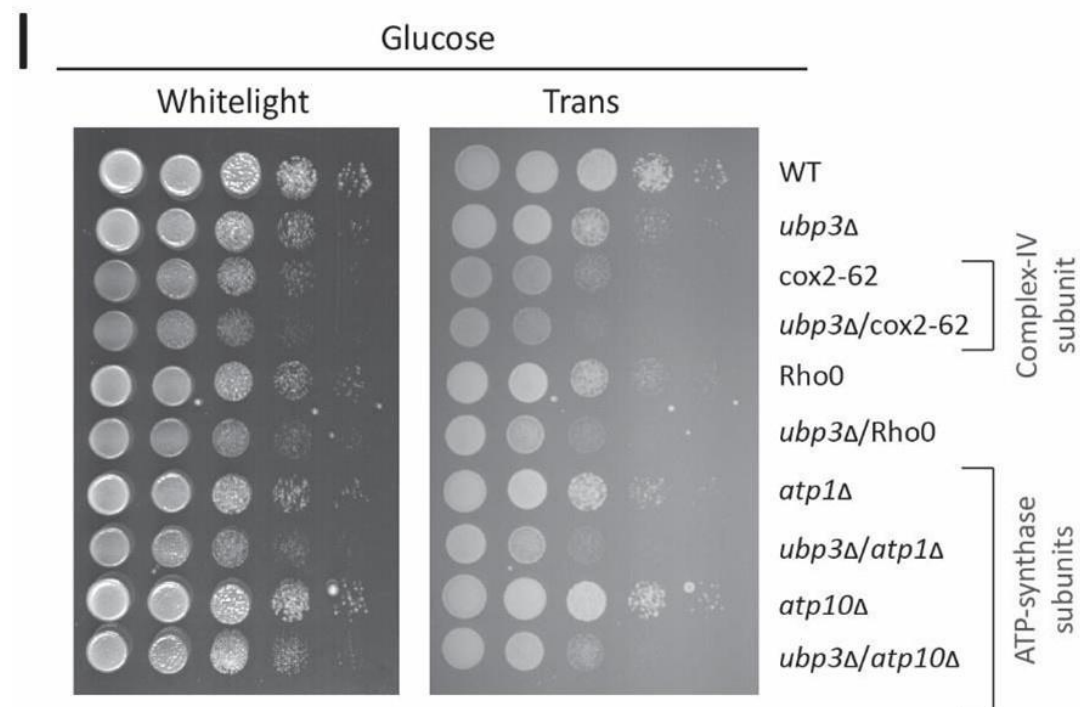


Since the mutants have decreased Cox2 levels, we went ahead and performed growth assays using this strain, in a WT or Ubp3 deletion background. Deletion of Ubp3 in the Cox2 mutant resulted in a more severe growth defect.

However, we fully agree that this strain is not a complete Cox2 knockout, and it is possible that the decrease in Cox2 is due to modifications in some other unrelated gene. In the text, we should also not have named this *cox2Δ*. Since we are not sure of the exact genetic modification in this mutant, we have removed this data from the revised manuscript.

Instead, we have now repeated all experiments, utilizing a fully characterised Cox2 mutant - *cox262*, described in (5) which has defective respiration. In this revised version, we find that deletion of Ubp3 in this strain retains the originally observed severe growth defect in glucose. This is consistent with our conclusion that a functional mitochondria is required for proper growth in *ubp3Δ* mutant. To separately validate this conclusion, we also utilized a Rho0 strain which does not have mitochondrial DNA and thereby cannot perform mitochondrial respiration. We show that deletion of Ubp3 results in a more severe growth defect in a Rho0 strain. These results are included in the revised manuscript as figure 1-figure supplement 1 I.

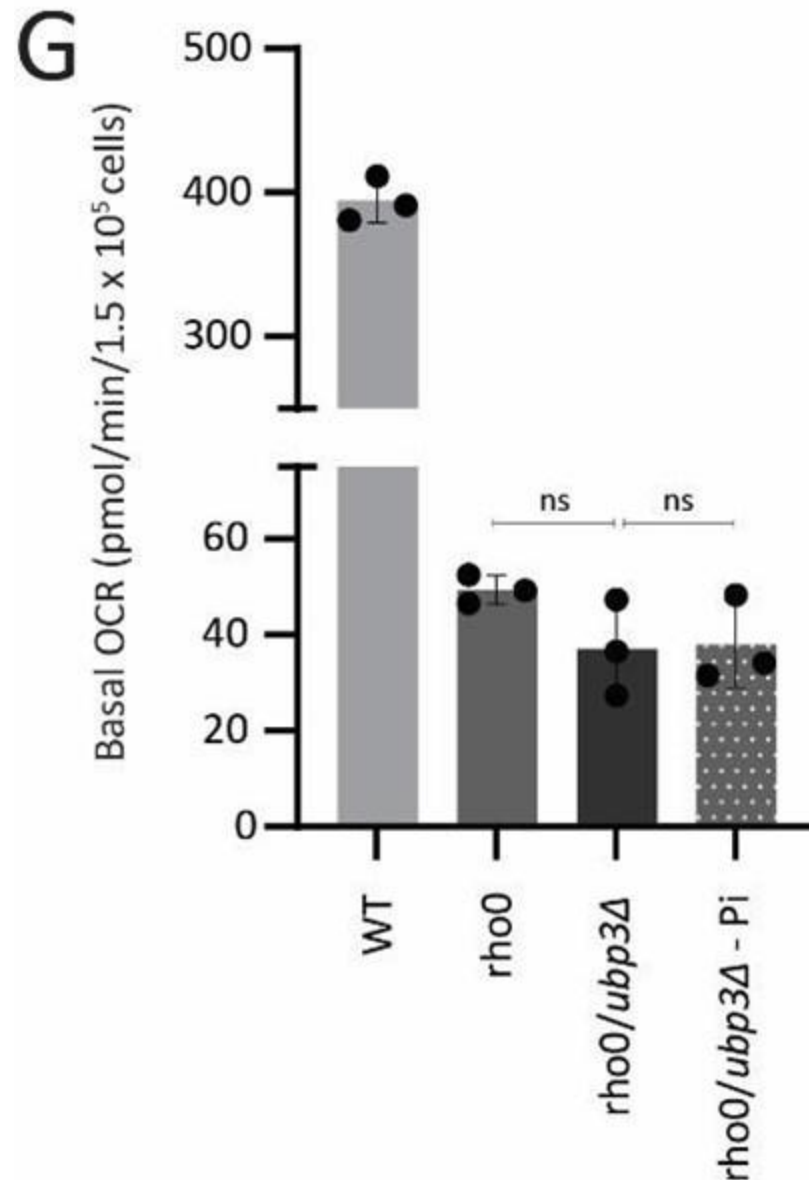
Author response image 15.



Also, we further confirmed that the Rho0 strain and Rho0 ubp3 strain is incapable of respiration, using seahorse assay. This data is included in the revised manuscript as Figure 4-figure supplement 1G.

Author response image 16.

Basal OCR in Rho0 cells



We hope that these new data address the reviewer's concerns about the Cox2 mutant.

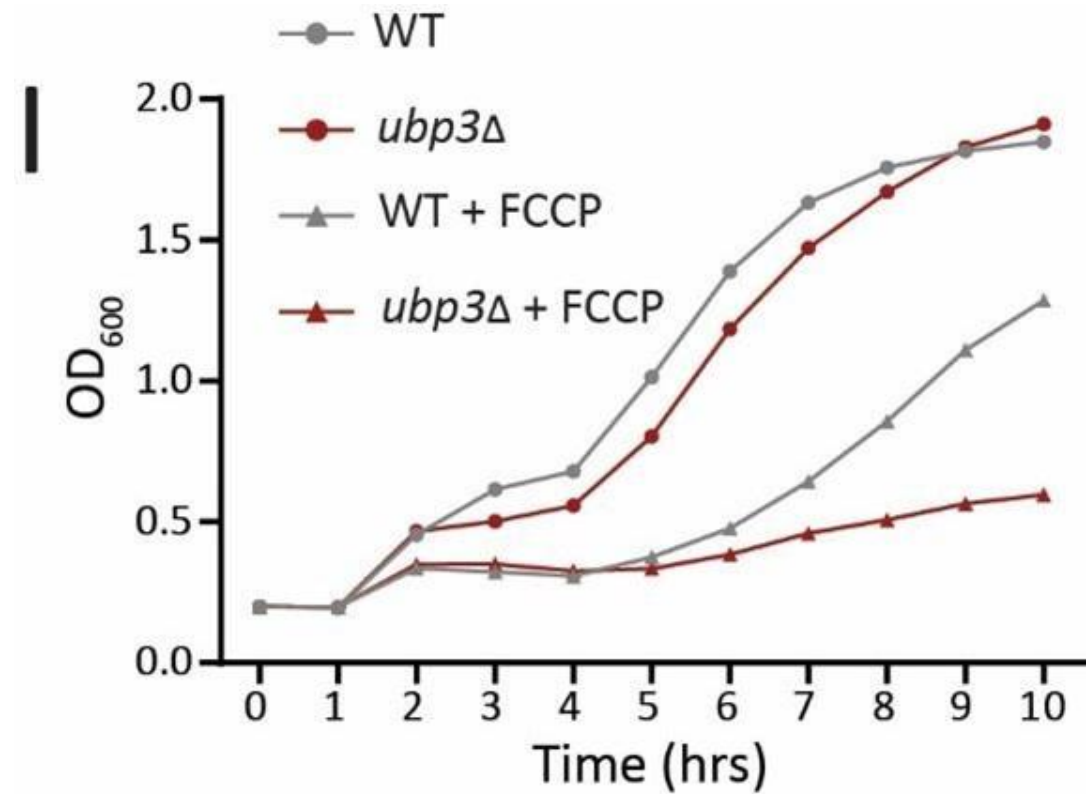
(c) They used sodium azide in a lot of assays to inhibit complex IV. However, this chemical is nonspecific and broadly affects many ATPases as well. Not sure why they do not use more specific inhibitors that are commonly used to assay OCR in seahorse.

We have now performed growth assays for WT and ubp3Δ cells in the presence of specific mitochondrial OXPHOS inhibitors - oligomycin and FCCP. We observe a more severe growth defect in ubp3Δ cells compared to WT cells in the presence of oligomycin and FCCP, similar to

the results observed with sodium azide. All these data are now included in the revised manuscript as Figure 1I, Figure1-figure supplement 1H, along with associated text.

Author response image 17.

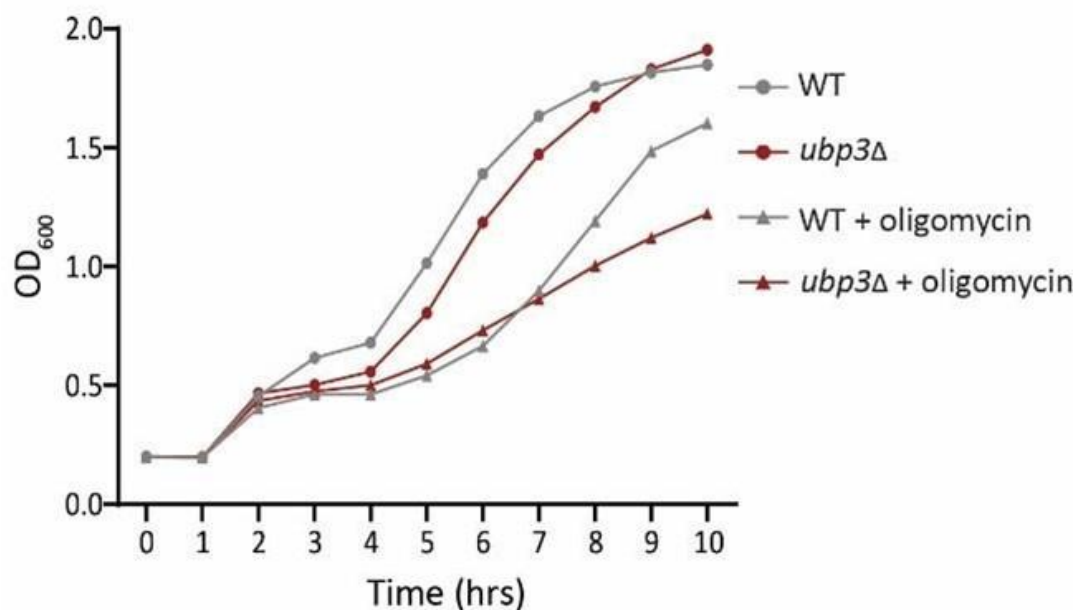
Growth rate in the presence of FCCP



Author response image 18.

Figure1-figure supplement 1H- Growth rate in the presence of oligomycin

H



We hope that these new data addresses the reviewer's concerns.

(d) The authors measured cellular Pi level by grinding the entire cells to release Pi. However, this will lead to a mix of cytosolic and vacuolar Pi. Related to this caveat, the cytosol has ~50mM Pi, while only 1-2mM of these glycolysis metabolites, I am not sure why the reduction of several glycolysis enzymes will cause significant changes in cytosolic Pi levels and make Pi the limiting factor for mitochondrial respiration. One possibility is that the observed cytosolic Pi level changes were caused by the measurement issue mentioned above.

The Pi estimation shown in figure 3 C, E, F and G is a measure of total Pi in the cells. The vacuole is a major storehouse of phosphate in cells. However, unlike plant cells where free phosphate is stored in vacuoles, yeast vacuoles store phosphate only in the form of polyphosphates (Yang et al., 2017, Hürlimann et al., 2007). The free Pi formed from the hydrolysis of polyphosphate is subsequently transported to cytosol via the exporter Pho91 (Hürlimann et al., 2007). This therefore makes cytosol and mitochondria the major storage of usable free Pi in yeast. Since the malachite green assay that we use for phosphate estimation is specific to free Pi, and not polyphosphate, the Pi estimates that we show in figure 3 come from a combination of cytosolic and mitochondrial Pi. As explained earlier, in order to specifically measure mitochondrial Pi, we have established methods to rapidly isolate mitochondria, and then followed this by estimating Pi in these isolated mitochondria (Figure 4B). Here we clearly see a large increase in mitochondrial Pi in the Ubp3 deletion cells. This allows us to estimate the changes in Pi levels that specific to mitochondria, without relying only on total Pi changes.

"the cytosol has ~50mM Pi, while only 1-2mM of these glycolysis metabolites, I am not sure why the reduction of several glycolysis enzymes will cause significant changes in cytosolic Pi levels and make Pi the limiting factor for mitochondrial respiration"

The reviewer has completely missed the fact that the glycolytic rate in yeast is the highest known for any cell. While the steady state levels of glycolytic metabolites might be ~2 mM, the process of glycolysis is not static but is rapid and continuous. Glucose is continuously broken down and converted to pyruvate, along with the consumption of Pi and generation of ATP. This is the reason for the rapid ¹³C label saturation (within seconds of ¹³C glucose addition) in yeast cells (Figure 2-figure supplement 1F). This instantaneous label saturation makes accurate flux measurements arduous because of which we had to optimize a method for measuring glycolytic flux in yeast cells (Figure 2-D, Figure 2-figure supplement 1F). Indeed, for that reason, our measurements of glycolytic flux in yeast are the first time this is being reported in the field. This in itself is an enormously challenging experiment, and establishes a new benchmark.

In highly glycolytic cells, most of the ATP is synthesized via glycolysis and the rate of glycolysis and ATP synthesis is very high. In the reaction catalysed by GAPDH, Pi and ADP is converted to ATP. This ATP formed acts as a Pi donor to most of the Pi consuming reactions in the cells. Some of these processes such as protein translation utilizes ATP, but releases Pi and ADP and this Pi enters the cellular Pi pool. Several other reactions such as nucleotide biosynthesis, polyphosphate biosynthesis and protein phosphorylation use ATP as a Pi donor and the Pi is fixed in biomolecules. Increasing the rates of these 'Pi sinks' therefore can result in a decrease in Pi pools. This is a concept we have earlier tried to clarify more elaborately in (Gupta and Laxman, 2021). In fact, increasing nucleotide biosynthesis and polyphosphate synthesis has earlier been suggested to decrease available free Pi (Austin and Mayer 2020, Desfougères et al., 2016). When glycolytic flux is high, this is coupled/tuned to the consumption of Pi which will be correspondingly high due to increased ATP, nucleotide and polyphosphate synthesis. Pi levels rapidly decrease upon glucose addition, due to the continuous Pi consumption during glycolysis (Hohmann et al., 1996, Van Heerden et al., 2014, Koobs et al., 1972). Therefore, changes in glycolytic rate due to change in glycolytic enzyme levels can result in significant changes in Pi levels due to changes in Pi consumption rate.

Our results also show that apart from Pi levels, the glycolytic state can regulate mitochondrial Pi transport as well. This is the reason for mitochondrial Pi levels and basal OCR not increasing merely by adding Pi to cells. We show that basal OCR can be increased by adding Pi in the presence of 2DG. This regulation of mitochondrial Pi transport is a major limiting factor for mitochondrial respiration and could be mediated partly by the regulating of Mir1 levels and also by the changes in the cytosolic pH which regulates the rate of mitochondrial Pi transport. We have discussed these points in the discussion section in our manuscript.

We hope that this clarifies the reviewer's concerns regarding how changes in glycolytic rate can regulate changes in cytosolic Pi levels.

(e) The authors used Δmir1 and MIR1 OE to show that Pi viability in the mitochondrial matrix is important for mitochondrial activity and biogenesis. This is not surprising as Pi is a key substrate required for OXPHOS activity. I doubt the approach of adding a control to determine whether Pi has a specific regulatory function, while other OXPHOS substrates, like ADP, O₂ etc do not have the same effect.

To clarify, we only used the mir1Δ cells to understand the requirement for Pi transport from cytosol to mitochondria in controlling respiration. The reviewer is correct in stating that deletion of Mir1 would reduce Pi import to mitochondria and thereby inhibit respiration. This is exactly the conclusion we suggest from this experiment as stated in the manuscript – "These data suggest that mitochondrial Pi transport (via Mir1) is critical for maintaining basal mitochondrial activity even in high glucose". We have only used these experiments to support the idea that even though glycolysis and mitochondria are in different compartments, a

change in Pi balance in one compartment (cytosol) can affect Pi levels in the other (mitochondria) since there is Pi transport between these two compartments. Since mitochondria has its own polyphosphate reserves, in the absence of these experiments with mir1Δ cells it can be imagined that mitochondria PolyP can be an additional source of Pi to support respiration, and therefore changes in cytosolic Pi may have only a minor effect on mitochondrial respiration. Our experiments with mir1Δ and Mir1-OE cells indubitably suggest that Pi transport to mitochondria from cytosol is important for respiration, and therefore changes in cytosolic Pi levels (or maintaining cytosolic Pi at a lower level due to the rate of glycolysis) will have rippling effects in mitochondrial Pi availability. Further, these data suggest that for example under glycolytic inhibition (low glucose, or 2DG), while all factors (signalling, substrate availability etc) favour respiration (and mitochondrial derepression), cells cannot achieve this in the absence of ample Pi transport from cytosol. This therefore places Pi at the centre stage in controlling mitochondrial respiration.

We conclude that Pi is a major, but not the only constraint for mitochondrial respiration. There certainly could be a role for ADP, oxygen availability etc in regulating respiration. However, these are beyond the scope of our study. We have discussed about the potential role of ADP in regulating mitochondrial repression in the discussion section. “An additional consideration is the possible contribution of changes in ADP in regulating mitochondrial activity, where the use of ADP in glycolysis might limit mitochondrial ADP. Therefore, when Pi changes as a consequence of glycolysis, it could be imagined that a change in ADP balance can coincidentally occur. However, prior studies show that even though cytosolic ADP decreases in the presence of glucose, this does not limit mitochondrial ADP uptake, or decrease respiration, due to the very high affinity of the mitochondrial ADP transporter.”

We hope that this clarifies the reviewer’s concerns regarding the use of Mir1 OE and mir1Δ strains.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

Some of the experiments should be repeated in other strain backgrounds for reproducibility and rigor.

As discussed in the response to point number 1, we have now utilized multiple strains of *S. cerevisiae* to test our findings. We now find that our discoveries regarding the role of altered Pi budgeting as a constraint for mitochondrial respiration, and the role of Ubp3 as a regulator of mitochondrial repression are conserved across multiple genetic backgrounds of *S. cerevisiae*. These results are included in the revised manuscript as a new figure- Figure 6 and associated text. We used the W303, Σ1278 and BY4742 strains of *S. cerevisiae* to show that deletion of Ubp3 increases mitochondrial activity (as shown by increased mitochondrial membrane potential and increased Cox2 levels). Using the W303 strain we show that the deletion of Ubp3 increases Pi levels and that the increased Pi is necessary for increasing mitochondrial respiration (Figure 6C, D). These added experiments have substantially broadened the generality of our findings.

The number of biological repeats needs to be increased in all experiments.

We have increased the number of biological repeats in key experiments that shows that the increased Pi levels are necessary for the increased mitochondrial respiration in *ubp3Δ* and *tdh2Δtdh3Δ* cells (revised Figure 4F). Apart from a few basal OCR measurements and mitotracker data in supplementary figure, all our experiments are performed for 3 biological repeats. In case of basal OCR measurements, yeast cells have to be aliquoted to poly-L-lysine coated Seahorse plates and centrifuged to ensure that the cells are properly settled. This is

due to the non-adherent nature of yeast cells. During the centrifugation step, the wells in the two end rows cannot be utilized due to uneven settling of cells which affects the basal OCR readings in these wells. In case of several experiments that involve multiple samples, we were therefore limited to restrict the number of biological replicates to 2 (repeated independently), so that all samples could be accommodated in the plate.

| *Full western blot images should be supplemented along with the other data.*

The complete western blot images are now included in the revised manuscript as supplementary file 2.

| *TCA cycle flux should be analyzed and presented in the study to conclude some of the findings.*

As discussed in detail in the response to point number 2, we have performed steady state and flux measurements for TCA cycle intermediates. This data is now included as a new supplement figure- Figure 2-figure supplement 2.

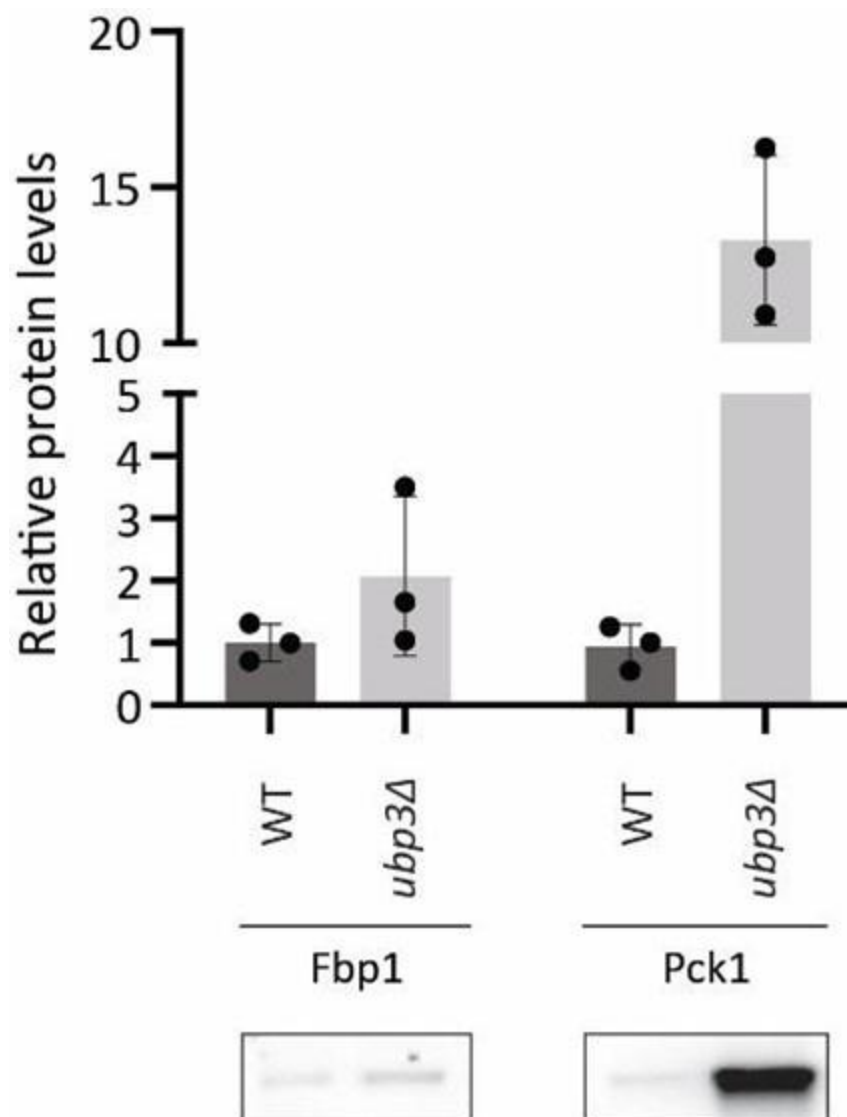
Reviewer #2 (Recommendations For The Authors):

(1) In Fig. 2A, they should also include the gluconeogenesis enzymes (fructose 1,6 biphosphatase, PEP carboxykinase, and pyruvate carboxylase) to exclude the possibility that glycolytic intermediates are not rerouted to gluconeogenesis.

We measured the protein levels of Fbp1 (fructose 1,6 bisphosphatase) and Pck1 (PEP carboxykinase). We observed an increase in the protein levels in both enzymes in *ubp3Δ*. The data is shown below.

Author response image 19.

Fbp1 and Pck1 protein levels



While we agree that this is an interesting observation which might help us in understanding the metabolic rewiring in *ubp3Δ*, we have not included this data in the current revised version of the manuscript due to two main reasons.

(1) Since *ubp3Δ* cells have a defective glycolysis and therefore a defective glucose repression, the mRNA and protein levels of gluconeogenic enzymes which are usually glucose-repressed might increase. This might be a response at the level of transcription and translation of these enzymes and might or might not change the rate of gluconeogenesis in these cells. This is because of multiple other factors that regulate gluconeogenic flux such as allosteric, mass action etc. Therefore, to avoid confounding our main points and since we cannot make a conclusive assumption on the gluconeogenic metabolism in these mutants, we don't include this data. The primary focus of our story is the mitochondrial repression component. Understanding the feedback controls that alter gluconeogenesis in these mutants is beyond the scope of this study and could be addressed in a separate future study.

(2) As we highlight extensively in the response letter and in the manuscript, our aim is not to understand the specific mechanistic role of Ubp3. In this manuscript, we identify the conserved constraints that control mitochondrial repression without focusing just on the role of Ubp3 in regulating this. Whether Ubp3 regulates gluconeogenesis is a question that could

be addressed in a future study that focuses on identifying the altered signalling mechanisms in *ubp3Δ* and the targets of Ubp3.

| (2) In line 292, page 10, there is a typo "dermine".

We apologize for this mistake. Corrected.

| (3) In Figure 5A, is there a reason why they chose 0.1% glucose condition as a low glucose condition? Also, is there a dose-dependent change in OCR or other mitochondrial functions according to the concentration of glucose?

The glucose concentration of 0.1% was selected to decrease (but not completely remove) the available glucose. 0.1% glucose is considered as a standard low glucose condition in *S. cerevisiae* (Yin et al., 2003) and the effect of this glucose concentration on cellular processes has been extensively studied (Yin et al., 2003, Takeda et al., 2015 etc). <0.2% glucose is the critical threshold for activating respiratory metabolism (Takeda et al., 2015) and shifting cells to 0.1% glucose in our experiments will activate respiration, as we show in our data. However, this is very different from completely removing glucose or using an alternate carbon source such as ethanol, because this would result in full activation of gluconeogenesis. We further find that when cells are grown in ethanol, the gluconeogenic activation will also change the Pi homeostasis. This will in part be a result of the fully reversed direction of the GAPDH catalysed reaction (Figure 3G). If such a condition is used, it could lead to misinterpretations, and confound the conclusions that we make from these set of experiments where Pi homeostasis play a major role. In 0.1% glucose it has been shown that gluconeogenesis is still partly repressed (Yin et al., 2003). The pathways utilizing alternate carbon sources still remain repressed (even though to a lower extend compared to 2% glucose) in 0.1% glucose (Yin et al., 2003). We hope that this clarifies the concerns regarding the rationale behind using 0.1% glucose in our experiments.

The extent of glucose repression is dependent on the concentration of glucose. Glucose concentration >1% has been shown to activate degradation of mRNAs involved in alternate carbon utilization. Different signaling pathways involved in growth under glucose and glucose repression is regulated by glucose concentration. This is discussed in detail in Yin et al., 2002. We (Figure 5figure supplement 1A) also observe a dose dependent increase in mitochondrial membrane potential in the presence of 2DG. This also suggests that the rate of glycolysis (which could be also mediated by changes in glucose concentration) can regulate the extent of mitochondrial derepression.